

# Supporting Materials for “Automatic Differentiation to Improve Parameter Estimation for Partially Observed Markov Processes”

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## Supplementary Content

|                                                                                                                                 |             |
|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>S1 Proof of Theorem 1 (DMOP-<math>\alpha</math> Functional Forms)</b>                                                        | <b>S-2</b>  |
| <b>S2 Proof of Theorem 2 (Optimization Convergence Analysis)</b>                                                                | <b>S-5</b>  |
| <b>S3 Feynman-Kac Models and Monte Carlo Approximations</b>                                                                     | <b>S-10</b> |
| <b>S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling</b> | <b>S-13</b> |
| <b>S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates</b>                                      | <b>S-19</b> |
| <b>S6 Proof of Theorem 5 (Bias-Variance Analysis)</b>                                                                           | <b>S-20</b> |
| <b>S7 Sharper Bias-Variance Analysis</b>                                                                                        | <b>S-24</b> |
| <b>S8 Figures for Bayesian Inference</b>                                                                                        | <b>S-67</b> |
| <b>S9 Justification for the Choice of Baseline Comparisons</b>                                                                  | <b>S-70</b> |
| <b>S10 Algorithmic Parameters</b>                                                                                               | <b>S-71</b> |

## S1 Proof of Theorem 1 (DMOP- $\alpha$ Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an  $A$  superscript is used to denote ancestral trajectories, so that  $x_{1:n,j}^{A,P,\theta}$  contains the history of the prediction particle  $x_{n,j}^{P,\theta}$  at observation times  $1:n$ . Similarly,  $g_{i,j}^{A,P,\theta}$  is the ancestral weight of  $x_{n,j}^{P,\theta}$  at observation time  $i$ .

**Lemma S1.** Write  $\nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$  for the DMOP- $\alpha$  gradient estimate (or, equivalently, the gradient of MOP- $\alpha$  when  $\theta = \phi$ ). Consider the case where we use the after-resampling conditional likelihood estimate so that  $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$ . When  $\alpha = 1$ ,

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta), \quad (\text{S1})$$

yielding the bootstrap filter estimator of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

*Proof.* Consider the case of MOP- $\alpha$  when  $\alpha = 1$  and  $\theta = \phi$ . We then have a nice telescoping product property for the after-resampling likelihood estimate:

$$\begin{aligned} \hat{\mathcal{L}}^1(\theta) &:= \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n-1,j}^{F,\theta}} \\ &= \left( \frac{1}{J} \sum_{j=1}^J w_{N,j}^{F,\theta} \right) \prod_{n=1}^N L_n^{\phi}, \end{aligned} \quad (\text{S2})$$

where the third equality follows from the choice of  $\alpha = 1$ , and the fourth equality is the resulting telescoping property. The log-derivative identity lets us decompose the score estimate as

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{\nabla_{\theta} \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{\nabla_{\theta} \left( \frac{1}{J} \sum_{j=1}^J w_{N,j}^{F,\theta} \right) \prod_{n=1}^N L_n^{\phi}}{\prod_{n=1}^N L_n^{\phi}} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta}. \quad (\text{S3})$$

From (S3), we see that the derivative of the log-likelihood estimate is

$$\nabla_{\theta} \hat{\ell}^1(\theta) := \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta}. \quad (\text{S4})$$

We proceed to decompose (S4). First, observe that as  $\alpha = 1$ ,

$$w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta} \frac{g_{n,j}^{\theta}}{g_{n,j}^{\phi}} = \prod_{i=1}^n \frac{g_{i,j}^{A,P,\theta}}{g_{i,j}^{A,P,\phi}}, \quad (\text{S5})$$

Note that the right hand side of (S5) is the cumulative product of measurement density ratios over the ancestral trajectory of the  $j$ -th prediction particle at timestep  $n$ . We then use the log-derivative identity again, yielding the following expression for the gradient of the log-weights as the sum of the log measurement densities over the ancestral trajectory,

$$\frac{\nabla_{\theta} w_{n,j}^{P,\theta}}{w_{n,j}^{P,\theta}} = \nabla_{\theta} \log w_{n,j}^{P,\theta} = \nabla_{\theta} \log \left( \prod_{i=1}^n \frac{g_{i,j}^{A,P,\theta}}{g_{i,j}^{A,P,\phi}} \right) \quad (\text{S6})$$

$$= \nabla_{\theta} \sum_{i=1}^n \left( \log g_{i,j}^{A,P,\theta} - \log g_{i,j}^{A,P,\phi} \right) \quad (\text{S7})$$

$$= \sum_{i=1}^n \nabla_{\theta} \log g_{i,j}^{A,P,\theta}. \quad (\text{S8})$$

This is equal to the gradient of the logarithm of the conditional density of the observed measurements given the ancestral trajectory of the  $j$ -th prediction particle up to timestep  $n$ ,

$$\nabla_{\theta} \sum_{n=1}^N \log g_{n,j}^{A,P,\theta} = \nabla_{\theta} \log \left( \prod_{n=1}^N g_{n,j}^{A,P,\theta} \right) \quad (\text{S9})$$

$$= \nabla_{\theta} \log \left( \prod_{n=1}^N f_{Y_n|X_n}(y_n^* | x_{n,j}^{A,P,\theta}; \theta) \right) \quad (\text{S10})$$

$$= \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta). \quad (\text{S11})$$

Multiplying both sides of the expression by  $w_{N,j}^{P,\theta}$  yields an expression for the gradient of the weights at timestep  $N$ ,

$$\nabla_{\theta} w_{N,j}^{P,\theta} = w_{N,j}^{P,\theta} \sum_{n=1}^N \nabla_{\theta} \log g_{n,j}^{A,P,\theta} = w_{N,j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,P,\theta}; \theta). \quad (\text{S12})$$

Substituting the above identity into the log-likelihood decomposition obtained earlier in Equation S3 yields

$$\begin{aligned} \nabla_{\theta} \hat{\ell}^1(\theta) &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,k_j}^{P,\theta} \\ &= \frac{1}{J} \sum_{j=1}^J w_{N,k_j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta). \end{aligned} \quad (\text{S13})$$

Finally, observing that  $\theta = \phi$  implies  $w_{N,j}^{F,\theta} = 1$ , we obtain

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,j}^{A,F,\theta}; \theta). \quad (\text{S14})$$

This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle.  $\square$

Note that the variance of the DMOP- $\alpha$  log-likelihood derivative estimate scales poorly with  $N$  when  $\theta \neq \phi$ . This can be seen by observing that

$$\nabla_{\theta} \hat{\ell}^1(\theta) = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,j}^{F,\theta} = \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} w_{N,k_j}^{P,\theta}$$

$$= \frac{1}{J} \sum_{j=1}^J w_{N,k_j}^{P,\theta} \nabla_{\theta} \log f_{Y_{1:N}|X_{1:N}}(y_{1:N}^* | x_{1:n,k_j}^{A,P,\theta}; \theta). \quad (\text{S15})$$

When  $\theta \neq \phi$ , we see that  $w_{N,k_j}^{P,\theta} = O(c^N)$ . When  $\theta = \phi$ , a generic normalized path-functional inequality gives the coarse upper bound  $O(N^4/J)$  after rescaling to the full unnormalized score. Section S7 gives an exact constant-weight calculation with order  $N^2/J$  at  $\alpha = 1$  and explains why this example does not by itself provide a general finite- $J$  upper bound.

**Lemma S2.** Write  $\nabla_{\theta} \hat{\ell}^{\alpha}(\theta)$  for the DMOP- $\alpha$  gradient estimate (or, equivalently, the gradient of MOP- $\alpha$  when  $\theta = \phi$ ). Consider the case where we use the after-resampling conditional likelihood estimate so that  $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$ . When  $\alpha = 0$ ,

$$\nabla_{\theta} \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left( f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S16})$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

*Proof.* First, write

$$s_{n,j} = \frac{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\phi}; \theta)} \quad (\text{S17})$$

as shorthand for the measurement density ratios. Observe that, when  $\alpha = 0$ , the likelihood estimate becomes

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^{\phi} \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \quad (\text{S18})$$

$$= \prod_{n=1}^N L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \quad (\text{S19})$$

$$= \prod_{n=1}^N L_n^{\phi} \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{P,\phi}; \theta)}. \quad (\text{S20})$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when  $\theta = \phi$  is therefore

$$\nabla_{\theta} \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_{\theta} \log \left( L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S21})$$

$$= \sum_{n=1}^N \frac{\nabla_{\theta} \left( L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left( L_n^{\phi} \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \quad (\text{S22})$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_{\theta} s_{n,j}}{\sum_{j=1}^J s_{n,j}} \quad (\text{S23})$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_{\theta} f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} \quad (\text{S24})$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left( f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S25})$$

where we use the log-derivative trick in the second equality, observe that  $\sum_{j=1}^J s_{n,j} = J$  when  $\theta = \phi$  in the fourth equality, and use the log-derivative trick again while noting that  $\theta = \phi$  in the fifth equality. This yields the desired result.  $\square$

## S2 Proof of Theorem 2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter  $\theta = \phi$ . Let  $Z_n$  contain the complete ancestral path through time  $n$ , the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^{\theta}(Z_n) = f_{Y_n|X_n}(y_n^* | X_n^{\theta}; \theta), \quad W_{\theta}(Z_N) = \prod_{n=1}^N G_n^{\theta}(Z_n), \quad (\text{S26})$$

$$\psi_{\theta}(Z_N) = \nabla_{\theta}^{\text{tot}} \log W_{\theta}(Z_N), \quad \Xi_{\theta}(Z_N) = \nabla_{\theta}^{2,\text{tot}} \log W_{\theta}(Z_N). \quad (\text{S27})$$

The superscript “tot” means that the derivative passes through the reparameterized simulated path while the base random numbers and the discrete resampling indices are held fixed.

Write  $\rho_N^J = J^{-1} \sum_{j=1}^J \delta_{Z_{N,j}}$  for the empirical measure of the after-resampling genealogical particles, and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_{\theta}(Z_N)F(Z_N)\}}{\mathbb{E}\{W_{\theta}(Z_N)\}}. \quad (\text{S28})$$

Assumptions (A3) and (A5) give  $|\partial_{\theta_i} W_{\theta}| \leq N \bar{G}^N G'(\theta)$ , so dominated convergence and the log-derivative identity yield

$$\nabla_{\theta} \ell(\theta) = \rho_N(\psi_{\theta}), \quad \nabla_{\theta} \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_{\theta}). \quad (\text{S29})$$

The second equality is Lemma S1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let  $Z_0$  be the augmented prediction path at observation 1; for  $q = 1, \dots, N-1$ , let  $Z_q$  be the prediction path at observation  $q+1$ ; and let  $Z_N$  be the terminal path after the final resampling. For  $0 \leq q \leq N$ , write  $\rho_q^J = J^{-1} \sum_{j=1}^J \delta_{Z_{q,j}}$ , where the  $Z_{0,j}$  are independent draws from the unweighted prediction law of  $Z_0$  and  $Z_{N,j} = Z_{N,j}$ . Write  $V_q^{\theta}(Z_q) = G_{q+1}^{\theta}(Z_q)$  for  $q = 0, \dots, N-1$ . Let the locally indexed augmented kernel  $\mathcal{M}_{q+1}^{\theta}$  append the next independent simulator innovation and its gradients for  $0 \leq q \leq N-2$ , and put  $\mathcal{M}_N^{\theta} = \text{Id}$  for the final selection. The one-step population map is

$$\Phi_{q+1}^{\theta}(\mu)(dz') = \frac{\int \mu(dz) V_q^{\theta}(z) \mathcal{M}_{q+1}^{\theta}(z, dz')}{\mu(V_q^{\theta})}, \quad 0 \leq q < N. \quad (\text{S30})$$

Conditionally on the current cloud, the  $J$  children are independent with common law  $\Phi_{q+1}^\theta(\rho_q^J)$ . The final observation is handled by the  $q = N - 1$  transition, namely multinomial selection with potential  $V_{N-1}^\theta = G_N^\theta$  followed by the identity mutation. Thus  $\rho_N^J$  is exactly the terminal empirical measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For  $0 \leq q \leq N$ , let

$$Q_{q,N}^\theta F(z_q) = \mathbb{E} \left[ F(Z_N) \prod_{r=q}^{N-1} V_r^\theta(Z_r) \middle| Z_q = z_q \right], \quad (\text{S31})$$

$$P_{q,N}^\theta F = \frac{Q_{q,N}^\theta F}{Q_{q,N}^\theta 1}, \quad g_{q,N} = \sup_{z,z'} \frac{Q_{q,N}^\theta 1(z)}{Q_{q,N}^\theta 1(z')}, \quad b_{q,N} = \sup_{\text{osc}(F) \leq 1} \text{osc}(P_{q,N}^\theta F), \quad (\text{S32})$$

where  $Q_{N,N}^\theta F = F$  and  $\text{osc}(F) = \sup_{z,z'} |F(z) - F(z')|$ . Thus  $0 \leq b_{q,N} \leq 1$ . With

$$R_G = \frac{\bar{G}}{\underline{G}}, \quad S_N = \sum_{q=0}^N g_{q,N}^3 b_{q,N}, \quad K_N = 32 S_N, \quad K_N^{\text{pf}} = 32 \sum_{r=0}^N R_G^{3r}, \quad (\text{S33})$$

the potential bounds give, term by term,

$$\underline{G}^{N-q} \leq Q_{q,N}^\theta 1(z_q) \leq \bar{G}^{N-q}, \quad g_{q,N} \leq R_G^{N-q}, \quad K_N \leq K_N^{\text{pf}}. \quad (\text{S34})$$

In particular, the constant used in the theorem is the closed-form quantity

$$K_N^{\text{pf}} = \begin{cases} 32(N+1), & R_G = 1, \\ 32 \frac{R_G^{3(N+1)} - 1}{R_G^3 - 1}, & R_G > 1. \end{cases} \quad (\text{S35})$$

Here is the constant calculation behind the tail bound used below. Let  $r_N^{\text{DM}}$  and  $\beta_N^{\text{DM}}$  denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r_N^{\text{DM}} \leq 4S_N, \quad (\beta_N^{\text{DM}})^2 \leq 4 \sum_{q=0}^N g_{q,N}^2 b_{q,N}^2 \leq 4S_N, \quad (\text{S36})$$

where the last inequality uses  $g_{q,N} \geq 1$  and  $0 \leq b_{q,N} \leq 1$ . If  $S_N = 0$ , the particle error is identically zero. Hence suppose  $S_N > 0$ . For  $u \in (0, 1/2]$ , set  $x = Ju^2/(32S_N)$ . If  $x \leq \log 6$ , then  $6e^{-x} \geq 1$  and the desired probability bound is trivial. If  $x > \log 6$ , use  $h^{-1}(x) \leq 2x + 2\sqrt{x}$  for  $h(t) = \{t - \log(1+t)\}/2$  and  $1 + 2x + 2\sqrt{x} \leq 8x$  to obtain

$$\begin{aligned} \frac{r_N^{\text{DM}}}{J} \{1 + h^{-1}(x)\} &\leq \frac{u^2}{8x} \{1 + 2x + 2\sqrt{x}\} \leq u^2 \leq \frac{u}{2}, \\ \beta_N^{\text{DM}} \sqrt{\frac{2x}{J}} &\leq 2\sqrt{S_N} \sqrt{\frac{2x}{J}} = \frac{u}{2}. \end{aligned}$$

Theorem 1.2 applied to both  $F$  and  $-F$ , after centering and scaling  $F$  to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr(|(\rho_N^J - \rho_N)(F)| > \text{osc}(F)u) \leq 6 \exp\left(-\frac{Ju^2}{K_N}\right) \leq 6 \exp\left(-\frac{Ju^2}{K_N^{\text{pf}}}\right). \quad (\text{S37})$$

for every  $F$  with  $\text{osc}(F) > 0$  and every  $u \in (0, 1/2]$ . If  $F$  is constant,  $(\rho_N^J - \rho_N)(F) = 0$  identically. Equations S33– S37 replace all unnamed model-dependent concentration constants.

**Lemma S3** (Explicit concentration of the DMOP-1 gradient). *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1 \leq i \leq p} \text{osc}(\psi_{\theta,i}), \quad u_g(J, \delta) = \sqrt{\frac{K_N^{\text{pf}}}{J} \log \frac{6p}{\delta}}. \quad (\text{S38})$$

If  $J \geq 4K_N^{\text{pf}} \log(6p/\delta)$ , then, with probability at least  $1 - \delta$ ,

$$\|\nabla_{\theta} \hat{\ell}_J^{A,1}(\theta) - \nabla_{\theta} \ell(\theta)\|_2 \leq A_N(\theta) \sqrt{p} u_g(J, \delta). \quad (\text{S39})$$

Moreover, Assumption (A3) gives  $A_N(\theta) \leq 2NG'(\theta)$ . Consequently, the explicit condition

$$J \geq K_N^{\text{pf}} \log \frac{6p}{\delta} \max \left\{ 4, \frac{4pN^2 G'(\theta)^2}{\epsilon^2} \right\} \quad (\text{S40})$$

implies that the left side of equation S39 is at most  $\epsilon$  with probability at least  $1 - \delta$ .

*Proof.* The lower bound on  $J$  makes  $u_g(J, \delta) \leq 1/2$ . Apply equation S37 to each normalized coordinate  $\psi_{\theta,i}/A_N(\theta)$ ; a constant coordinate has zero error and needs no normalization. A union bound over the  $p$  coordinates gives

$$\max_{1 \leq i \leq p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \leq A_N(\theta) u_g(J, \delta)$$

with probability at least  $1 - \delta$ . Now use  $\|v\|_2 \leq \sqrt{p}\|v\|_{\infty}$  and equation S29. Finally,  $|\psi_{\theta,i}| \leq NG'(\theta)$ , so  $A_N(\theta) \leq 2NG'(\theta)$ ; substituting this inequality and solving  $2NG'(\theta) \sqrt{p} u_g(J, \delta) \leq \epsilon$  gives equation S40.  $\square$

The next result treats the actual particle Hessian. It retains the outer-product and empirical-mean terms created by differentiating the logarithm of the terminal empirical average.

**Lemma S4** (Explicit concentration of the DMOP-1 particle Hessian). *Suppose in addition that Assumption (A3) gives the entrywise bound  $|\Xi_{\theta,ab}(Z_N)| \leq NG''(\theta)$  for  $1 \leq a, b \leq p$ . Define*

$$d_p = \frac{p(p+3)}{2}, \quad H_N(\theta) = 2p\{NG''(\theta) + 3N^2 G'(\theta)^2\}, \quad u_H(J, \delta) = \sqrt{\frac{K_N^{\text{pf}}}{J} \log \frac{6d_p}{\delta}}. \quad (\text{S41})$$

If  $J \geq 4K_N^{\text{pf}} \log(6d_p/\delta)$ , then, with probability at least  $1 - \delta$ ,

$$\|\nabla_{\theta}^2 \hat{\ell}_J^{A,1}(\theta) - \nabla_{\theta}^2 \ell(\theta)\|_{\text{op}} \leq H_N(\theta) u_H(J, \delta). \quad (\text{S42})$$

Fix  $0 < \kappa < \gamma$ . If

$$\gamma I_p \preceq -\nabla_{\theta}^2 \ell(\theta) \preceq \Gamma I_p \quad \text{and} \quad J \geq K_N^{\text{pf}} \log \frac{6d_p}{\delta} \max \left\{ 4, \frac{H_N(\theta)^2}{\kappa^2} \right\}, \quad (\text{S43})$$

then, with probability at least  $1 - \delta$ ,

$$(\gamma - \kappa) I_p \preceq -\nabla_{\theta}^2 \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma + \kappa) I_p. \quad (\text{S44})$$

In particular, the raw particle curvature is then positive definite and invertible.

*Proof.* For fixed reference-filter random numbers and resampling indices, the telescoping identity in the proof of Lemma S1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_{\phi} + \log \left\{ \frac{1}{J} \sum_{j=1}^J \exp(a_j(\theta)) \right\}, \quad a_j(\theta) = \log w_{N,j}^{F,\theta}, \quad a_j(\phi) = 0,$$

where  $C_{\phi}$  is constant in  $\theta$ . Twice differentiating this log-sum-exp at  $\theta = \phi$  gives the exact finite-particle identity

$$\nabla_{\theta}^2 \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_{\theta} + \psi_{\theta} \psi_{\theta}^T) - \rho_N^J(\psi_{\theta}) \rho_N^J(\psi_{\theta})^T. \quad (\text{S45})$$

Assumption (A3) also gives the entrywise dominating bound

$$|\partial_{\theta_a \theta_b}^2 W_{\theta}| = W_{\theta} |\Xi_{\theta,ab} + \psi_{\theta,a} \psi_{\theta,b}| \leq \bar{G}^N \{NG''(\theta) + N^2 G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating  $\log \mathbb{E} W_{\theta}$  gives the population identity

$$\nabla_{\theta}^2 \ell = \rho_N(\Xi_{\theta} + \psi_{\theta} \psi_{\theta}^T) - \rho_N(\psi_{\theta}) \rho_N(\psi_{\theta})^T. \quad (\text{S46})$$

Let  $A_1 = 2NG'(\theta)$  and  $A_2 = 2\{NG''(\theta) + N^2 G'(\theta)^2\}$ . The  $p$  scalar functions  $\psi_a$  have oscillation at most  $A_1$ , and the symmetric entries of  $\Xi + \psi \psi^T$  have oscillation at most  $A_2$ . Apply equation S37 jointly to these functions, using  $d_p = p + p(p+1)/2$  scalar events. On the resulting event,

$$\begin{aligned} \|(\rho_N^J - \rho_N)(\psi)\|_2 &\leq A_1 \sqrt{p} u_H(J, \delta), \\ \|(\rho_N^J - \rho_N)(\Xi + \psi \psi^T)\|_{\text{op}} &\leq p A_2 u_H(J, \delta). \end{aligned}$$

Both  $\|\rho_N^J(\psi)\|_2$  and  $\|\rho_N(\psi)\|_2$  are at most  $\sqrt{p} NG'(\theta)$ . Therefore

$$\begin{aligned} \|\rho_N^J(\psi) \rho_N^J(\psi)^T - \rho_N(\psi) \rho_N(\psi)^T\|_{\text{op}} \\ \leq 2\sqrt{p} NG'(\theta) \|(\rho_N^J - \rho_N)(\psi)\|_2 \leq 4p N^2 G'(\theta)^2 u_H(J, \delta). \end{aligned}$$

Combining the last three displays with equation S45 and equation S46 proves equation S42. Condition equation S43 makes its right side at most  $\kappa$ ; Weyl's eigenvalue inequality then gives equation S44.  $\square$

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If  $\tilde{B} = Q \text{diag}(\lambda_1, \dots, \lambda_p) Q^T$  is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q \text{diag}([\lambda_1]_c^C, \dots, [\lambda_p]_c^C) Q^T, \quad [x]_c^C = \min\{C, \max\{c, x\}\}. \quad (\text{S47})$$

For every  $v \in \mathbb{R}^p$ , direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \leq v^T B v \leq C\|v\|_2^2, \quad C^{-1}\|v\|_2^2 \leq v^T B^{-1} v \leq c^{-1}\|v\|_2^2. \quad (\text{S48})$$

Thus  $cI_p \preceq B \preceq CI_p$  and  $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$ . Lemma S4 shows that the unclipped particle Hessian has these properties with  $c = \gamma - \kappa$  and  $C = \Gamma + \kappa$  under equation S43; equation S47 guarantees the chosen  $c, C$  bounds for every realization.

*Proof of Theorem 2.* Let  $\tau$  and  $\mathcal{E}_M$  be the stopping time and simultaneous gradient event defined in Theorem 2. If  $\mathcal{F}_m$  is the history before the  $m$ th particle-filter call, then  $\theta_m$  and  $\{m \leq \min(\tau, M-1)\}$  are  $\mathcal{F}_m$ -measurable. Lemma S3, with failure probability  $\delta/M$ , gives

$$\begin{aligned} \Pr(\mathcal{E}_M^c \mid \theta_0) &\leq \sum_{m=0}^{M-1} \mathbb{E} [\mathbf{1}_{\{m \leq \min(\tau, M-1)\}} \Pr(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m) \mid \theta_0] \\ &\leq \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta. \end{aligned} \quad (\text{S49})$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on  $\mathcal{E}_M$  and write  $p_m = -B_m^{-1}g_m$ . If  $\|g_m\|_2 \leq \sigma\epsilon$ , then

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon \leq (\sigma + 1)\epsilon. \quad (\text{S50})$$

Strong convexity implies  $f(\theta) - f(\theta^*) \leq \|\nabla f(\theta)\|_2^2/(2\gamma)$ ; therefore

$$f(\theta_m) - f(\theta^*) \leq \frac{(\sigma + 1)^2 \epsilon^2}{2\gamma}. \quad (\text{S51})$$

Suppose instead that  $\|g_m\|_2 > \sigma\epsilon$ . Since  $g_m = -B_m p_m$  and  $cI_p \preceq B_m \preceq CI_p$ ,

$$c\|p_m\|_2 \leq \|g_m\|_2 \leq C\|p_m\|_2, \quad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \quad (\text{S52})$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned} f(\theta_m + \eta p_m) - f(\theta_m) &\leq \eta \nabla f(\theta_m)^T p_m + \frac{\Gamma \eta^2}{2} \|p_m\|_2^2 \\ &\leq -\eta p_m^T B_m p_m + \eta \epsilon \|p_m\|_2 + \frac{\Gamma \eta^2}{2} \|p_m\|_2^2. \end{aligned} \quad (\text{S53})$$

The choices of  $\sigma$  and  $\eta$  imply

$$\frac{C}{\sigma} \|p_m\|_2^2 \leq \frac{c(1-\beta)}{2} \|p_m\|_2^2, \quad \frac{\Gamma \eta}{2} \|p_m\|_2^2 \leq \frac{c(1-\beta)}{2} \|p_m\|_2^2.$$

Using  $p_m^T B_m p_m \geq c \|p_m\|_2^2$  in equation S53 now gives

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta\beta p_m^T B_m p_m. \quad (\text{S54})$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon < \frac{\sigma + 1}{\sigma} \|g_m\|_2. \quad (\text{S55})$$

Consequently, equation S48 and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\geq \frac{1}{C} \left( \frac{\sigma}{\sigma + 1} \right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\geq \frac{2\gamma}{C} \left( \frac{\sigma}{\sigma + 1} \right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \quad (\text{S56})$$

Combining equation S54 and equation S56 proves

$$f(\theta_{m+1}) - f(\theta^*) \leq q \{f(\theta_m) - f(\theta^*)\}, \quad q = 1 - \frac{2\eta\beta\gamma}{C} \left( \frac{\sigma}{\sigma + 1} \right)^2. \quad (\text{S57})$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1 - \beta) \frac{c\gamma}{C\Gamma} \leq \frac{1}{2}, \quad \text{so} \quad \frac{1}{2} < q < 1.$$

Iteration gives the  $q^m$  conclusion in the theorem, while equation S51 gives its stopping conclusion after substituting  $f = -\ell$ .  $\square$

The result is deliberately restricted to DMOP-1. For  $\alpha < 1$ , the population limit is a discounted gradient. A theorem for that case must add a finite-sample bound for the discount bias relative to the exact score; choosing  $\alpha$  close to one does not by itself provide such a bound.

### S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let  $(\eta_n)_{n=1}^N, (\pi_n)_{n=1}^N, (\rho_n)_{n=1}^N$  be sequences of probability measures on the state space  $\mathcal{X}$ . This is the sequence of prediction distributions  $f_{X_n|Y_{1:n-1}}$ , filtering distributions  $f_{X_n|Y_{1:n}}$ , and posterior distributions  $f_{X_{1:n}|Y_{1:n}}$  that we seek to approximate with the particle filter. For any measurable bounded functional  $h$ , we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

**Markov kernels and the process model:** A Markov kernel  $M$  with source  $\mathcal{X}_1$  and target  $\mathcal{X}_2$  is a map  $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \rightarrow [0, 1]$  such that for every set  $A \in \mathcal{B}(\mathcal{X}_2)$  and every point  $x \in \mathcal{X}_1$ , the map  $x \mapsto M(x, A)$  is a measurable function of  $x$ , and the map  $A \mapsto M(x, A)$  is a probability measure on  $\mathcal{X}_2$ . The quantity  $M(x, A)$  can be thought of as the probability of transitioning to the set  $A$  given that we are at the point  $x$ . If this yields a density, this then corresponds to the process density  $f_{X_2|X_1}$  conditional on  $x$  and integrated over  $A$ .

**Markov kernels and measures:** For any measure  $\eta$ , any Markov kernel  $M$  on  $\mathcal{X}$ , any point  $x \in \mathcal{X}$  and any measurable subset  $A \subseteq \mathcal{X}$ , let

$$\eta(h) = \int h d\eta = \int h(x)\eta(dx), \quad (\text{S58})$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \quad (\text{S59})$$

$$(Mh)(x) = \int M(x, dy)h(y). \quad (\text{S60})$$

**Compositions of Markov kernels:** The composition of a Markov kernel  $M_1$  with another Markov kernel  $M_2$  is another Markov kernel, given by

$$(M_1 M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \quad (\text{S61})$$

**Total variation distance:** The total variation distance between two measures  $\mu$  and  $\nu$  on  $\mathcal{X}$  is

$$\|\mu - \nu\|_{\text{TV}} = \sup_{\|h\|_{\infty} \leq 1/2} |\mu(h) - \nu(h)| = \sup_{\text{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \quad (\text{S62})$$

**Dobrushin contraction:** The Dobrushin contraction coefficient  $\beta_{\text{TV}}$  of a Markov kernel  $M$  is given by

$$\beta_{\text{TV}}(M) = \sup_{x, y \in \mathcal{X}} \|M(x, \cdot) - M(y, \cdot)\|_{\text{TV}} = \sup_{\mu, \nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\text{TV}}}{\|\mu - \nu\|_{\text{TV}}}. \quad (\text{S63})$$

**Potential functions and the measurement model:** A potential function  $G : \mathcal{X} \rightarrow [0, \infty)$  is a non-negative function of an element of the state space  $x \in \mathcal{X}$ . In our case, this corresponds to the measurement model, and in our previous notation is written as  $g_{n,j} = f_{Y_n|X_n}(y_n^*|x_{n,j}^F; \theta) = G_n(x_{n,j}^F)$ , where we suppress the dependence on  $\theta$  for notational simplicity. Note that  $G_n(\cdot) = f_{Y_n|X_n}(y_n^*|\cdot)$  is the conditional density of the observed measurement at time  $n$ , where we condition on the filtering particle  $x_{n,j}^F$  as an element of the state space.

**Feynman-Kac models:** A Feynman-Kac model on  $\mathcal{X}$  is a tuple  $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$  of an initial probability measure on the state space  $\pi_0$ , a sequence of transition kernels  $(M_n)_{n=1}^N$ , and a sequence of potential functions  $(G_n)_{n=1}^N$ . In the notation used in the main text, this corresponds to the starting distribution  $f_{X_0}$ , the sequence of transition densities  $f_{X_n|X_{n-1}}$ , and the measurement densities  $f_{Y_n|X_n}$ . This induces a set of mappings from the set of probability measures on  $\mathcal{X}$  to itself,  $\mathcal{P}(\mathcal{X}) \rightarrow \mathcal{P}(\mathcal{X})$ , as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \quad (\text{S64})$$

- The map from the prediction distribution at timestep  $n$  to timestep  $n + 1$  is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n)M_{n+1}. \quad (\text{S65})$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time  $k$  to the prediction distribution at time  $n$  where  $k \leq n$ ,

$$\Phi_{k,n} = \Phi_n \circ \dots \circ \Phi_{k+1}. \quad (\text{S66})$$

**The particle filter:** The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep  $n$  is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^P}, \quad (\text{S67})$$

and the filtering measure at timestep  $n$  is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^J \delta_{x_{n,j}^F}. \quad (\text{S68})$$

In a slight abuse of notation, we will identify  $x_{n,1:J}^P \equiv \eta_n^J$ , and  $x_{n,1:J}^F \equiv \pi_n^J$ . As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on  $\mathcal{X}^J$ . The corresponding Markov transition kernel is then

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = (\Phi_n(\eta_{n-1}^J))^{\otimes J} = \left( \Phi_n \left( \frac{1}{J} \sum_{j=1}^J \delta_{x_{n-1,j}^P} \right) \right)^{\otimes J}, \quad (\text{S69})$$

and the composition of Markov kernels on particles from timestep  $n$  to timestep  $n + k$  is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \quad (\text{S70})$$

One may wonder why Karjalainen et al. (2023) require this process to evolve on  $\mathcal{X}^J$ . This is because at every timestep  $n$ , we in fact draw  $X_{n,j}^P | \{X_{n-1,1:J}^P = x_{n-1,1:J}^P\} \sim \mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \eta_0^{\otimes J} \mathbf{M}_{0,n}$  for  $j = 1, \dots, J$ .

**Forgetting of the particle filter:** The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP- $\alpha$ . That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}(\mathbf{M}_{n,n+k}) \leq (1 - \epsilon)^{\lfloor k/(O(\log J)) \rfloor}, \quad (\text{S71})$$

for some  $\epsilon$  dependent on  $\bar{G}, \underline{G}, \bar{M}, \underline{M}$  in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of  $O(\log(J))$  timesteps.

Equipped with the above formalisms and results, we are now in a position to provide guarantees on the performance of MOP- $\alpha$  itself.

## S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general result, from which it will be clear that MOP- $\alpha$  targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

**Definition S1** (Off-Parameter Resampled Particle Filters). *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model  $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ , where we inductively define given some  $x_{n-1,j}^F, w_{n-1,j}^F$  comprising the filtering measure approximation  $\pi_n^J$ :*

1. *The prediction particles at timestep  $n$ ,  $x_{n,j}^P$ , are drawn from  $X_{n,j}^P \sim \pi_n^J M_n$ , and the prediction weights are given by  $w_{n,j}^P = w_{n-1,j}^F$ .*
2. *The prediction measure at timestep  $n$  is given by*

$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \quad (\text{S72})$$

3. *The particles are resampled at every timestep  $n$ , yielding indices  $k_j$  according to some arbitrary probabilities  $(p_{n,j})_{j=1}^J$ .*
4. *The filtering measure at timestep  $n$  is given by*

$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P g_{n,j}}{\sum_{j=1}^J w_{n,j}^P g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \quad (\text{S73})$$

5. *The prediction weights at the next timestep are given by  $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}$ .*

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

**Definition S2** (Targeting). *A pair of random vectors  $(X, W)$  drawn from some measure  $g$  **targets** another measure  $\pi$  if, for any measurable and bounded functional  $h$ ,*

$$E_g[h(X) \cdot W] = E_\pi[h(X)]. \quad (\text{S74})$$

*A particle approximation is a triangular array of pairs of random vectors  $(X_j^J, W_j^J), j = 1, 2, \dots, J$ . The particle approximation **targets**  $\pi$  if, for any measurable and bounded functional  $h$ ,*

$$\frac{\sum_{j=1}^J h(X_j^J) W_j^J}{\sum_{j=1}^J W_j^J} \xrightarrow{\text{a.s.}} E_\pi[h(X)] \quad (\text{S75})$$

as  $J \rightarrow \infty$ .

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

**Lemma S5** (Change of Weight Measure). *Suppose that  $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$  targets  $f_X$ . Now, let  $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$  be a multinomial sample with indices  $k_j$  drawn from  $\{(\tilde{X}_j^J, U_j^J)\}$  where  $(\tilde{X}_j^J, U_j^J)$  is represented, on average, proportional to  $\pi_j^J J$  times. Write*

$$(Y_j^J, V_j^J) = (\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J). \quad (\text{S76})$$

*If the importance sampling weights  $U_j / \pi_j$  are bounded, then  $\{(Y_j^J, V_j^J), j = 1, \dots, J\}$  targets  $f_X$ .*

*Proof.* Note that as the  $Y_j^J$  are a subsample from  $X_j^J$ ,  $h$  can be a function of  $Y$  as well as it is one for  $X$ . We then expand

$$\frac{\sum_j h(Y_j^J) V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \quad (\text{S77})$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[f(X)]. \quad (\text{S78})$$

We want to show

$$\frac{\sum_j h(X_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} 0. \quad (\text{S79})$$

For this, it is sufficient to show that

$$\sum_j h(\tilde{X}_{k_j}^J) \frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J} \xrightarrow{a.s.} 0 \quad (\text{S80})$$

since an application of this result with  $h(x) = 1$  provides almost sure convergence of the denominator in (S79). Write  $g(\tilde{X}_j^J) = h(\tilde{X}_j^J) \frac{U_j^J}{\pi_j^J}$ . We therefore need to show that

$$\sum_j Z_j^J := \sum_j \left( g(\tilde{X}_{k_j}^J) - g(\tilde{X}_j^J) \pi_j^J \right) \xrightarrow{a.s.} 0. \quad (\text{S81})$$

Because the functional  $h$  and importance sampling weights  $u_{k_j}^J/\pi_{k_j}^J$  are bounded, we have that  $\mathbb{E}[(Z_j^J)^4] < \infty$ . We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the  $Z_j^J$  are conditionally independent given the  $\tilde{X}_j^J$  and  $U_j^J$ ,

$$\begin{aligned} \mathbb{E} \left[ \left( \sum_j Z_j^J \right)^4 \middle| (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] &= J \mathbb{E} \left[ (Z_1^J)^4 | (\tilde{X}_j^J, U_j^J)_{j=1}^J \right] \\ &\quad + 3J(J-1) \left( \mathbb{E}[(Z_j^J)^2 | (\tilde{X}_j^J, U_j^J)_{j=1}^J] \right) \end{aligned} \quad (\text{S82})$$

$$\leq CJ^2, \quad (\text{S83})$$

for some  $C > 0$ . Taking expectations on both sides yields

$$\mathbb{E} \left[ \left( \sum_j Z_j^J \right)^4 \right] \leq CJ^2 \quad (\text{S84})$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P} \left( \left| \frac{1}{J} \sum_{j=1}^J Z_j^J \right| > \epsilon \right) \leq \frac{1}{\epsilon^4 J^4} \mathbb{E} \left[ \left( \sum_{j=1}^J Z_j^J \right)^4 \right] \leq \frac{C}{\epsilon^4 J^2}, \quad (\text{S85})$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every  $\epsilon > 0$ , giving us the almost-sure convergence for

$$\sum_j h(X_{k_j}^J) \frac{u_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(X_j^J) \frac{u_j^J}{\pi_j^J} \xrightarrow{a.s.} 0. \quad (\text{S86})$$

Similarly, we also have that

$$\sum_j \frac{u_j^J}{\pi_j^J} \pi_j^J - \sum_j \frac{u_{k_j}^J}{\pi_{k_j}^J} \pi_{k_j}^J \xrightarrow{a.s.} 0, \quad (\text{S87})$$

and the result is proved.  $\square$

**Remark:** Note that Lemma S5 permits  $\pi_{1:J}$  to depend on  $\{(X_j, u_j)\}$  as long as the resampling is carried out independently of  $\{(X_j, u_j)\}$ , conditional on  $\pi_{1:J}$ .

**Lemma S6** (Particle Marginals). *Suppose that  $\{(\tilde{X}_j^J, U_j^J), j = 1, \dots, J\}$  targets  $f_X$ . Also suppose that  $\tilde{Z}_j^J \sim f_{Z|X}(\cdot | \tilde{X}_j^J)$  where  $f_{Z|X}$  is a conditional probability density function corresponding to a joint density  $f_{X,Z}$  with marginal densities  $f_X$  and  $f_Z$ . Then, if the  $U_j^J$  are bounded,  $\{(\tilde{Z}_j^J, U_j^J)\}$  targets  $f_Z$ .*

*Proof.* We want to show that, for any measurable bounded  $h$ ,

$$\frac{\sum_j h(\tilde{Z}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}[\mathbb{E}_{f_{Z|X}}[h(Z)|X]]. \quad (\text{S88})$$

By assumption, for any measurable and bounded functional  $g$  with domain  $\mathcal{X}$ ,

$$\frac{\sum_j g(\tilde{X}_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S89})$$

Let  $\bar{U}_j^J = \frac{J U_j^J}{\sum_j U_j^J}$ . Examine the numerator and denominator of the quantity

$$\frac{J^{-1} \sum_j h(\tilde{Z}_j^J) U_j^J}{J^{-1} \sum_j U_j^J} = \frac{J^{-1} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J}{J^{-1} \sum_j \bar{U}_j^J}. \quad (\text{S90})$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J, \quad (\text{S91})$$

and by the same fourth moment argument to the above lemma, it converges almost surely to the limit of its expectation,

$$\lim_{J \rightarrow \infty} \mathbb{E} \left[ \frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \right] = \lim_{J \rightarrow \infty} \mathbb{E} \left[ \frac{1}{J} \sum_j \mathbb{E} [h(\tilde{Z}_j^J) \bar{U}_j^J | \tilde{X}_j^J, \bar{U}_j^J] \right] \quad (\text{S92})$$

$$= \lim_{J \rightarrow \infty} \mathbb{E} \left[ \frac{1}{J} \sum_j \mathbb{E} [h(Z) | X = \tilde{X}_j^J] \bar{U}_j^J \right]. \quad (\text{S93})$$

Applying (S89) with  $g(x) = \mathbb{E}[h(Z)|X = x]$ , the average on the right hand side converges almost surely to  $\mathbb{E}\{\mathbb{E}[h(Z)|X]\} = \mathbb{E}[h(Z)]$ . It remains to swap the limit and expectations. We can do so with the bounded convergence theorem, and therefore obtain

$$\frac{1}{J} \sum_j h(\tilde{Z}_j^J) \bar{U}_j^J \xrightarrow{a.s.} \mathbb{E}_{f_Z}[h(Z)]. \quad (\text{S94})$$

□

**Lemma S7** (Particle Posteriors). *Suppose that  $\{(X_j^J, U_j^J), j = 1, \dots, J\}$  targets  $f_X$ . Also suppose that  $(X_j^J, U_j^J) = (X_j^J, U_j^J f_{Z|X}(z^*|X_j^J))$ . Then, if  $U_j^J f_{Z|X}(z^*|X_j^J)$  and  $U_j^J f_{Z|X}(z^*|X_j^J)/f_Z(z^*)$  are bounded,  $\{(X_j^J, U_j^J)\}$  targets  $f_{X|Z}(\cdot|z^*)$ .*

*Proof.* Again, we want to show that

$$\frac{\sum_j h(X_j^J) \cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \xrightarrow{a.s.} \mathbb{E}_{f_{X|Z}}[h(X)|z^*]. \quad (\text{S95})$$

We already have that for any measurable bounded  $g$ ,

$$\frac{\sum_j g(X_j^J) U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}[g(X)]. \quad (\text{S96})$$

Consider the following:

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \times \left( \frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \right)^{-1}. \quad (\text{S97})$$

We will apply Equation (S96) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1} \sum_j h(X_j^J) f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [h(X) f_{Z|X}(z^*|X)], \quad (\text{S98})$$

while the reciprocal of the denominator converges to

$$\frac{J^{-1} \sum_j f_{Z|X}(z^*|X_j^J) U_j^J}{J^{-1} \sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X} [f_{Z|X}(z^*|X)] = f_Z(z^*). \quad (\text{S99})$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X} [h(X) f_{Z|X}(z^*|X)]}{f_Z(z^*)} = \mathbb{E}_{f_X} \left[ \frac{h(X) f_{Z|X}(z^*|X)}{f_Z(z^*)} \right] \quad (\text{S100})$$

$$= \mathbb{E}_{f_X} \left[ h(X) \frac{f_{X|Z}(X|z^*)}{f_X(X)} \right] = \mathbb{E}_{f_{X|Z}} [h(X)|z^*], \quad (\text{S101})$$

to give the desired result.  $\square$

**Theorem S1** (Off-Parameter Resampled Particle Filters Target the Filtering Distribution). *The off-parameter resampled particle filter as outlined in Definition S1 targets the filtering distribution.*

*Proof.* Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that  $\{(X_{n-1,j}^F, w_{n-1,j}^F)\}$  targets  $\pi_{n-1}$ . Then, Lemma S6 tells us that  $\{(X_{n,j}^P, w_{n,j}^P)\}$  targets  $\eta_n$ . Lemma S7 tells us that  $\{(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta)\}$  therefore targets  $\pi_n$ . Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^F, w_{n,j}^F) = (X_{n,k_j}^P, w_{n,k_j}^P g_{n,k_j} / p_{n,k_j}), \quad (\text{S102})$$

with resampling probabilities proportional to  $p_{n,j}$ , therefore also targets  $\pi_n$ .  $\square$

**Proposition S1** (MOP-1 Targets the Filtering Distribution). *When  $\alpha = 1$  or  $\phi = \theta$ , MOP- $\alpha$  targets the filtering distribution.*

*Proof.* When  $\theta = \phi$ , regardless of the value of  $\alpha$ , the ratio  $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$ , and this reduces to the vanilla particle filter estimate.

When  $\alpha = 1$ , and  $\theta \neq \phi$ , the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that  $\{(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta})\}$  is properly weighted for  $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*; \theta)$ . Then, Lemma S6 tells

us that  $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$  targets  $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$ . Lemma S7 tells us that  $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta} g_{n,j}^\theta)\}$  therefore targets  $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$ . Lemma S5 guarantees that the resampling rule, given by

$$(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}) = (X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta} g_{n,k_j}^\theta / g_{n,k_j}^\phi), \quad (\text{S103})$$

with resampling weights proportional to  $g_{n,j}^\phi$ , therefore also targets  $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*; \theta)$ .  $\square$

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

**Lemma S8** (Likelihood Proper Weighting).  *$f_{Y_n|Y_{1:n-1}}(y_n^*|y_{1:n-1}^*; \theta)$  is strongly consistently estimated by either the before-resampling estimate,*

$$L_n^{B,\theta} = \frac{\sum_{j=1}^J g_{n,j}^\theta w_{n,j}^{P,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}, \quad (\text{S104})$$

*or by the after-resampling estimate,*

$$L_n^{A,\theta} = L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}. \quad (\text{S105})$$

where  $L_n^\phi$  is as defined in the various algorithms.

Here, (S104) is a direct consequence of our earlier result that  $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$  targets  $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*; \theta)$ . To see (S105), we write the numerator of (S5) as

$$L_n^\phi \sum_{j=1}^J \left[ \frac{g_{n,j}^\theta}{g_{n,j}^\phi} w_{n,j}^{P,\theta} \right] \frac{g_{n,j}^\phi}{L_n^\phi} = L_n^\phi \sum_{j=1}^J w_{n,j}^{FC,\theta} \frac{g_{n,j}^\phi}{L_n^\phi}. \quad (\text{S106})$$

Using Lemma S5, we resample according to probabilities  $\frac{g_{n,j}^\phi}{L_n^\phi}$  to see this is properly estimated by

$$L_n^\phi \sum_{j=1}^J w_{n,j}^{F,\theta}, \quad (\text{S107})$$

from which we obtain (S105). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^N \left( L_n^\phi \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right). \quad (\text{S108})$$

Since  $w_{n,j}^{F,\theta} = w_{n+1,j}^{P,\theta}$ , this is a telescoping product. The remaining terms are  $\sum_{j=1}^J w_{0,j}^{P,\theta} = J$  on the denominator and  $\sum_{j=1}^J w_{N,j}^{F,\theta}$  on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$  should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

## S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now provide a result showing that under sufficient regularity conditions, one can interchange the order of differentiation and expectation of functionals of off-parameter resampled particle estimates, showing that if an off-parameter resampled particle estimate is consistent for some estimand, its derivative is consistent for the derivative of the estimand as well.

One may wonder why this may be necessary, given that we have already shown strong consistency for measurable bounded functionals in Theorem S1. If we require the gradient to be bounded as well, then the gradient is also a bounded functional. The answer is that the strong consistency is for expectations under the filtering distribution, so Theorem S1 only establishes strong consistency of the gradient of the estimate to its expectation under the filtering distribution,  $\nabla_\theta \eta_n^J(h_\theta) \xrightarrow{a.s.} \eta_n(\nabla_\theta h_\theta)$ , which is not a-priori the gradient of the estimand  $\nabla_\theta \eta_n(h_\theta)$ . We require an interchange of the derivative and expectation, which we show is possible when the particle estimates have two continuous uniformly bounded derivatives over all  $J$  below.

**Theorem S2** (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals). *Let  $h_\theta : \mathcal{X} \rightarrow \mathbb{R}$  be a measurable bounded functional of particles, where  $\eta_n^J(h_\theta)$  has two continuous derivatives uniformly bounded over all  $J$  by  $H^*$  for almost every  $\omega \in \Omega$  and  $\theta \in \Theta$ . If it holds that  $\eta_n^J(h_\theta) \xrightarrow{a.s.} \eta(h_\theta) = h_\theta^*$ , then we also have that  $\nabla_\theta \eta_n^J(h_\theta) \xrightarrow{a.s.} \eta_n(\nabla_\theta h_\theta) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*$ .*

*Proof.* Fix  $\omega \in \Omega$ . The sequence  $(\nabla_\theta \eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$  is uniformly bounded over all  $J$ , by assumption. The sequence is also uniformly equicontinuous. To see this, by assumption, the second derivative of  $\eta_n^J(h_\theta)(\omega)|_{\theta=\theta'}$  is also uniformly bounded over all  $J$  for almost every  $\omega \in \Omega$  and every  $\theta' \in \Theta$ . A set of functions with derivatives bounded by the same constant is uniformly Lipschitz, and therefore uniformly equicontinuous. So the sequence  $(\eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$  is uniformly equicontinuous over  $\theta$  for almost every  $\omega \in \Omega$ . Explicitly, for almost every  $\omega \in \Omega$  and every  $\epsilon > 0$ , there exists some  $\delta(\omega) > 0$  such that for every  $\|\theta - \theta'\|_\infty < \delta$  and every  $J \in \mathbb{N}$  we have that

$$\|\eta_n^J(h_\theta)(\omega) - \eta_n^J(h_{\theta'})(\omega)\|_\infty < \epsilon. \quad (\text{S109})$$

Then, by Arzela-Ascoli, there exists a uniformly convergent subsequence. We claim that there is only one subsequential limit. When the gradient is bounded, we can treat the gradient as a bounded functional. So by Theorem S1 the sequence  $(\eta_n^J(h_\theta)(\omega))_{J \in \mathbb{N}}$  converges pointwise for  $\theta = \phi$  and almost every  $\omega \in \Omega$ , and there is therefore only one subsequential limit. The sequence therefore converges uniformly to its limit  $\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega)$ . Therefore, with uniform convergence for the derivatives established, we can swap the limit and derivative, and obtain that for almost every  $\omega \in \Omega$ ,

$$\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega). \quad (\text{S110})$$

Again from Theorem S1, we know that

$$\eta_n^J(h_\theta)(\omega) \rightarrow \eta_n(h_\theta) = h_\theta^* \quad (\text{S111})$$

for almost every  $\omega \in \Omega$ . We then have that for almost every  $\omega \in \Omega$ ,

$$\lim_{J \rightarrow \infty} \eta_n^J(h_\theta)(\omega) = \nabla_\theta \eta_n(h_\theta) = \nabla_\theta h_\theta^*, \quad (\text{S112})$$

as we wanted.  $\square$

Theorem 4 is proved as a corollary of Theorem S2, where we apply  $\eta_n^J(h_\theta) = \hat{\mathcal{L}}_J^1(\theta)$ ,  $\eta_n(h_\theta) = \mathcal{L}(\theta) = h_\theta^*$ , and then use the continuous mapping theorem.

## S6 Proof of Theorem 5 (Bias-Variance Analysis)

In this section, we prove various rates on the bias, variance, and MSE of DMOP- $\alpha$ . The earlier truncation argument is retained below for comparison. The current DMOP variables are the particle location and the accumulated weight gradient. Forgetting of the location alone does not immediately control the accumulated genealogical score. We prove the DMOP-0 result below and retain the former extension to  $\alpha > 0$  inactive for comparison. Section S7 gives an exact finite- $J$  genealogy audit, the bounds that remain valid for every  $\alpha$ , and the centered genealogical lag-moment estimate used in the sharper variance theorem.

First, we note the following relation between the Dobrushin contraction coefficient and the alpha-mixing coefficients in our context below.

**Lemma S9.** *Setting  $X$  to be the particle collection at time  $m$ , and  $Y$  at time  $n$ , we have that the alpha mixing coefficients,*

$$\int |f_{XY}(x, y) - f_X(x) f_Y(y)| dx dy < \alpha, \quad (\text{S113})$$

*are bounded by the Dobrushin coefficient, i.e we have*

$$\int |f_{Y|X}(y|x_1) - f_{Y|X}(y|x_2)| dy < \alpha \quad (\text{S114})$$

*for all  $x_1, x_2$ .*

*Proof.* We rewrite the alpha-mixing assertion as

$$\int \int |f_{Y|X}(y|x) - f_Y(y)| dy f_X(x) dx. \quad (\text{S115})$$

We claim that the Dobrushin coefficient implies

$$\int |f_{Y|X}(y|x_1) - f_Y(y)| dy < \alpha \quad (\text{S116})$$

for all  $x_1$ . This is shown as follows:

$$\int |f_{Y|X}(y|x) - f_Y(y)| dy = \int \left| \int [f_{Y|X}(y|x_1) - f_{Y|X}(y|x)] f_X(x) dx \right| dy \quad (\text{S117})$$

$$< \int \int |f_{Y|X}(y|x_1) - f_{Y|X}(y|x)| dy f_X(x) dx \quad (\text{S118})$$

$$< \int \alpha f_X(x) dx \quad (\text{S119})$$

$$= \alpha \quad (\text{S120})$$

We then have the desired result.  $\square$

As a warm-up, we consider a variance bound for DMOP-0. Note that we made Assumptions (A2) and (A3) to leverage the results from Karjalainen et al. (2023) on the forgetting of the particle filter. Together with the current-location premise stated in the theorem below, this is used to recover the  $O(N/J)$  variance bound for DMOP-0. The former truncation extension to  $\alpha > 0$  is retained inactive for comparison. Section S7 gives the exact genealogical audit showing why the current-filter mixing argument cannot be transferred to the accumulated score functional. We start by decomposing the MOP- $\alpha$  estimator as follows, for the case  $\theta = \phi$ ,

$$\hat{\mathcal{L}}(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} = \prod_{n=1}^N L_n^{A,\theta,\alpha} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J (w_{n-1,j}^{F,\theta})^\alpha}. \quad (\text{S121})$$

Now, to illustrate the proof strategy for the general case in an easier context, we first analyze the special case when  $\alpha = 0$ . We now prove the variance bound for this case, presented below. To our knowledge, this result for the variance of the gradient estimate of Naesseth et al. (2018) is novel.

**Theorem S3** (Variance of MOP-0 Gradient Estimate). *Suppose the observations are fixed,  $J \geq 2$ ,  $\theta = \phi$ , Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Assume additionally that the total fixed-random-number derivative of each log measurement density is a bounded measurable function of the current physical particle location. Then the variance of DMOP-0, i.e. the algorithm of Naesseth et al. (2018), is*

$$\text{Var}(\nabla_\theta \hat{\ell}^0(\theta)) \lesssim \frac{Np G'(\theta)^2}{J}. \quad (\text{S122})$$

*Proof.* When  $\alpha = 0$ , we have

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,0} = \prod_{n=1}^N L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta,0}}{\sum_{j=1}^J w_{n,j}^{P,\theta,0}} \quad (\text{S123})$$

$$= \prod_{n=1}^N L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} = \prod_{n=1}^N L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\phi}; \phi)}. \quad (\text{S124})$$

Similarly to the proof of Lemma S2, we define

$$s_{n,j} = \frac{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{P,\phi}; \phi)}, \quad (\text{S125})$$

and from the exact same arguments conclude that its gradient when  $\theta = \phi$  is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_\theta \log \left( L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \quad (\text{S126})$$

$$= \sum_{n=1}^N \frac{\nabla_\theta \left( L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left( L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \quad (\text{S127})$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} \quad (\text{S128})$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_{\theta} f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta)}{f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\phi}; \phi)} \quad (\text{S129})$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left( f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right), \quad (\text{S130})$$

where we use the derivative of the logarithm, observe that  $\sum_{j=1}^J s_{n,j} = J$  when  $\theta = \phi$ , and use the derivative of the logarithm where  $\theta = \phi$  again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all  $N$  and  $J$ :

$$\nabla_{\theta} \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_{\theta} \log \left( f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right). \quad (\text{S131})$$

Therefore, for a given  $\theta_i$  in the parameter vector  $\theta$ ,

$$\text{Var} \left( \frac{\partial}{\partial \theta_i} \hat{\ell}^0(\theta) \right) = \frac{1}{J^2} \text{Var} \left( \sum_{n=1}^N \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left( f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \quad (\text{S132})$$

$$\begin{aligned} &= \frac{1}{J^2} \sum_{n=1}^N \text{Var} \left( \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left( f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \\ &\quad + 2 \sum_{m < n} \text{Cov} \left( \frac{1}{J} \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left( f_{Y_m|X_m}(y_m^*|x_{m,j}^{F,\theta}; \theta) \right), \frac{1}{J} \sum_{j=1}^J \frac{\partial}{\partial \theta_i} \log \left( f_{Y_n|X_n}(y_n^*|x_{n,j}^{F,\theta}; \theta) \right) \right) \\ &= \sum_{n=1}^N \text{Var} \left( \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m < n} \text{Cov} \left( \frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right). \end{aligned} \quad (\text{S133})$$

Here, we use Assumptions (A2) and (A3) that ensure strong mixing. We know from Theorem 3.1 of Karjalainen et al. (2023) that when  $\mathbf{M}_{n,n+k}$  is the  $k$ -step Markov operator from timestep  $n$  and  $\beta_{\text{TV}}(M) = \sup_{x,y \in E} \|M(x, \cdot) - M(y, \cdot)\|_{\text{TV}} = \sup_{\mu, \nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\text{TV}}}{\|\mu - \nu\|_{\text{TV}}}$  is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\text{TV}}(\mathbf{M}_{n,n+k}) \leq (1 - \epsilon)^{\lfloor k/(c \log(J)) \rfloor}, \quad (\text{S134})$$

i.e. the mixing time of the particle filter is  $O(\log(J))$ , where  $\epsilon$  and  $c$  depend on  $\bar{M}, \underline{M}, \bar{G}, \underline{G}$  in (A2) and (A3).

By the theorem's current-location premise, each time- $n$  contribution is a function of the current physical particle cloud. Lemma 2.3 of Karjalainen et al. (2023), followed by conditional multinomial resampling and the bounded Bayes-ratio step, gives the centered  $L^4$  bound below. Theorem 3.1 of the same paper gives the displayed particle-cloud mixing coefficient. At  $\alpha = 0$  these two results give

$$\begin{aligned} \left\| \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) - \mathbb{E} \left[ \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right] \right\|_{L^4} &\leq \frac{CG'(\theta)}{\sqrt{J}}, \\ a(k) &\leq (1 - \epsilon)^{\lfloor (k-1)/[c\{1+\log J\}] \rfloor}. \end{aligned} \quad (\text{S135})$$

Davydov's inequality must be applied to the centered variables. Since covariance is unchanged by centering, Equation (S135) gives

$$\text{Cov} \left( \frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) \leq (1 - \epsilon)^{\frac{1}{2} \lfloor \frac{|n-m|-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J}. \quad (\text{S136})$$

Putting it all together, we see that

$$\text{Var} \left( \frac{\partial}{\partial \theta_i} \hat{\ell}^0(\theta) \right) = \sum_{n=1}^N \text{Var} \left( \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) + 2 \sum_{m < n} \text{Cov} \left( \frac{\partial}{\partial \theta_i} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial \theta_i} \hat{\ell}_n^0(\theta) \right) \quad (\text{S137})$$

$$\leq \frac{NG'(\theta)^2}{J} + 2 \sum_{n=1}^{N-1} (N-n)(1 - \epsilon)^{\frac{1}{2} \lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \frac{G'(\theta)^2}{J} \quad (\text{S138})$$

$$\leq \frac{G'(\theta)^2}{J} \left( N + 2 \sum_{n=1}^{N-1} (N-n)(1 - \epsilon)^{\frac{1}{2} \lfloor \frac{n-1}{c\{1+\log(J)\}} \rfloor} \right) \quad (\text{S139})$$

$$\lesssim \frac{G'(\theta)^2 N}{J}. \quad (\text{S140})$$

It then follows that as  $\theta \in \Theta \subseteq \mathbb{R}^p$ ,

$$\text{Var}(\nabla_{\theta} \hat{\ell}^0(\theta)) \lesssim \frac{NpG'(\theta)^2}{J}. \quad (\text{S141})$$

□

Note that the factor of  $N$  that pops up here is due to the use of the unnormalized gradient. If one divides the gradient estimate by  $\sqrt{N}$  before usage, the variance does not depend on the horizon. If one divides by  $N$ , the error is  $O(1/\sqrt{NJ})$ .

Under the stated current-location premise, the order  $NpG'(\theta)^2/J$  is the verified DMOP-0 baseline. It is not asserted to be a lower bound for every model or every value of  $\alpha$ .

**Theorem S4** (MSE and variance bounds for DMOP- $\alpha$ ). *Suppose the observations are fixed,  $J \geq 2$ ,  $\alpha \in [0, 1)$ ,  $\theta = \phi$ , Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Assume additionally that there is a bounded measurable function  $h_n$  on the physical state space such that, under the fixed-random-number construction,*

$$\nabla_{\theta}^{\text{tot}} \log g_{n,k_j}^{\theta} \Big|_{\theta=\phi} = h_n(X_{n,j}^{F,\phi}) \quad \text{almost surely.}$$

Then

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} \lesssim pG'(\theta)^2 \left( \frac{N}{J} + \frac{N^2}{J^2} \right) \frac{1 + \alpha \varrho}{(1 - \alpha^2)(1 - \alpha \varrho)}, \quad (\text{S142})$$

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 \left[ \left( \frac{N}{J} + \frac{N^2}{J^2} \right) \frac{1 + \alpha \varrho}{(1 - \alpha^2)(1 - \alpha \varrho)} + N^2 \left\{ \frac{1 - \alpha}{1 - \alpha \varrho} \right\}^2 \right]. \quad (\text{S143})$$

The population component of the fixed-observation bias satisfies

$$\left\| \nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2 \leq CN\sqrt{p}G'(\theta) \frac{1 - \alpha}{1 - \alpha \varrho}. \quad (\text{S144})$$

Equation (S144) does not require the additional current-location premise: Lemma S10 first projects each bounded genealogical input onto its creation-time physical location. If  $J \geq N$ , the  $N^2/J^2$  term in Equation (S142) is absorbed by the  $N/J$  term, so

$$\text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} \lesssim \frac{NpG'(\theta)^2}{J} \frac{1 + \alpha \varrho}{(1 - \alpha^2)(1 - \alpha \varrho)}. \quad (\text{S145})$$

Section S7 gives the complete proof and an exact constant-weight calculation that checks the  $N$  and  $J$  dependence and explains the singularity of the closed  $\alpha < 1$  bound as  $\alpha \uparrow 1$ .

*Proof.* Lemma S13 proves Equation (S142). Theorem S5 proves Equation (S143) and Equation (S144).  $\square$

## S7 Sharper Bias-Variance Analysis

We now establish the estimates used in Theorem S4. Write  $\nabla_{\theta} \ell_n^{\alpha}(\theta)$  for the  $n$ th conditional score produced by the deterministic differentiated population recursion. At  $\theta = \phi$ , its  $\alpha = 1$  version is the exact conditional score  $\nabla_{\theta} \ell_n(\theta)$ . Three quantities must be distinguished. The physical location follows the ordinary particle transition to which Assumption (A2) applies. Its fixed-random-number simulator gradient is carried by the differentiation of that transition. The log-weight gradient obeys, along each selected lineage, the existing recursion

$$\nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \alpha \nabla_{\theta} \log w_{n-1,k_j}^{F,\theta} \Big|_{\theta=\phi} + \nabla_{\theta}^{\text{tot}} \log g_{n,k_j}^{\theta} \Big|_{\theta=\phi}. \quad (\text{S146})$$

The final term is the one-time measurement-gradient input called  $h_n$  below. It is the total derivative through the recursively simulated state, not merely the partial derivative with the state held fixed. Thus it is in general carried by the fixed-random-number simulator genealogy. The sharper results below state separately the additional case in which this input is a bounded function of the current physical location. Equation (S146) is the derivative at  $\theta = \phi$  of the log-space recursion

$$\begin{aligned} \log w_{n,j}^{P,\theta} &= \alpha \log w_{n-1,j}^{F,\theta}, \\ \log w_{n,j}^{F,\theta} &= \alpha \log w_{n-1,k_j}^{F,\theta} + \log g_{n,k_j}^{\theta} - \log g_{n,k_j}^{\phi}. \end{aligned} \quad (\text{S147})$$

At  $\theta = \phi$ , the log weights are zero but their gradients need not be. The factor  $\alpha$  contracts the inherited log-weight gradient. It does not contract the simulator gradient inside the newly created input  $h_n$ . Consequently, Assumption (A2) alone does not imply that the computational state consisting of the location, its simulator gradient, and its log-weight gradient is mixing in total variation. When  $h_n$  is a current-location function, the proof instead expands the log-weight gradient into its single-time inputs, applies location forgetting to those inputs, and sums their powers of  $\alpha$ . Throughout this section, the observations are fixed. Expectations and variances are over the simulator and particle-filter random variables, conditional on those observations.

We use Assumptions (A3) and (A5) in the precise pathwise sense already required by the differentiated MOP construction. For fixed simulator random numbers and fixed resampling indices, Assumption (A5) says that the recursively simulated state path and the composed log measurement densities are continuously differentiable in  $\theta$  near  $\phi$ , and that the joint law of the base simulator random numbers does not depend on  $\theta$ . The derivative in the last clause of Assumption (A3) is

the resulting total derivative: uniformly in  $n, j$  and in a neighborhood of  $\phi$ , after enlarging  $G'(\theta)$  to bound that neighborhood if necessary,

$$\left\| \nabla_{\theta} \log g_{n,j}^{A,F,\theta} \right\|_{\infty} \leq G'(\theta). \quad (\text{S148})$$

These two existing assumptions also justify differentiation under the finite-horizon simulator expectations used below. Indeed, the product rule and Equation (S148) give, pathwise,

$$\begin{aligned} & \left\| \nabla_{\theta} \prod_{s=1}^t G_s(X_s) \right\|_{\infty} \\ &= \left\| \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \nabla_{\theta} \log G_s(X_s) \right\|_{\infty} \\ &\leq \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \left\| \nabla_{\theta} \log G_s(X_s) \right\|_{\infty} \\ &\leq t \bar{G}^t G'(\theta). \end{aligned} \quad (\text{S149})$$

The final expression is deterministic and finite for each fixed  $t$ . For each coordinate of  $\theta$ , the mean-value theorem therefore bounds every difference quotient about  $\phi$  by the same quantity  $t \bar{G}^t G'(\theta)$ . The dominated convergence theorem gives

$$\nabla_{\theta} \mathbb{E}_U \left[ \prod_{s=1}^t G_s(X_s) \right] = \mathbb{E}_U \left[ \nabla_{\theta} \prod_{s=1}^t G_s(X_s) \right]. \quad (\text{S150})$$

Thus the required interchange follows from the clarified Assumptions (A3) and (A5); it is not an additional assumption. Geometric population forgetting follows from the backward-kernel contraction under Assumption (A2). Lemma S11 gives a uniform weak-error bound for a current filtering functional. Its proof is not used for a historical functional carried by the genealogy; that separate issue is audited below.

For the functionals below, evaluated at  $\theta = \phi$ , let  $\rho_t$  be the likelihood-tilted law of the complete fixed-random-number simulator history through time  $t$ , including the base simulator variables. Its  $X_{0:t}$ -marginal is the usual posterior path law. Thus, for every integrable functional  $h$  of that complete history,

$$\rho_t(h) = \frac{\mathbb{E} [h \prod_{s=1}^t G_s(X_s)]}{\mathbb{E} [\prod_{s=1}^t G_s(X_s)]}. \quad (\text{S151})$$

Here  $G_s(X_s)$  is the existing measurement potential. Future base simulator variables are fresh and conditionally independent of the complete past given the current physical location, as in the original POMP construction. The accumulated log-weight gradient is distinct from both the physical location and its simulator gradient; neither gradient is included in the transition density appearing in Assumption (A2).

Unrolling the existing filtered-weight recursion at  $\theta = \phi$  gives

$$\nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r \nabla_{\theta} \log g_{n-r,j}^{A,F,\theta} \Big|_{\theta=\phi}. \quad (\text{S152})$$

For a complete ancestral path, write  $h_k$ , only in this section, for the measurement-score contribution

$$h_k = \nabla_{\theta} \log g_k^{A,F,\theta} \Big|_{\theta=\phi} \quad (\text{S153})$$

carried on that path. In the path-space notation of Chan et al. (2018), the notation used below means explicitly

$$m(\xi_t) = \frac{1}{J} \sum_{j=1}^J \delta_{\xi_{t,j}}, \quad m(\xi_t)(h_k) = \frac{1}{J} \sum_{j=1}^J h_k(\xi_{t,j}), \quad 1 \leq k \leq t, \quad (\text{S154})$$

where  $\xi_{t,j}$  is the complete ancestral particle path carried by particle  $j$  at time  $t$ . Thus  $h_k(\xi_{t,j})$  is the single time- $k$  measurement-score contribution on that particle, not an arbitrary functional denoted by  $m$ . Equation (S151) gives its population counterpart.

We now identify the lag coefficients in the conditional score. At  $\theta = \phi$ , all weights equal one, so differentiating the two sample averages in the definition of  $L_n^{A,\theta,\alpha}$  gives

$$\begin{aligned} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} \\ &= \frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{F,\theta} \Big|_{\theta=\phi} - \frac{\alpha}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \end{aligned} \quad (\text{S155})$$

where the second equality uses  $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^{\alpha}$ . In the MOP- $\alpha$  recursion, the discounted-weight accumulation and process propagation preserve the index  $j$  and occur before the time- $n$  resampling step. The resampling index  $k_j$  enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta} \frac{g_{n,k_j}^{\theta}}{g_{n,k_j}^{\phi}}. \quad (\text{S156})$$

Consequently, the second average in Equation (S155) is pathwise

$$\frac{1}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n,j}^{P,\theta} \Big|_{\theta=\phi} = \frac{\alpha}{J} \sum_{j=1}^J \nabla_{\theta} \log w_{n-1,j}^{F,\theta} \Big|_{\theta=\phi}, \quad (\text{S157})$$

which is the empirical average over the time- $(n-1)$  filtered genealogy. Substituting Equation (S152) into Equation (S155) and changing the index in the second sum give

$$\begin{aligned} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r}) \\ &= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \{m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r})\}. \end{aligned} \quad (\text{S158})$$

Consequently, define

$$\Delta_{n,0}^J = m(\xi_n)(h_n), \quad \Delta_{n,0} = \rho_n(h_n),$$

$$\Delta_{n,r}^J = m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), \quad \Delta_{n,r} = \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \leq r < n. \quad (\text{S159})$$

We now verify that the right-hand population quantities are the lag coefficients of the deterministic differentiated recursion. Equations (S150) and (S153) give, at  $\theta = \phi$ ,

$$\begin{aligned} \nabla_\theta \log \mathbb{E}_U \left[ \prod_{s=1}^n G_s(X_s) \right] &= \frac{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s) \sum_{s=1}^n h_s]}{\mathbb{E}_U [\prod_{s=1}^n G_s(X_s)]} \\ &= \sum_{s=1}^n \rho_n(h_s). \end{aligned} \quad (\text{S160})$$

Subtracting the same identity at time  $n-1$  gives

$$\begin{aligned} \nabla_\theta \ell_n(\theta) &= \sum_{s=1}^n \rho_n(h_s) - \sum_{s=1}^{n-1} \rho_{n-1}(h_s) \\ &= \rho_n(h_n) + \sum_{r=1}^{n-1} \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\ &= \sum_{r=0}^{n-1} \Delta_{n,r}. \end{aligned} \quad (\text{S161})$$

Finally, differentiating the two population weighted averages in the deterministic MOP- $\alpha$  recursion and using Equation (S150) gives

$$\begin{aligned} \nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1} \alpha^r \rho_n(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r \rho_{n-1}(h_{n-r}) \\ &= \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\ &= \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}. \end{aligned} \quad (\text{S162})$$

**Lemma S10** (Lagwise first-derivative forgetting). *Suppose  $\theta = \phi$ . Under Assumptions (A2), (A3), and (A5), there are constants  $C < \infty$  and  $\varrho \in (0, 1)$  such that, for  $0 \leq r < n$ ,*

$$\|\Delta_{n,r}\|_2 \leq C\sqrt{p} G'(\theta) \varrho^r. \quad (\text{S163})$$

*Proof.* We derive the result from the posterior update and the contraction of the posterior backward kernels.

*Step 1: one-step posterior update.* First suppose that  $1 \leq r < n$ , and put  $k = n - r$ . Fix a coordinate  $i \in \{1, \dots, p\}$ , and let  $h_{k,i}$  be the corresponding coordinate of  $h_k$  in Equation (S153). By Equation (S148),

$$|h_{k,i}| \leq G'(\theta). \quad (\text{S164})$$

Because  $h_{k,i}$  is measurable with respect to the canonical extended path through time  $k \leq n-1$ , Equation (S151) and conditional expectation with respect to the new base simulator variable give

$$\begin{aligned}\rho_n(h_{k,i}) &= \frac{\mathbb{E}_U \left[ h_{k,i} \prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{ G_n(X_n) \mid X_{n-1} \} \right]}{\mathbb{E}_U \left[ \prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U \{ G_n(X_n) \mid X_{n-1} \} \right]} \\ &= \frac{\rho_{n-1} \{ h_{k,i} M_n(G_n) \}}{\rho_{n-1} M_n(G_n)},\end{aligned}\tag{S165}$$

where the first equality uses the independence of the new base simulator variable from the path through time  $n-1$ . The second equality first uses the identity below and then divides its numerator and denominator by  $\mathbb{E}_U \{ \prod_{s=1}^{n-1} G_s(X_s) \}$ . Using the manuscript's kernel notation, the required identity is

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n) G_n(x_n).\tag{S166}$$

The bounds on  $G_n$  and the fact that  $M_n$  is a Markov kernel imply for every  $x_{n-1}$  that

$$M_n(G_n)(x_{n-1}) \geq \underline{G} \int M_n(x_{n-1}, dx_n) = \underline{G},\tag{S167}$$

$$M_n(G_n)(x_{n-1}) \leq \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}.\tag{S168}$$

*Step 2: backward-kernel contraction.* We next control how much information about the path through time  $k$  can remain at time  $n-1$ . For  $s \geq 1$ , the posterior backward kernel from  $X_s$  to  $X_{s-1}$ , denoted below by  $B_s$ , is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)}.\tag{S169}$$

Assumption (A2) gives the following minorization for every measurable set  $A$ :

$$\begin{aligned}B_s(x_s, A) &= \frac{\int_A \pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)} \\ &\geq \frac{\underline{M} \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\ &= \frac{\underline{M}}{\bar{M}} \pi_{s-1}(A).\end{aligned}\tag{S170}$$

Put  $\epsilon = \underline{M}/\bar{M}$ . If  $\epsilon < 1$ , Equation (S170) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1 - \epsilon) R_s(x_s, \cdot),\tag{S171}$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1 - \epsilon}\tag{S172}$$

is a Markov kernel. Hence, for every bounded real function  $f$  and any  $x_s, x'_s$ ,

$$\begin{aligned} B_s(f)(x_s) - B_s(f)(x'_s) &= (1 - \epsilon)\{R_s(f)(x_s) - R_s(f)(x'_s)\}, \\ |B_s(f)(x_s) - B_s(f)(x'_s)| &\leq (1 - \epsilon) \operatorname{osc}(f). \end{aligned} \quad (\text{S173})$$

If  $\epsilon = 1$ , Equation (S170) implies  $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$ , so the left-hand side is zero. Taking the supremum over  $x_s, x'_s$  gives

$$\operatorname{osc}\{B_s(f)\} \leq (1 - \epsilon) \operatorname{osc}(f). \quad (\text{S174})$$

*Step 3: project and propagate the time- $k$  contribution.* The total derivative  $h_{k,i}$  may depend on the complete simulator genealogy through time  $k$ . Its conditional projection onto the physical location is

$$\bar{h}_{k,i}(X_k) = \mathbb{E}_{\rho_k}(h_{k,i} \mid X_k). \quad (\text{S175})$$

We first verify that the same projection applies under every later posterior path law. For  $t \geq k$ , let

$$L_{k,t}(X_k) = \mathbb{E}_U \left\{ \prod_{s=k+1}^t G_s(X_s) \middle| X_k \right\}, \quad L_{k,k} = 1. \quad (\text{S176})$$

For every bounded function  $q$  of  $X_k$ , the Markov property and Equation (S151) give

$$\begin{aligned} \rho_t\{h_{k,i}q(X_k)\} &= \frac{\mathbb{E}_U \left[ h_{k,i}q(X_k) \prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]}{\mathbb{E}_U \left[ \prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]} \\ &= \frac{\mathbb{E}_U \left[ \bar{h}_{k,i}(X_k)q(X_k) \prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]}{\mathbb{E}_U \left[ \prod_{s=1}^k G_s(X_s) L_{k,t}(X_k) \right]} \\ &= \rho_t\{\bar{h}_{k,i}(X_k)q(X_k)\}. \end{aligned} \quad (\text{S177})$$

The second equality is the defining conditional-expectation identity in Equation (S175), multiplied by the  $X_k$ -measurable function  $q(X_k)L_{k,t}(X_k)$ . Hence

$$\mathbb{E}_{\rho_t}(h_{k,i} \mid X_k) = \bar{h}_{k,i}(X_k), \quad t \geq k. \quad (\text{S178})$$

Repeated conditioning through the physical backward kernels therefore gives

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = x_{n-1}\} = (B_{n-1}B_{n-2} \cdots B_{k+1}\bar{h}_{k,i})(x_{n-1}). \quad (\text{S179})$$

There are  $n - 1 - k = r - 1$  backward kernels in this display. Moreover, Equation (S164) gives

$$\begin{aligned} \|\bar{h}_{k,i}\|_\infty &\leq \|h_{k,i}\|_\infty \leq G'(\theta), \\ \operatorname{osc}(\bar{h}_{k,i}) &\leq 2\|\bar{h}_{k,i}\|_\infty \\ &\leq 2G'(\theta). \end{aligned} \quad (\text{S180})$$

If  $r = 1$ , Equation (S179) contains no backward kernel. If  $r \geq 2$ , applying Equation (S174) successively gives

$$\operatorname{osc}[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}]$$

$$\leq 2G'(\theta) \begin{cases} 1, & r = 1, \\ (1 - \epsilon)^{r-1}, & r \geq 2. \end{cases} \quad (\text{S181})$$

*Step 4: covariance identity and geometric bound.* To finish, abbreviate the conditional expectation on the left of Equation (S181) by  $H$ , only for the next calculation. Since the  $X_{n-1}$ -marginal of  $\rho_{n-1}$  is  $\pi_{n-1}$ , the tower property gives

$$\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H), \quad (\text{S182})$$

$$\rho_{n-1}\{h_{k,i}M_n(G_n)\} = \pi_{n-1}\{HM_n(G_n)\}. \quad (\text{S183})$$

Substituting Equations (S182)–(S183) into Equation (S165) and then subtracting  $\rho_{n-1}(h_{k,i})$  gives the exact covariance identity

$$\begin{aligned} \Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i}) \\ &= \frac{\pi_{n-1}\{HM_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1}M_n(G_n)}{\pi_{n-1}M_n(G_n)} \\ &= \frac{\pi_{n-1}[\{H - \pi_{n-1}(H)\}\{M_n(G_n) - \pi_{n-1}M_n(G_n)\}]}{\pi_{n-1}M_n(G_n)}. \end{aligned} \quad (\text{S184})$$

For any probability measure  $\mu$  and bounded real  $f$ ,

$$|f(x) - \mu(f)| \leq \text{osc}(f). \quad (\text{S185})$$

The first positive lag also inherits one contraction from the forward transition. To see this directly, fix  $x^*$ . Assumption (A2) gives, for every measurable set  $A$ ,

$$\begin{aligned} M_n(x, A) &= \int_A f_{X_n|X_{n-1}}(x_n | x; \theta) dx_n \\ &\geq \underline{M} \int_A dx_n \\ &\geq \frac{\underline{M}}{\overline{M}} \int_A f_{X_n|X_{n-1}}(x_n | x^*; \theta) dx_n \\ &= \epsilon M_n(x^*, A). \end{aligned} \quad (\text{S186})$$

If  $\epsilon < 1$ , Equation (S186) permits the decomposition

$$M_n(x, \cdot) = \epsilon M_n(x^*, \cdot) + (1 - \epsilon)R_n(x, \cdot), \quad (\text{S187})$$

where  $R_n$  is a Markov kernel. Therefore, for any  $x, x'$ ,

$$\begin{aligned} M_n(G_n)(x) - M_n(G_n)(x') &= (1 - \epsilon)\{R_n(G_n)(x) - R_n(G_n)(x')\}, \\ |M_n(G_n)(x) - M_n(G_n)(x')| &\leq (1 - \epsilon) \text{osc}(G_n) \\ &\leq (1 - \epsilon)(\bar{G} - \underline{G}). \end{aligned} \quad (\text{S188})$$

Taking the supremum over  $x, x'$  gives

$$\text{osc}\{M_n(G_n)\} \leq (1 - \epsilon)(\bar{G} - \underline{G}). \quad (\text{S189})$$

If  $\epsilon = 1$ , Equation (S186) implies that  $M_n(x, \cdot)$  does not depend on  $x$ , so the left side of Equation (S189) is zero and the same bound holds. Taking absolute values in Equation (S184), using Equations (S185)–(S189), and then using Equation (S181), yields

$$\begin{aligned} |\Delta_{n,r,i}| &\leq \frac{\text{osc}(H) \text{osc}\{M_n(G_n)\}}{\underline{G}} \\ &\leq \frac{2(\bar{G} - \underline{G})}{\underline{G}} G'(\theta) \begin{cases} (1 - \epsilon), & r = 1, \\ (1 - \epsilon)^r, & r \geq 2. \end{cases} \end{aligned} \quad (\text{S190})$$

For both cases, take

$$\varrho = 1 - \frac{\epsilon}{2} = 1 - \frac{\underline{M}}{2\bar{M}} \in (0, 1). \quad (\text{S191})$$

Since  $1 - \epsilon \leq \varrho$ , the two cases in Equation (S190) are both bounded by  $\varrho^r$ . Summing the squared coordinate bounds and taking square roots give

$$\begin{aligned} \|\Delta_{n,r}\|_2 &= \left\{ \sum_{i=1}^p |\Delta_{n,r,i}|^2 \right\}^{1/2} \\ &\leq C\sqrt{p} G'(\theta) \varrho^r, \quad 1 \leq r < n. \end{aligned} \quad (\text{S192})$$

*Step 5: the zero-lag case.* Finally, when  $r = 0$ , Equation (S159) and Equation (S148) give

$$\begin{aligned} \|\Delta_{n,0}\|_2 &= \|\rho_n(h_n)\|_2 \\ &\leq \rho_n(\|h_n\|_2) \\ &\leq \sqrt{p} G'(\theta) \\ &\leq C\sqrt{p} G'(\theta) \varrho^0. \end{aligned} \quad (\text{S193})$$

Equations (S192) and (S193) prove Equation (S163). If  $\epsilon = 1$ , Equation (S184) and the zero oscillation in Equation (S189) show that  $\Delta_{n,r} = 0$  for every  $r \geq 1$ , so any fixed  $\varrho \in (0, 1)$  proves the same result. Thus  $\varrho$  depends only on the model minorization bounds and is independent of  $N, J, \alpha$ , and any lag truncation  $k$ . In particular, the factor  $\varrho^r$  includes the contraction of the first positive lag; it is not only a tail bound. The projection in Equation (S175) is sufficient here because  $\Delta_{n,r}$  is a difference of two linear expectations of one historical input. It does not eliminate conditional covariance between two genealogical inputs, and therefore does not supply the particle covariance bound in Equation (S279). □

**Lemma S11** (Uniform weak bias of current-particle averages). *Suppose  $\theta = \phi$ . Under Assumptions (A2), (A3), and (A5), with multinomial resampling, there is  $C < \infty$  such that, for every bounded real function  $f$  of the current particle location and every  $t \geq 1$ ,*

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \leq \frac{C}{J} \text{osc}(f). \quad (\text{S194})$$

*The constant is independent of  $t$  and  $J$ . Here “current” means measurable with respect to the time- $t$  physical particle location to which Assumption (A2) applies. Under the additional premise of Lemma S13,  $h_t(X_t)$  is an admissible choice of  $f$ . Without that premise, the total measurement derivative can depend on the simulator genealogy and is not an admissible current-location function.*

*Proof.* The proof has four stages: reduce the filtered average to the prediction cloud, establish contraction of the normalized semigroup, compute the  $J^{-1}$  bias of one particle update, and telescope those updates.

*Step 1: prediction-to-filtering reduction.* The predictor cloud consists of the current particle locations. A bounded function of the current location is therefore a current-state functional in the Feynman–Kac model already defined in Section S3. This statement does not turn an accumulated weight gradient or an individual ancestral contribution into a function of the current location. The current-location marginal of  $\rho_t$  is the filtering distribution, and hence

$$\rho_t(f) = \pi_t(f) = \Psi_t(\eta_t)(f). \quad (\text{S195})$$

Conditionally on the prediction cloud  $\eta_t^J$ , multinomial resampling draws the filtering particles independently from  $\Psi_t(\eta_t^J)$ . Therefore

$$\begin{aligned} \mathbb{E}\{m(\xi_t)(f) \mid \eta_t^J\} &= \frac{1}{J} \sum_{j=1}^J \mathbb{E}\{f(x_{t,j}^F) \mid \eta_t^J\} \\ &= \frac{1}{J} \sum_{j=1}^J \Psi_t(\eta_t^J)(f) \\ &= \Psi_t(\eta_t^J)(f). \end{aligned} \quad (\text{S196})$$

*Step 2: normalized-semigroup contraction.* Take  $\Phi_{t,t}$  to be the identity. For  $1 \leq p \leq t$ , define

$$\begin{aligned} Q_{t,t}(f)(x) &= G_t(x)f(x), \\ Q_{p,t}(f)(x) &= G_p(x)M_{p+1}\{Q_{p+1,t}(f)\}(x), \quad p < t. \end{aligned} \quad (\text{S197})$$

The definitions of  $\Phi_{p,t}$  and  $\Psi_t$  give, for every probability measure  $\mu$ ,

$$\Psi_t\{\Phi_{p,t}(\mu)\}(f) = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}. \quad (\text{S198})$$

For  $p < t$ ,

$$Q_{p,t}(1)(x) = G_p(x) \int f_{X_{p+1}|X_p}(u \mid x; \theta) Q_{p+1,t}(1)(u) du. \quad (\text{S199})$$

Thus, for every  $x, x'$ ,

$$\begin{aligned} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} &\leq \frac{\bar{G}\bar{M} \int Q_{p+1,t}(1)(u) du}{\underline{G}\underline{M} \int Q_{p+1,t}(1)(u) du} \\ &= \frac{\bar{G}\bar{M}}{\underline{G}\underline{M}}. \end{aligned} \quad (\text{S200})$$

For  $p = t$ , the same ratio is at most  $\bar{G}/\underline{G}$ . Put

$$R = \frac{\bar{G}\bar{M}}{\underline{G}\underline{M}} \geq 1. \quad (\text{S201})$$

Then

$$\sup_{x, x'} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} \leq R, \quad 1 \leq p \leq t. \quad (\text{S202})$$

For  $p \leq s < t$ , the normalized forward kernel is

$$P_{s,s+1}^{(t)}(x, A) = \frac{\int_A f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u) du}{\int f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u) du}. \quad (\text{S203})$$

Define, only within this proof,

$$\nu_{s,t}(A) = \frac{\int_A Q_{s+1,t}(1)(u) du}{\int Q_{s+1,t}(1)(u) du}. \quad (\text{S204})$$

Assumption (A2) gives

$$\begin{aligned} P_{s,s+1}^{(t)}(x, A) &\geq \frac{\underline{M} \int_A Q_{s+1,t}(1)(u) du}{\bar{M} \int Q_{s+1,t}(1)(u) du} \\ &= \frac{\underline{M}}{\bar{M}} \nu_{s,t}(A). \end{aligned} \quad (\text{S205})$$

Consequently, by Equation (S191), the Dobrushin coefficient of every kernel in Equation (S203) is at most  $\varrho$ . The factor  $G_s(x)$  cancels from the normalized conditional expectation, and hence

$$\frac{Q_{p,t}(f)}{Q_{p,t}(1)} = P_{p,p+1}^{(t)} P_{p+1,p+2}^{(t)} \cdots P_{t-1,t}^{(t)} f. \quad (\text{S206})$$

It follows that

$$\text{osc} \left\{ \frac{Q_{p,t}(f)}{Q_{p,t}(1)} \right\} \leq \varrho^{t-p} \text{osc}(f). \quad (\text{S207})$$

*Step 3: one complete particle update.* We next bound one complete selection–mutation update. For  $p = 1$ , put  $\mu = \eta_1 = \pi_0 M_1$ . For  $p \geq 2$ , condition on  $\eta_{p-1}^J$  and put

$$\mu = \Phi_p(\eta_{p-1}^J). \quad (\text{S208})$$

In either case, the particles forming  $\eta_p^J$  are conditionally independent with common law  $\mu$ . Here  $Q_{p,t}$  propagates the local error to time  $t$ :  $q$  isolates its random normalizer, while  $d$  is the centered empirical fluctuation. Within this calculation, put

$$q = \frac{Q_{p,t}(1)}{\mu Q_{p,t}(1)}, \quad a = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}, \quad d = \frac{Q_{p,t}(f - a)}{\mu Q_{p,t}(1)}. \quad (\text{S209})$$

Then

$$\mu(q) = 1, \quad \mu(d) = 0, \quad (\text{S210})$$

and Equation (S198) gives

$$\begin{aligned}
& \Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\mu)\}(f) \\
&= \frac{\eta_p^J Q_{p,t}(f)}{\eta_p^J Q_{p,t}(1)} - a \\
&= \frac{\eta_p^J Q_{p,t}(f - a)}{\eta_p^J Q_{p,t}(1)} \\
&= \frac{\eta_p^J(d)}{\eta_p^J(q)}.
\end{aligned} \tag{S211}$$

Equation (S202) implies

$$R^{-1} \leq q(x) \leq R, \quad \eta_p^J(q) \geq R^{-1}, \quad \|q - 1\|_\infty \leq R. \tag{S212}$$

Writing

$$F = \frac{Q_{p,t}(f)}{Q_{p,t}(1)}, \tag{S213}$$

we have  $a = \mu(qF)$  and  $\mu(q) = 1$ . Thus  $a$  is an average of  $F$  under the probability measure  $q \, d\mu$ , so

$$\inf F \leq a \leq \sup F. \tag{S214}$$

Since  $d = q(F - a)$ , Equations (S212), (S214), and (S207) give

$$\begin{aligned}
\|d\|_\infty &\leq R \operatorname{osc}(F) \\
&\leq R \varrho^{t-p} \operatorname{osc}(f).
\end{aligned} \tag{S215}$$

Conditional independence and  $\mu(d) = 0$  give

$$\begin{aligned}
\mathbb{E}\{\eta_p^J(d) \mid \mu\} &= 0, \\
\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} &= \frac{1}{J} \mu(d^2), \\
\mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\} &= \frac{1}{J} \mu\{(q - 1)^2\}.
\end{aligned} \tag{S216}$$

Consequently,

$$\begin{aligned}
& \mathbb{E} \left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \\
&= -\mathbb{E} \left\{ \frac{\eta_p^J(d) \{\eta_p^J(q) - 1\}}{\eta_p^J(q)} \middle| \mu \right\}.
\end{aligned} \tag{S217}$$

Conditional Cauchy–Schwarz and Equations (S212)–(S216) yield

$$\left| \mathbb{E} \left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \right|$$

$$\begin{aligned}
&\leq R [\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} \mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\}]^{1/2} \\
&= \frac{R}{J} [\mu(d^2) \mu\{(q-1)^2\}]^{1/2} \\
&\leq \frac{C}{J} \varrho^{t-p} \text{osc}(f).
\end{aligned} \tag{S218}$$

*Step 4: telescope over update times.* It remains to write the exact full-update telescope. Since  $\eta_t = \Phi_{1,t}(\eta_1)$ ,

$$\begin{aligned}
&\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f) \\
&= \mathbb{E} [\Psi_t\{\Phi_{1,t}(\eta_1^J)\}(f) - \Psi_t\{\Phi_{1,t}(\eta_1)\}(f)] \\
&\quad + \sum_{p=2}^t \mathbb{E} [\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\Phi_p(\eta_{p-1}^J))\}(f)].
\end{aligned} \tag{S219}$$

Indeed, for  $2 \leq p \leq t$ ,

$$\Phi_{p,t} \circ \Phi_p = \Phi_{p-1,t}, \tag{S220}$$

so consecutive terms cancel. The only terms left are  $\Psi_t(\eta_t^J)(f)$  and  $-\Psi_t(\eta_t)(f)$ .

Apply Equation (S218) to every summand in Equation (S219). Then

$$\begin{aligned}
|\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f)| &\leq \frac{C \text{osc}(f)}{J} \sum_{p=1}^t \varrho^{t-p} \\
&\leq \frac{C}{J(1-\varrho)} \text{osc}(f).
\end{aligned} \tag{S221}$$

Absorb  $(1-\varrho)^{-1}$  into  $C$  and combine this display with Equations (S195) and (S196).  $\square$

**Lemma S12** (Full-score summation-by-parts identities). *Suppose  $\theta = \phi$  and  $\alpha \in [0, 1]$ . Then the particle and population scores satisfy Equations (S225) and (S226) below. Moreover, for every coordinate  $i$  and  $t \leq N$ ,*

$$\left\| \sum_{k=1}^t \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^t \alpha^{t-k}. \tag{S222}$$

*Proof. Step 1: particle summation by parts.* We first record the summation-by-parts identity used again in the variance proof. Sum Equation (S158) over  $n = 1, \dots, N$  and put  $k = n - r$ . Since the sums are finite,

$$\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) = \sum_{k=1}^N \left[ m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\} \right]. \tag{S223}$$

For each fixed  $k$ , collecting the coefficient of every  $m(\xi_t)(h_k)$  gives

$$m(\xi_k)(h_k) + \sum_{t=k+1}^N \alpha^{t-k} \{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\}$$

$$= (1 - \alpha) \sum_{t=k}^{N-1} \alpha^{t-k} m(\xi_t)(h_k) + \alpha^{N-k} m(\xi_N)(h_k). \quad (\text{S224})$$

Indeed, the coefficients are  $1 - \alpha$  at  $t = k$ ,  $\alpha^{t-k} - \alpha^{t+1-k} = (1 - \alpha)\alpha^{t-k}$  for  $k < t < N$ , and  $\alpha^{N-k}$  at  $t = N$ . Interchanging the finite sums once more yields

$$\nabla_{\theta} \hat{\ell}^{\alpha}(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} m(\xi_t) \left( \sum_{k=1}^t \alpha^{t-k} h_k \right) + m(\xi_N) \left( \sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S225})$$

*Step 2: population identity and uniform bound.* The same algebra applied to Equation (S162) gives

$$\nabla_{\theta} \ell^{\alpha}(\theta) = (1 - \alpha) \sum_{t=1}^{N-1} \rho_t \left( \sum_{k=1}^t \alpha^{t-k} h_k \right) + \rho_N \left( \sum_{k=1}^N \alpha^{N-k} h_k \right). \quad (\text{S226})$$

For every coordinate  $i$ , Equation (S148) gives

$$\left\| \sum_{k=1}^t \alpha^{t-k} h_{k,i} \right\|_{\infty} \leq G'(\theta) \sum_{k=1}^t \alpha^{t-k}. \quad (\text{S227})$$

Equation (S227) is identical to Equation (S222), completing the proof.  $\square$

**Claim S1** (Conditional empirical moments). *Let  $\mathcal{F}$  be a sigma-field and let  $\mu$  be an  $\mathcal{F}$ -measurable probability measure. Suppose that, conditionally on  $\mathcal{F}$ ,  $X_1, \dots, X_J$  are independent with common law  $\mu$ . The following bounds hold for bounded  $\mathcal{F}$ -measurable real functions  $f, f_1, f_2$ :*

$$\mathbb{E} \left\{ \left. \frac{1}{J} \sum_{j=1}^J \{f(X_j) - \mu(f)\} \right| \mathcal{F} \right\} = 0, \quad (\text{S228})$$

$$\mathbb{E} \left\{ \left. \left[ \frac{1}{J} \sum_{j=1}^J \{f(X_j) - \mu(f)\} \right]^2 \right| \mathcal{F} \right\} = \frac{\mu\{(f - \mu f)^2\}}{J} \leq \frac{\text{osc}(f)^2}{4J}, \quad (\text{S229})$$

$$\begin{aligned} \mathbb{E} \left\{ \left. \left[ \frac{1}{J} \sum_{j=1}^J \{f(X_j) - \mu(f)\} \right]^4 \right| \mathcal{F} \right\} &= \frac{\mu\{(f - \mu f)^4\}}{J^3} + \frac{3(J-1)}{J^3} \mu\{(f - \mu f)^2\}^2 \\ &\leq \frac{\mu\{(f - \mu f)^4\}}{J^3} + \frac{3}{J^2} \mu\{(f - \mu f)^2\}^2 \\ &\leq \frac{3 \text{osc}(f)^4}{J^2}. \end{aligned} \quad (\text{S230})$$

Consequently, for bounded  $f_1$  and  $f_2$ ,

$$\begin{aligned} &\left\| \left[ \frac{1}{J} \sum_{j=1}^J \{f_1(X_j) - \mu(f_1)\} \right] \left[ \frac{1}{J} \sum_{j=1}^J \{f_2(X_j) - \mu(f_2)\} \right] \right\|_{L^2|\mathcal{F}} \\ &\leq \frac{\sqrt{3}}{J} \text{osc}(f_1) \text{osc}(f_2). \end{aligned} \quad (\text{S231})$$

*Proof.* Put  $Y_j = f(X_j) - \mu(f)$ . Conditional independence and  $\mathbb{E}(Y_j \mid \mathcal{F}) = 0$  give Equations (S228) and (S229). They also give

$$\begin{aligned} & \mathbb{E} \left\{ \left( \frac{1}{J} \sum_{j=1}^J Y_j \right)^4 \middle| \mathcal{F} \right\} \\ &= \frac{1}{J^4} \{ J\mu\{(f - \mu f)^4\} + 3J(J-1)\mu\{(f - \mu f)^2\}^2 \} \\ &= \frac{\mu\{(f - \mu f)^4\}}{J^3} + \frac{3(J-1)}{J^3} \mu\{(f - \mu f)^2\}^2 \\ &\leq \frac{3}{J^2} \text{osc}(f)^4, \end{aligned} \tag{S232}$$

which is Equation (S230). Conditional Cauchy–Schwarz applied to the two fourth moments proves Equation (S231).  $\square$

**Claim S2** (Normalized-ratio perturbations). *Let  $\mu$  and  $\nu$  be probability measures. For  $l \in \{0, 1\}$ , let  $a_l > 0$  and  $b_l$  be bounded functions such that  $\mu(a_l) = 1$ , and put*

$$\mathcal{R}_l(\lambda) = \frac{\lambda(b_l)}{\lambda(a_l)}, \quad D_\mu \mathcal{R}_l = b_l - \mathcal{R}_l(\mu) a_l. \tag{S233}$$

*Then*

$$\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) = (\nu - \mu) D_\mu \mathcal{R}_l - \frac{(\nu - \mu)(a_l)}{\nu(a_l)} (\nu - \mu) D_\mu \mathcal{R}_l, \tag{S234}$$

*and hence*

$$\begin{aligned} & \{\mathcal{R}_1(\nu) - \mathcal{R}_0(\nu)\} - \{\mathcal{R}_1(\mu) - \mathcal{R}_0(\mu)\} \\ &= (\nu - \mu) \{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\} \\ &= -\frac{(\nu - \mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)} (\nu - \mu) D_\mu \mathcal{R}_1 \\ &\quad - \frac{(\nu - \mu)(a_0)}{\nu(a_0)} (\nu - \mu) \{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}. \end{aligned} \tag{S235}$$

*If  $\inf_{l \in \{0, 1\}} \nu(a_l) \geq C^{-1}$ , then the absolute value of the left side of Equation (S235) is at most*

$$\begin{aligned} & C^2 |(\nu - \mu)(a_1 - a_0)| |(\nu - \mu) D_\mu \mathcal{R}_1| \\ &+ C |(\nu - \mu)(a_0)| |(\nu - \mu) \{D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0\}|. \end{aligned} \tag{S236}$$

*Proof.* Since  $\mu(a_l) = 1$ ,

$$\begin{aligned} \mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) &= \frac{\nu(b_l) - \mathcal{R}_l(\mu)\nu(a_l)}{\nu(a_l)} \\ &= \frac{(\nu - \mu) D_\mu \mathcal{R}_l}{\nu(a_l)} \\ &= (\nu - \mu) D_\mu \mathcal{R}_l - \frac{(\nu - \mu)(a_l)}{\nu(a_l)} (\nu - \mu) D_\mu \mathcal{R}_l, \end{aligned} \tag{S237}$$

which proves Equation (S234). Moreover,

$$\begin{aligned}
& \frac{(\nu - \mu)(a_1)}{\nu(a_1)} - \frac{(\nu - \mu)(a_0)}{\nu(a_0)} \\
&= \left(1 - \frac{1}{\nu(a_1)}\right) - \left(1 - \frac{1}{\nu(a_0)}\right) \\
&= \frac{\nu(a_1) - \nu(a_0)}{\nu(a_0)\nu(a_1)} \\
&= \frac{(\nu - \mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)}. \tag{S238}
\end{aligned}$$

Subtracting the cases  $l = 0$  and  $l = 1$  in Equation (S234) now proves Equation (S235). Taking absolute values there and using  $\nu(a_0)^{-1} \leq C$  and  $\nu(a_1)^{-1} \leq C$  proves Equation (S236). This exact identity is the paired specialization of the normalized Feynman–Kac perturbation calculation in Equations (3.1)–(3.4) and Appendix A.3 of Del Moral and Rio (2011).  $\square$

**Lemma S13** (Particle error and variance of the DMOP- $\alpha$  score). *Suppose  $J \geq 2$ ,  $\theta = \phi$ ,  $\alpha \in [0, 1]$ , Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Assume additionally that there is a bounded measurable function  $h_n$  on the physical state space such that the total derivative in Equation (S146) equals  $h_n(X_n)$  almost surely under the fixed-random-number construction. Then*

$$\begin{aligned}
& \max \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2, \text{Var} \left( \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \right\} \\
& \lesssim pG'(\theta)^2 \left[ \frac{1}{J} \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right. \\
& \quad \left. + \frac{1}{J^2} \left\{ \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2} \right\}^2 \right] \\
& \lesssim pG'(\theta)^2 \left( \frac{N}{J} + \frac{N^2}{J^2} \right) \left( \sum_{r=0}^{N-1} \alpha^r \right)^2. \tag{S239}
\end{aligned}$$

Here the multiplicative constants suppressed by  $\lesssim$  depend on the fixed model bounds, but are independent of  $N, J$ , and  $\alpha$ , and  $\text{Var}(Z) = \mathbb{E} \|Z - \mathbb{E}Z\|_2^2$ . For  $\alpha < 1$ , the first upper bound gives

$$\begin{aligned}
& \max \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2, \text{Var} \left( \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right) \right\} \\
& \lesssim pG'(\theta)^2 \left( \frac{N}{J} + \frac{N^2}{J^2} \right) \frac{1 + \alpha \varrho}{(1 - \alpha^2)(1 - \alpha \varrho)}. \tag{S240}
\end{aligned}$$

At  $\alpha = 1$ , it gives

$$\max \left\{ \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^1(\theta) - \nabla_{\theta} \ell^1(\theta) \right\|_2^2, \text{Var} \left( \nabla_{\theta} \hat{\ell}^1(\theta) \right) \right\} \lesssim pG'(\theta)^2 \min \left\{ N^2, \frac{N^2}{J} + \frac{N^3}{J^2} \right\}. \tag{S241}$$

*Proof.* The proof keeps three time indices explicit:  $s$  is the time at which  $h_{s,i}$  is created,  $t$  is the local particle-sampling time, and  $n$  is the conditional-score time. We first separate terminal resampling, then control predictable contributions for  $s \leq t$  and  $t < s$ , and only then sum the local particle errors over  $t$ . Since  $\theta = \phi$ , every particle location, measurement weight, and transition density in this proof is evaluated at that common parameter; redundant  $\phi$  superscripts are suppressed.

*Step 1: terminal multinomial resampling.* Fix a coordinate  $i$  and write

$$Z_i^J = \partial_{\theta_i} \hat{\ell}^\alpha(\theta), \quad z_i^\alpha = \partial_{\theta_i} \ell^\alpha(\theta). \quad (\text{S242})$$

Equation (S158) and Equation (S148) give

$$\begin{aligned} |Z_i^J| &\leq \sum_{n=1}^N \left\{ G'(\theta) + 2G'(\theta) \sum_{r=1}^{n-1} \alpha^r \right\} \\ &\leq NG'(\theta) \left\{ 1 + 2 \sum_{r=1}^{N-1} \alpha^r \right\} \leq 2N^2 G'(\theta) < \infty. \end{aligned} \quad (\text{S243})$$

Thus all random variables used below are square integrable.

We first separate the error created by the time- $n$  multinomial resampling. For the prediction paths at time  $n$ , write

$$p_{n,a} = \frac{g_{n,a}}{\sum_{b=1}^J g_{n,b}}, \quad a = 1, \dots, J, \quad (\text{S244})$$

and put

$$R_{n,i}^J = \frac{1}{J} \sum_{j=1}^J \left[ \sum_{s=1}^n \alpha^{n-s} h_{s,i}(\xi_{n,k_j}^P) - \sum_{a=1}^J p_{n,a} \sum_{s=1}^n \alpha^{n-s} h_{s,i}(\xi_{n,a}^P) \right]. \quad (\text{S245})$$

Conditionally on the prediction paths, the  $J$  summands in Equation (S245) are independent and centered. Moreover,

$$\begin{aligned} \text{osc} \left( \sum_{s=1}^n \alpha^{n-s} h_{s,i} \right) &\leq 2G'(\theta) \sum_{s=1}^n \alpha^{n-s} \\ &\leq 2G'(\theta) \sum_{r=0}^{N-1} \alpha^r. \end{aligned} \quad (\text{S246})$$

Claim S1 gives the exact conditional variance

$$\mathbb{E} \left\{ (R_{n,i}^J)^2 \mid \xi_{n,1:J}^P \right\} = \frac{1}{J} \Psi_n \{ m(\xi_n^P) \} \left[ \left\{ \sum_{s=1}^n \alpha^{n-s} h_{s,i} - \Psi_n \{ m(\xi_n^P) \} \left( \sum_{s=1}^n \alpha^{n-s} h_{s,i} \right) \right\}^2 \right]. \quad (\text{S247})$$

Taking expectations and applying the genealogical lag bound in Equation (S273), proved in Step 5, gives

$$\mathbb{E} (R_{n,i}^J)^2 \leq \frac{CG'(\theta)^2}{J} \sum_{r=0}^{n-1} \sum_{u=0}^{n-1} \alpha^{r+u} \varrho^{|r-u|}. \quad (\text{S248})$$

*Step 2: exact predictable-score decomposition.* We next derive the local perturbation bound for the predictable part. Fix  $s \leq t \leq n$  and retain the manuscript's  $h_{s,i}$ . Define

$$\begin{aligned} q_{t,n}^1 &= Q_{t,n}(1), \\ q_{t,n}^0 &= \begin{cases} Q_{t,n-1}(1), & t < n, \\ 1, & t = n, \end{cases} \quad F_{t,n} = \frac{q_{t,n}^1}{q_{t,n}^0}, \end{aligned} \quad (\text{S249})$$

and, for a probability measure  $\mu$  on prediction paths through time  $t$ ,

$$C_{t,n}(\mu) = \frac{\mu(h_{s,i} q_{t,n}^1)}{\mu(q_{t,n}^1)} - \frac{\mu(h_{s,i} q_{t,n}^0)}{\mu(q_{t,n}^0)}. \quad (\text{S250})$$

At  $t = n$ , this is precisely the predictable part of the one-lag difference:

$$\begin{aligned} C_{n,n}\{m(\xi_n^P)\} &= \sum_{a=1}^J p_{n,a} h_{s,i}(\xi_{n,a}^P) - m(\xi_n^P)(h_{s,i}), \\ C_{n,n}(\eta_n) &= \begin{cases} \rho_n(h_{s,i}) - \rho_{n-1}(h_{s,i}), & s < n, \\ \rho_n(h_{n,i}) - \eta_n(h_{n,i}), & s = n. \end{cases} \end{aligned} \quad (\text{S251})$$

When several values of  $s$  occur in the same display below, write  $C_{t,n}^{(s)}$  for the same functional with the displayed  $h_{s,i}$ ; the superscript records no additional object. The conditional-score decomposition gives the exact identities

$$Z_i^J = \sum_{n=1}^N R_{n,i}^J + \sum_{n=1}^N m(\xi_n^P)(h_{n,i}) + \sum_{n=1}^N \sum_{s=1}^n \alpha^{n-s} C_{n,n}^{(s)}\{m(\xi_n^P)\}, \quad (\text{S252})$$

$$z_i^\alpha = \sum_{n=1}^N \eta_n(h_{n,i}) + \sum_{n=1}^N \sum_{s=1}^n \alpha^{n-s} C_{n,n}^{(s)}(\eta_n). \quad (\text{S253})$$

Indeed, for  $s < n$ , the  $s$ th summand of  $R_{n,i}^J$  plus  $\alpha^{n-s} C_{n,n}^{(s)}\{m(\xi_n^P)\}$  is

$$\alpha^{n-s} \{m(\xi_n)(h_{s,i}) - m(\xi_{n-1})(h_{s,i})\}, \quad (\text{S254})$$

while for  $s = n$  the three terms are

$$\begin{aligned} &\left[ m(\xi_n)(h_{n,i}) - \sum_{a=1}^J p_{n,a} h_{n,i}(\xi_{n,a}^P) \right] + m(\xi_n^P)(h_{n,i}) \\ &+ \left[ \sum_{a=1}^J p_{n,a} h_{n,i}(\xi_{n,a}^P) - m(\xi_n^P)(h_{n,i}) \right] = m(\xi_n)(h_{n,i}). \end{aligned} \quad (\text{S255})$$

This proves Equation (S252); the population identity follows by the same cancellation.

*Step 3: continuation when  $s \leq t$ .* The recursion for  $Q_{t,n}$  and preservation of  $h_{s,i}$  after time  $s$  give, for  $s < t \leq n$ ,

$$C_{t,n}\{\Phi_t(\mu)\} = C_{t-1,n}(\mu). \quad (\text{S256})$$

Indeed, for  $l \in \{0, 1\}$ ,

$$\begin{aligned}
\frac{\Phi_t(\mu)(h_{s,i}q_{t,n}^l)}{\Phi_t(\mu)(q_{t,n}^l)} &= \frac{\mu\{G_{t-1}M_t(h_{s,i}q_{t,n}^l)\}}{\mu\{G_{t-1}M_t(q_{t,n}^l)\}} \\
&= \frac{\mu\{h_{s,i}G_{t-1}M_t(q_{t,n}^l)\}}{\mu\{G_{t-1}M_t(q_{t,n}^l)\}} \\
&= \frac{\mu(h_{s,i}q_{t-1,n}^l)}{\mu(q_{t-1,n}^l)}. \tag{S257}
\end{aligned}$$

Subtracting the cases  $l = 1$  and  $l = 0$  proves Equation (S256).

Assumptions (A2) and (A3) imply, exactly as in Equations (S202) and (S207), that

$$\begin{aligned}
\sup_{x,x'} \frac{q_{t,n}^0(x)}{q_{t,n}^0(x')} &\leq C, \\
\underline{G} \leq F_{t,n}(x) &\leq \bar{G}, \\
\text{osc}(F_{t,n}) &\leq C\varrho^{n-t}. \tag{S258}
\end{aligned}$$

Normalize  $q_{t,n}^0$  so that  $\mu(q_{t,n}^0) = 1$  and write

$$\bar{\mu}(f) = \mu(q_{t,n}^0 f). \tag{S259}$$

Then

$$C_{t,n}(\mu) = \frac{\bar{\mu}(h_{s,i}F_{t,n})}{\bar{\mu}(F_{t,n})} - \bar{\mu}(h_{s,i}). \tag{S260}$$

For constants  $h_0$  and  $F_0$  in the respective ranges of  $h_{s,i}$  and  $F_{t,n}$ ,

$$\begin{aligned}
&\bar{\mu}(h_{s,i}F_{t,n}) - \bar{\mu}(h_{s,i})\bar{\mu}(F_{t,n}) \\
&= \bar{\mu}\{(h_{s,i} - h_0)(F_{t,n} - F_0)\} - \bar{\mu}(h_{s,i} - h_0)\bar{\mu}(F_{t,n} - F_0). \tag{S261}
\end{aligned}$$

Equations (S148) and (S258) therefore give

$$|C_{t,n}(\mu)| \leq CG'(\theta)\varrho^{n-t}. \tag{S262}$$

To control the particle perturbation, put

$$\mathcal{R}_l(\lambda) = \frac{\lambda(h_{s,i}q_{t,n}^l)}{\lambda(q_{t,n}^l)}, \quad a_l = \frac{q_{t,n}^l}{\mu(q_{t,n}^l)}, \quad l \in \{0, 1\}. \tag{S263}$$

Thus  $C_{t,n} = \mathcal{R}_1 - \mathcal{R}_0$ . With the normalization  $\mu(q_{t,n}^0) = 1$ , direct differentiation gives

$$\begin{aligned}
a_0 &= q_{t,n}^0, & a_1 &= \frac{q_{t,n}^0 F_{t,n}}{\bar{\mu}(F_{t,n})}, \\
D_\mu \mathcal{R}_0 &= q_{t,n}^0 \{h_{s,i} - \bar{\mu}(h_{s,i})\}, & D_\mu \mathcal{R}_1 &= a_1 \left\{ h_{s,i} - \frac{\bar{\mu}(h_{s,i}F_{t,n})}{\bar{\mu}(F_{t,n})} \right\}. \tag{S264}
\end{aligned}$$

Subtracting the two expressions in Equation (S264) gives the exact first variation

$$\begin{aligned} D_\mu C_{t,n} &= D_\mu \mathcal{R}_1 - D_\mu \mathcal{R}_0 \\ &= (a_1 - a_0)\{h_{s,i} - \mathcal{R}_1(\mu)\} + a_0\{\mathcal{R}_0(\mu) - \mathcal{R}_1(\mu)\}. \end{aligned} \quad (\text{S265})$$

Equations (S258), (S148), and (S262) therefore imply

$$\begin{aligned} \text{osc}(a_0) &\leq C, \\ \text{osc}(a_1 - a_0) &\leq C\varrho^{n-t}, \\ \text{osc}(D_\mu \mathcal{R}_1) &\leq CG'(\theta), \\ \text{osc}(D_\mu C_{t,n}) &\leq CG'(\theta)\varrho^{n-t}. \end{aligned} \quad (\text{S266})$$

The uniform ratio bounds also give  $\inf_l \nu(a_l) \geq C^{-1}$ . Apply Claim S2 with  $b_l = a_l h_{s,i}$ . Equations (S236) and (S266) give

$$\begin{aligned} &|C_{t,n}(\nu) - C_{t,n}(\mu) - (\nu - \mu)D_\mu C_{t,n}| \\ &\leq C^2 |(\nu - \mu)(a_1 - a_0)| |(\nu - \mu)D_\mu \mathcal{R}_1| \\ &\quad + C |(\nu - \mu)(a_0)| |(\nu - \mu)D_\mu C_{t,n}|. \end{aligned} \quad (\text{S267})$$

In each product on the right, one oscillation is bounded by  $C\varrho^{n-t}$  and the other by  $CG'(\theta)$ . Thus the factor  $\varrho^{n-t}$  follows from the displayed two-ratio cancellation, rather than from a mixing assertion for the historical function  $h_{s,i}$ .

*Step 4: telescope over particle updates.* We now write the full local telescope before summing any norms. Put

$$\mu_t^J = \begin{cases} \eta_1, & t = 1, \\ \Phi_t\{m(\xi_{t-1}^P)\}, & t \geq 2. \end{cases} \quad (\text{S268})$$

In this display,  $\Phi_t$  denotes the canonical path-space lift of the same predictor update: the potential is evaluated at the last location, multinomial selection chooses a complete path, and  $M_t$  appends the new location. Its last-coordinate marginal is the predictor update already defined in Section S3. When  $H_{t,n}$  or  $C_{t,n}^{(s)}$  is applied to  $\eta_t$  below,  $\eta_t$  is read through the same canonical path-space lift. Thus, conditionally on the preceding prediction cloud, the  $J$  paths whose empirical measure is  $m(\xi_t^P)$  are independent with common law  $\mu_t^J$ . For the current predictor term in Equation (S252), define

$$H_{t,n}(\mu) = \Phi_{t,n}(\mu)(h_{n,i}), \quad 1 \leq t \leq n, \quad (\text{S269})$$

where  $\Phi_{n,n}$  is the identity. For  $t < s$ , extend the already defined continuation functional by

$$C_{t,n}^{(s)}(\mu) = C_{s,n}^{(s)}\{\Phi_{t,s}(\mu)\}. \quad (\text{S270})$$

Equations (S256) and (S220) then give the exact telescopes

$$\begin{aligned} &H_{n,n}\{m(\xi_n^P)\} - H_{n,n}(\eta_n) \\ &= \sum_{t=1}^n [H_{t,n}\{m(\xi_t^P)\} - H_{t,n}(\mu_t^J)], \\ &C_{n,n}^{(s)}\{m(\xi_n^P)\} - C_{n,n}^{(s)}(\eta_n) \end{aligned} \quad (\text{S271})$$

$$= \sum_{t=1}^n \left[ C_{t,n}^{(s)} \{m(\xi_t^P)\} - C_{t,n}^{(s)} (\mu_t^J) \right]. \quad (\text{S272})$$

For example, the second term in the  $t$ th summand of Equation (S272) is  $C_{t-1,n}^{(s)} \{m(\xi_{t-1}^P)\}$  for  $t \geq 2$ , so consecutive terms cancel. At  $t = 1$ ,  $\mu_1^J = \eta_1$ , and the remaining endpoints are exactly those displayed on the left. The same verification proves Equation (S271).

*Step 5: lag moments along a sampled genealogy.* We first prove the particle-genealogy estimate used below. For  $q \in \{2, 4\}$  and every  $t \leq N$ ,

$$\begin{aligned} & \max \left\{ \mathbb{E} \Psi_t \{m(\xi_t^P)\} \left[ \left| \sum_{s=1}^t \alpha^{t-s} h_{s,i} - \Psi_t \{m(\xi_t^P)\} \left( \sum_{s=1}^t \alpha^{t-s} h_{s,i} \right) \right|^q \right], \right. \\ & \quad \left. \mathbb{E} \mu_t^J \left[ \left| \sum_{s=1}^t \alpha^{t-s} h_{s,i} - \mu_t^J \left( \sum_{s=1}^t \alpha^{t-s} h_{s,i} \right) \right|^q \right] \right\} \\ & \leq C G'(\theta)^q \left\{ \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right\}^{q/2}. \end{aligned} \quad (\text{S273})$$

To verify Equation (S273), use the manuscript's prediction-cloud notation and put

$$\mathcal{X}_t^J = \sigma(X_{0,1:J}^P, \dots, X_{t,1:J}^P). \quad (\text{S274})$$

Thus we condition on all prediction clouds through time  $t$ , but not on the complete predicted genealogies  $\xi_{t,1:J}^P$ . The exact backward representation of a randomly selected ancestral line in Equation (3.3) of Chan et al. (2018) has, from time  $s$  to time  $s-1$ , the transition

$$\sum_{j=1}^J \frac{g_{s-1,j} f_{X_s|X_{s-1}}(x_s | x_{s-1,j}^P; \theta)}{\sum_{a=1}^J g_{s-1,a} f_{X_s|X_{s-1}}(x_s | x_{s-1,a}^P; \theta)} \delta_{x_{s-1,j}^P}(dx_{s-1}). \quad (\text{S275})$$

In the notation of Chan et al. (2018), the density in their backward transition is  $H'_s(x_{s-1}, x_s) = G'_{s-1}(x_{s-1}) M'_s(x_{s-1}, x_s)$ . Taking  $m(\xi_{s-1}^P)$  to be the empirical prediction cloud,  $G'_{s-1}(x_{s-1,j}^P) = g_{s-1,j}$ , and

$$M'_s(x_{s-1,j}^P, x_s) = f_{X_s|X_{s-1}}(x_s | x_{s-1,j}^P; \theta)$$

gives exactly Equation (S275). Equation (3.3) of Chan et al. (2018) uses the uniform terminal empirical measure. Selecting its terminal state instead with probability proportional to  $g_{t,j}$  gives  $\Psi_t \{m(\xi_t^P)\}$  and changes only the terminal distribution, not the backward transitions in Equation (S275). Thus the displayed transition is an exact specialization of their backward-law identity.

For every measurable set  $A$ , Assumption (A2) gives the pointwise minorization

$$\begin{aligned} & \sum_{j=1}^J \frac{g_{s-1,j} f_{X_s|X_{s-1}}(x_s | x_{s-1,j}^P; \theta)}{\sum_{a=1}^J g_{s-1,a} f_{X_s|X_{s-1}}(x_s | x_{s-1,a}^P; \theta)} \mathbf{1}\{x_{s-1,j}^P \in A\} \\ & \geq \frac{\underline{M}}{\overline{M}} \frac{\sum_{j=1}^J g_{s-1,j} \mathbf{1}\{x_{s-1,j}^P \in A\}}{\sum_{a=1}^J g_{s-1,a}}. \end{aligned} \quad (\text{S276})$$

Consequently, the Dobrushin coefficient of every transition in Equation (S275) is at most  $1 - \underline{M}/\bar{M} \leq \varrho$ . If  $1 \leq s \leq u \leq t$ , composition of these backward transitions and Equation (S148) therefore give

$$\text{osc} \{ \mathbb{E}(h_{s,i} \mid X_u, \mathcal{X}_t^J) \} \leq 2G'(\theta)\varrho^{u-s}. \quad (\text{S277})$$

This is the point at which the additional current-function premise in Lemma S13 is used. It makes  $h_{u,i}$  measurable with respect to  $X_u$ , and it makes  $h_{s,i}$  a bounded function at the endpoint of the backward composition. Equation (3.3) of Chan et al. (2018) supplies the conditional genealogy law, but by itself it would not give the following covariance decay for arbitrary functionals of the ancestral path. In particular, the factor  $\alpha$  does not repair this step: it weights the inputs only after the covariance of  $h_{s,i}$  and  $h_{u,i}$  has been formed. For real random variables  $U, V$ ,

$$|\text{Cov}(U, V)| \leq \frac{1}{4} \text{osc}(U) \text{osc}(V). \quad (\text{S278})$$

Using the backward Markov property, followed by Equation (S277), yields

$$\begin{aligned} & |\text{Cov}\{h_{s,i}(X_s), h_{u,i}(X_u) \mid \mathcal{X}_t^J\}| \\ &= |\text{Cov}\{\mathbb{E}(h_{s,i} \mid X_u, \mathcal{X}_t^J), h_{u,i}(X_u) \mid \mathcal{X}_t^J\}| \\ &\leq G'(\theta)^2 \varrho^{u-s}. \end{aligned} \quad (\text{S279})$$

Equation (S279) concerns two single-time measurement-gradient inputs, not two accumulated log-weight gradients. Their realized values are carried by the sampled genealogy. Nevertheless, at  $\theta = \phi$ , the reference genealogy and the single-time input maps are independent of  $\alpha$ , because every MOP weight ratio equals one. The factors of  $\alpha$  enter only when Equation (S152) reassembles the accumulated log-weight gradient. Expanding the conditional variance of that sum and putting  $r = t - s$  and  $v = t - u$  now gives

$$\begin{aligned} & \text{Var} \left( \sum_{s=1}^t \alpha^{t-s} h_{s,i}(X_s) \mid \mathcal{X}_t^J \right) \\ & \leq G'(\theta)^2 \sum_{r=0}^{t-1} \sum_{v=0}^{t-1} \alpha^{r+v} \varrho^{|r-v|}. \end{aligned} \quad (\text{S280})$$

For completeness, the fourth moment follows from the same backward transitions. Put  $c_s = \alpha^{t-s}$  and expose the selected path in the order  $X_t, X_{t-1}, \dots, X_1$ . The conditional expectations of the fixed bounded sum  $\sum_{s=1}^t c_s h_{s,i}(X_s)$  along these nested sigma-fields form its Doob martingale; this is an identity, not an additional stochastic assumption on the model or observations. At the step revealing  $X_u$ , the oscillation of the corresponding Doob martingale difference is at most

$$2G'(\theta) \sum_{s=1}^u c_s \varrho^{u-s}. \quad (\text{S281})$$

Indeed, conditional expectations of the terms with  $s < u$  are obtained by the backward transitions from  $u$  to  $s$ , and each has oscillation at most  $2G'(\theta)c_s\varrho^{u-s}$ ; terms with  $s > u$  have already been revealed and cancel from that martingale difference. This fourth-moment step is needed for the products of two empirical errors in Step 8; the second-moment covariance calculation alone does not control those products.

Conditional Hoeffding's lemma applied successively to these centered differences gives, for every  $z > 0$ ,

$$\begin{aligned} & \Pr \left\{ \left| \sum_{s=1}^t c_s h_{s,i}(X_s) - \mathbb{E} \left( \sum_{s=1}^t c_s h_{s,i}(X_s) \middle| \mathcal{X}_t^J \right) \right| \geq z \middle| \mathcal{X}_t^J \right\} \\ & \leq 2 \exp \left[ - \frac{z^2}{2G'(\theta)^2 \sum_{u=1}^t \left\{ \sum_{s=1}^u c_s \varrho^{u-s} \right\}^2} \right]. \end{aligned} \quad (\text{S282})$$

Young's convolution inequality gives

$$\begin{aligned} \sum_{u=1}^t \left\{ \sum_{s=1}^u c_s \varrho^{u-s} \right\}^2 & \leq \left( \sum_{d=0}^{\infty} \varrho^d \right)^2 \sum_{s=1}^t c_s^2 \\ & \leq \frac{1}{(1-\varrho)^2} \sum_{r=0}^{t-1} \sum_{v=0}^{t-1} \alpha^{r+v} \varrho^{|r-v|}. \end{aligned} \quad (\text{S283})$$

Using  $\mathbb{E}|U|^4 = 4 \int_0^\infty z^3 \Pr(|U| \geq z) dz$  in Equations (S282) and (S283) proves

$$\begin{aligned} & \mathbb{E} \left[ \left| \sum_{s=1}^t \alpha^{t-s} h_{s,i}(X_s) - \mathbb{E} \left( \sum_{s=1}^t \alpha^{t-s} h_{s,i}(X_s) \middle| \mathcal{X}_t^J \right) \right|^4 \middle| \mathcal{X}_t^J \right] \\ & \leq CG'(\theta)^4 \left\{ \sum_{r=0}^{t-1} \sum_{v=0}^{t-1} \alpha^{r+v} \varrho^{|r-v|} \right\}^2. \end{aligned} \quad (\text{S284})$$

It remains only to pass from a selected ancestral line to the empirical conditional laws in Equation (S273). For the finer conditioning, add the complete predicted genealogies  $\xi_{t,1:J}^P$  to  $\mathcal{X}_t^J$ ; the coarser conditioning is only on  $\mathcal{X}_t^J$ . For the second moment, the conditional variance identity gives

$$\begin{aligned} & \mathbb{E} \left[ \text{Var} \left\{ \sum_{s=1}^t \alpha^{t-s} h_{s,i}(X_s) \middle| \mathcal{X}_t^J, \xi_{t,1:J}^P \right\} \middle| \mathcal{X}_t^J \right] \\ & \leq \text{Var} \left\{ \sum_{s=1}^t \alpha^{t-s} h_{s,i}(X_s) \middle| \mathcal{X}_t^J \right\}. \end{aligned} \quad (\text{S285})$$

For the fourth moment,  $(a+b)^4 \leq 8a^4 + 8b^4$  and conditional Jensen give

$$\begin{aligned} & \mathbb{E} \left[ \left| S - \mathbb{E}(S \mid \mathcal{X}_t^J, \xi_{t,1:J}^P) \right|^4 \middle| \mathcal{X}_t^J \right] \\ & \leq 16 \mathbb{E} \left[ \left| S - \mathbb{E}(S \mid \mathcal{X}_t^J) \right|^4 \middle| \mathcal{X}_t^J \right], \end{aligned} \quad (\text{S286})$$

where  $S$  in this display is the finite sum appearing in Equation (S273). Equations (S280), (S284), (S285), and (S286) prove the  $\Psi_t\{m(\xi_t^P)\}$  part of Equation (S273).

We now verify the  $\mu_t^J$  part separately. For  $t \geq 2$ , draw one auxiliary predictor path conditionally on the complete time- $(t-1)$  prediction genealogies, with the conditional law  $\mu_t^J$  in Equation (S268).

If one conditions more coarsely only on  $\mathcal{X}_{t-1}^J$ , and then conditions on the newly propagated terminal location  $X_t$ , the parent-selection probability proportional to  $g_{t-1,j}$  and the transition density to  $X_t$  give exactly Equation (S275). Therefore the conditional variance and fourth-moment bounds in Equations (S280) and (S284) also hold under this coarser conditioning. The conditional variance identity now gives

$$\begin{aligned} & \mathbb{E} \mu_t^J \left[ |S - \mu_t^J(S)|^2 \right] \\ &= \mathbb{E} \left[ \text{Var}\{S \mid \mathcal{X}_{t-1}^J, \xi_{t-1,1:J}^P\} \right] \\ &\leq \mathbb{E} \left[ \text{Var}(S \mid \mathcal{X}_{t-1}^J) \right] \\ &\leq CG'(\theta)^2 \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|}. \end{aligned} \quad (\text{S287})$$

The same decomposition used in Equation (S286) gives

$$\begin{aligned} & \mathbb{E} \mu_t^J \left[ |S - \mu_t^J(S)|^4 \right] \\ &= \mathbb{E} \left[ |S - \mathbb{E}(S \mid \mathcal{X}_{t-1}^J, \xi_{t-1,1:J}^P)|^4 \right] \\ &\leq 16 \mathbb{E} \left[ |S - \mathbb{E}(S \mid \mathcal{X}_{t-1}^J)|^4 \right] \\ &\leq CG'(\theta)^4 \left\{ \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right\}^2. \end{aligned} \quad (\text{S288})$$

For  $t = 1$ , the same two bounds follow directly from  $\text{osc}(h_{1,i}) \leq 2G'(\theta)$  and the fact that the  $r = u = 0$  summand equals one. Equations (S285), (S286), (S287), and (S288) therefore prove both parts of Equation (S273).

*Step 6: continuation when  $t < s$ .* We now propagate a current function from its creation time. For  $t \leq n$ , define

$$\tilde{Q}_{t,n}(f) = \begin{cases} f, & t = n, \\ Q_{t,n-1}\{M_n(f)\}, & t < n. \end{cases} \quad (\text{S289})$$

The definition of  $\Phi_{t,n}$  gives

$$\Phi_{t,n}(\mu)(f) = \frac{\mu \tilde{Q}_{t,n}(f)}{\mu \tilde{Q}_{t,n}(1)}. \quad (\text{S290})$$

For  $t < n$ ,  $\tilde{Q}_{t,n}(1) = Q_{t,n-1}(1)$ , so Equation (S202) gives the first bound below. For every  $t \leq r < n$ , the normalized one-step kernel below satisfies

$$\begin{aligned} & \frac{M_{r+1}(x, du) \tilde{Q}_{r+1,n}(1)(u)}{M_{r+1}\{\tilde{Q}_{r+1,n}(1)\}(x)} \\ &\geq \frac{\underline{M}}{\overline{M}} \frac{\tilde{Q}_{r+1,n}(1)(u) du}{\int \tilde{Q}_{r+1,n}(1)(v) dv}. \end{aligned} \quad (\text{S291})$$

Its Dobrushin coefficient is therefore at most  $\varrho$ . Composing these kernels exactly as in Equations (S206) and (S207) gives

$$\begin{aligned} \sup_{x,x'} \frac{\tilde{Q}_{t,n}(1)(x)}{\tilde{Q}_{t,n}(1)(x')} &\leq C, \\ \text{osc} \left\{ \frac{\tilde{Q}_{t,n}(f)}{\tilde{Q}_{t,n}(1)} \right\} &\leq \varrho^{n-t} \text{osc}(f). \end{aligned} \quad (\text{S292})$$

The same display is immediate at  $t = n$ .

For the current predictor continuation in Equation (S269), differentiating its normalized ratio gives

$$D_\mu H_{t,n}(x) = \frac{\tilde{Q}_{t,n}(1)(x)}{\mu \tilde{Q}_{t,n}(1)} \left\{ \frac{\tilde{Q}_{t,n}(h_{n,i})(x)}{\tilde{Q}_{t,n}(1)(x)} - H_{t,n}(\mu) \right\}. \quad (\text{S293})$$

Equations (S292) and (S148) imply

$$\text{osc}\{D_\mu H_{t,n}\} \leq C G'(\theta) \varrho^{n-t}. \quad (\text{S294})$$

Applying the single-ratio identity (S234) to  $H_{t,n}$  shows that

$$\begin{aligned} |H_{t,n}(\nu) - H_{t,n}(\mu) - (\nu - \mu) D_\mu H_{t,n}| \\ \leq C |(\nu - \mu)(a_H)| |(\nu - \mu) D_\mu H_{t,n}|, \end{aligned} \quad (\text{S295})$$

where the normalized denominator  $a_H = \tilde{Q}_{t,n}(1)/\{\mu \tilde{Q}_{t,n}(1)\}$  satisfies  $\mu(a_H) = 1$  and  $C^{-1} \leq a_H \leq C$ , so  $\nu(a_H) \geq C^{-1}$ . The two empirical factors in Equation (S234) have oscillations at most  $C$  and  $C G'(\theta) \varrho^{n-t}$ , respectively, by Equations (S292) and (S294).

For  $t < s \leq n$ , put  $\mu_s = \Phi_{t,s}(\mu)$  and

$$d_{s,n} = D_{\mu_s} C_{s,n}^{(s)}. \quad (\text{S296})$$

At time  $s$ , the functions  $h_{s,i}$ ,  $q_{s,n}^0$ , and  $q_{s,n}^1$  depend on the current location. Hence Equation (S265) shows that  $d_{s,n}$  is a current function satisfying

$$\text{osc}(d_{s,n}) \leq C G'(\theta) \varrho^{n-s}. \quad (\text{S297})$$

Differentiating Equation (S270) and using Equation (S290) give

$$D_\mu C_{t,n}^{(s)}(x) = \frac{\tilde{Q}_{t,s}(1)(x)}{\mu \tilde{Q}_{t,s}(1)} \left\{ \frac{\tilde{Q}_{t,s}(d_{s,n})(x)}{\tilde{Q}_{t,s}(1)(x)} - \mu_s(d_{s,n}) \right\}. \quad (\text{S298})$$

Equation (S292) applied from  $t$  to  $s$ , followed by Equation (S297), gives

$$\begin{aligned} \text{osc} \left\{ \frac{\tilde{Q}_{t,s}(d_{s,n})}{\tilde{Q}_{t,s}(1)} \right\} &\leq \varrho^{s-t} \text{osc}(d_{s,n}) \\ &\leq C G'(\theta) \varrho^{s-t} \varrho^{n-s} \end{aligned}$$

$$= CG'(\theta)\varrho^{n-t}. \quad (\text{S299})$$

The normalizer bound in Equation (S292) now proves

$$\begin{aligned} \text{osc}\{D_\mu H_{t,n}\} &\leq CG'(\theta)\varrho^{n-t}, \\ \text{osc}\{D_\mu C_{t,n}^{(s)}\} &\leq CG'(\theta)\varrho^{n-t}, \quad t < s \leq n. \end{aligned} \quad (\text{S300})$$

It remains to verify the quadratic remainder for  $C_{t,n}^{(s)}$  when  $t < s$ . Its two normalized ratios are

$$\mathcal{R}_{l,t,s}(\lambda) = \frac{\lambda \tilde{Q}_{t,s}(h_{s,i}q_{s,n}^l)}{\lambda \tilde{Q}_{t,s}(q_{s,n}^l)}, \quad l \in \{0, 1\}, \quad (\text{S301})$$

and  $C_{t,n}^{(s)} = \mathcal{R}_{1,t,s} - \mathcal{R}_{0,t,s}$ . The semigroup recursion gives

$$\tilde{Q}_{t,s}(q_{s,n}^l) = q_{t,n}^l, \quad l \in \{0, 1\}. \quad (\text{S302})$$

Consequently, the normalized denominator factors are precisely the  $a_l$  in Equation (S263), and

$$\text{osc}(a_0) \leq C, \quad \text{osc}(a_1 - a_0) \leq C\varrho^{n-t}, \quad \inf_l \nu(a_l) \geq C^{-1}. \quad (\text{S303})$$

The first variation of the difference is  $D_\mu C_{t,n}^{(s)}$ , already bounded in Equation (S300). For the first single ratio, the chain rule gives

$$D_\mu \mathcal{R}_{1,t,s}(x) = \frac{\tilde{Q}_{t,s}(1)(x)}{\mu \tilde{Q}_{t,s}(1)} \left[ \frac{\tilde{Q}_{t,s}(D_{\mu_s} \mathcal{R}_{1,s,s})(x)}{\tilde{Q}_{t,s}(1)(x)} - \mu_s(D_{\mu_s} \mathcal{R}_{1,s,s}) \right]. \quad (\text{S304})$$

Consequently,

$$\begin{aligned} \text{osc}(D_\mu \mathcal{R}_{1,t,s}) &\leq C\varrho^{s-t} \text{osc}(D_{\mu_s} \mathcal{R}_{1,s,s}) \\ &\leq CG'(\theta). \end{aligned} \quad (\text{S305})$$

The last inequality follows directly from Equation (S233), the bounded normalizer ratios, and  $\text{osc}(h_{s,i}) \leq 2G'(\theta)$ . Apply Claim S2 with

$$a_l = \frac{\tilde{Q}_{t,s}(q_{s,n}^l)}{\mu \tilde{Q}_{t,s}(q_{s,n}^l)}, \quad b_l = \frac{\tilde{Q}_{t,s}(h_{s,i}q_{s,n}^l)}{\mu \tilde{Q}_{t,s}(q_{s,n}^l)}. \quad (\text{S306})$$

Equations (S236), (S303), (S300), and (S305) give

$$\begin{aligned} &\left| C_{t,n}^{(s)}(\nu) - C_{t,n}^{(s)}(\mu) - (\nu - \mu)D_\mu C_{t,n}^{(s)} \right| \\ &\leq C^2 |(\nu - \mu)(a_1 - a_0)| |(\nu - \mu)D_\mu \mathcal{R}_{1,t,s}| \\ &\quad + C |(\nu - \mu)(a_0)| \left| (\nu - \mu)D_\mu C_{t,n}^{(s)} \right|. \end{aligned} \quad (\text{S307})$$

In each product on the right, one oscillation is bounded by  $C\varrho^{n-t}$  and the other by  $CG'(\theta)$ . Together with Equations (S267) and (S295), this proves the required first- and second-order continuation bounds for every  $t, s, n$ .

*Step 7: current and future coefficient sums.* After collecting all score contributions at their local sampling time  $t$ , those with  $t < s \leq n$  have total coefficient

$$\begin{aligned} CG'(\theta) & \sum_{s=t+1}^N \sum_{n=s}^N \alpha^{n-s} \varrho^{s-t} \varrho^{n-s} \\ & \leq CG'(\theta) \left( \sum_{a=1}^{\infty} \varrho^a \right) \left( \sum_{b=0}^{\infty} (\alpha \varrho)^b \right) \\ & = \frac{CG'(\theta) \varrho}{(1-\varrho)(1-\alpha \varrho)}. \end{aligned} \quad (\text{S308})$$

The current predictor terms in Equation (S271) contribute

$$CG'(\theta) \sum_{n=t}^N \varrho^{n-t} \leq \frac{CG'(\theta)}{1-\varrho}. \quad (\text{S309})$$

*Step 8: local particle errors.* We next define the two local terms rather than citing a fixed-terminal formula for their sum. For the signed empirical error  $m(\xi_t^P) - \mu_t^J$ , put

$$\Gamma_{t,i}^J = \{m(\xi_t^P) - \mu_t^J\} \left[ \sum_{n=t}^N D_{\mu_t^J} H_{t,n} + \sum_{n=t}^N \sum_{s=1}^n \alpha^{n-s} D_{\mu_t^J} C_{t,n}^{(s)} \right], \quad (\text{S310})$$

$$\begin{aligned} B_{t,i}^J &= \sum_{n=t}^N \left[ H_{t,n} \{m(\xi_t^P)\} - H_{t,n}(\mu_t^J) - \{m(\xi_t^P) - \mu_t^J\} D_{\mu_t^J} H_{t,n} \right] \\ &+ \sum_{n=t}^N \sum_{s=1}^n \alpha^{n-s} \left[ C_{t,n}^{(s)} \{m(\xi_t^P)\} - C_{t,n}^{(s)}(\mu_t^J) - \{m(\xi_t^P) - \mu_t^J\} D_{\mu_t^J} C_{t,n}^{(s)} \right]. \end{aligned} \quad (\text{S311})$$

Substitute the telescopes Equations (S271) and (S272) into the difference of Equations (S252) and (S253), and then use Equations (S310) and (S311). Term-by-term collection gives the exact identity

$$Z_i^J - z_i^\alpha = \sum_{n=1}^N R_{n,i}^J + \sum_{t=1}^N \Gamma_{t,i}^J + \sum_{t=1}^N B_{t,i}^J, \quad (\text{S312})$$

which uses the same local perturbation construction as Section 5, in particular Equation (5.5), and Appendix A.1 of Del Moral and Rio (2011), with the DMOP sums taken only after the local fields have been formed. Conditionally on the information just before the combined time- $t$  multinomial-selection and mutation block,  $m(\xi_t^P)$  is the empirical measure of  $J$  independent draws with common law  $\mu_t^J$ . We now use the lag moments before summing over the historical index. Within the next calculation only, abbreviate

$$S = \sum_{s=1}^t \alpha^{t-s} h_{s,i}. \quad (\text{S313})$$

Because the functional in Equation (S250) is linear in its displayed  $h_{s,i}$ , for every  $n \geq t$ ,

$$\sum_{s=1}^t \alpha^{n-s} D_{\mu_t^J} C_{t,n}^{(s)} = \alpha^{n-t} D_{\mu_t^J} C_{t,n}^S, \quad (\text{S314})$$

where the superscript  $S$  means only that  $S$  replaces  $h_{s,i}$  in Equation (S250).

Write  $\mu = \mu_t^J$  in this paragraph. With the  $a_l$  from Equation (S263), put  $\mathcal{R}_l^S(\mu) = \mu(a_l S)$ . Since  $\mu(a_l) = 1$  and the  $a_l$  are uniformly bounded,

$$\begin{aligned} |\mathcal{R}_l^S(\mu) - \mu(S)| &= |\mu\{(a_l - 1)(S - \mu S)\}| \\ &\leq C\mu\{(S - \mu S)^2\}^{1/2}, \end{aligned} \quad (\text{S315})$$

while Equation (S265) and  $\|a_1 - a_0\|_\infty \leq C\varrho^{n-t}$  give

$$\begin{aligned} |\mathcal{R}_1^S(\mu) - \mathcal{R}_0^S(\mu)| &= |\mu\{(a_1 - a_0)(S - \mu S)\}| \\ &\leq C\varrho^{n-t}\mu\{(S - \mu S)^2\}^{1/2}, \end{aligned} \quad (\text{S316})$$

Here  $\mu(a_1 - a_0) = 0$  by  $\mu(a_0) = \mu(a_1) = 1$ , so the asserted supremum bound follows from  $\text{osc}(a_1 - a_0) \leq C\varrho^{n-t}$  in Equation (S266), and hence

$$\mu\{(D_\mu C_{t,n}^S)^2\}^{1/2} \leq C\varrho^{n-t}\mu\{(S - \mu S)^2\}^{1/2}, \quad (\text{S317})$$

$$\mu\{|D_\mu C_{t,n}^S|^4\}^{1/4} \leq C\varrho^{n-t}\mu\{|S - \mu S|^4\}^{1/4}. \quad (\text{S318})$$

The fourth-moment inequality follows from the same two displayed identities and Lyapunov's inequality,

$$\{\mu((S - \mu S)^2)\}^{1/2} \leq \{\mu(|S - \mu S|^4)\}^{1/4}. \quad (\text{S319})$$

Minkowski's inequality, Equations (S314) and (S317), and the geometric sum give

$$\begin{aligned} \mu_t^J \left[ \left\{ \sum_{n=t}^N \sum_{s=1}^t \alpha^{n-s} D_{\mu_t^J} C_{t,n}^{(s)} \right\}^2 \right]^{1/2} \\ \leq C\mu_t^J \{(S - \mu_t^J S)^2\}^{1/2} \sum_{n=t}^N (\alpha\varrho)^{n-t} \\ \leq C\mu_t^J \{(S - \mu_t^J S)^2\}^{1/2}. \end{aligned} \quad (\text{S320})$$

All remaining first-order terms are current functions at time  $t$ . Equations (S294), (S300), (S308), and (S309) give their combined  $L^2(\mu_t^J)$  norm at most  $CG'(\theta)$ . Every first variation is centered under  $\mu_t^J$ . Therefore Claim S1 gives

$$\begin{aligned} \mathbb{E}(\Gamma_{t,i}^J \mid \mathcal{F}_{t-1}) &= 0, \\ \mathbb{E}(\Gamma_{t,i}^J)^2 &= \frac{1}{J} \mathbb{E} \mu_t^J \left[ \left\{ \sum_{n=t}^N D_{\mu_t^J} H_{t,n} + \sum_{n=t}^N \sum_{s=1}^n \alpha^{n-s} D_{\mu_t^J} C_{t,n}^{(s)} \right\}^2 \right] \\ &\leq \frac{CG'(\theta)^2}{J} \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|}. \end{aligned} \quad (\text{S321})$$

Here  $\mathcal{F}_{t-1}$  is the sigma-field just before that combined selection–mutation block:  $m(\xi_{t-1}^P)$  is measurable, but the parent selections and mutations used to construct  $m(\xi_t^P)$  have not yet been revealed.

The last inequality is Equation (S273) with  $q = 2$ ; its right side also dominates the squared  $CG'(\theta)$  current-function bound because its  $r = u = 0$  summand equals one.

The same reordering sharpens the quadratic remainder. For each  $n \geq t$ , first sum the terms with  $s \leq t$  inside the two products in Equation (S267). By linearity, the resulting products are the same products with  $\alpha^{n-t}S$  in place of  $h_{s,i}$ . Thus each product has an outside factor  $\alpha^{n-t}$ . Equations (S315)–(S318) give the conditional fourth-moment bound

$$C\varrho^{4(n-t)}\mu_t^J\{|S - \mu_t^J S|^4\}. \quad (\text{S322})$$

To write out the empirical-product step, condition on  $\mathcal{F}_{t-1}$  and apply Cauchy–Schwarz. The exact fourth-moment identity in Equation (S230) gives, for each product on the right of Equation (S267),

$$\begin{aligned} & \mathbb{E} \left[ \left| \left[ \{m(\xi_t^P) - \mu_t^J\}(a_1 - a_0) \right] \left[ \{m(\xi_t^P) - \mu_t^J\} D_{\mu_t^J} \mathcal{R}_1^S \right] \right|^2 \middle| \mathcal{F}_{t-1} \right] \\ & \leq \mathbb{E} \left[ \left| \{m(\xi_t^P) - \mu_t^J\}(a_1 - a_0) \right|^4 \middle| \mathcal{F}_{t-1} \right]^{1/2} \mathbb{E} \left[ \left| \{m(\xi_t^P) - \mu_t^J\} D_{\mu_t^J} \mathcal{R}_1^S \right|^4 \middle| \mathcal{F}_{t-1} \right]^{1/2} \\ & \leq \frac{C\varrho^{2(n-t)}}{J^2} [\mu_t^J\{|S - \mu_t^J S|^4\}]^{1/2}. \end{aligned} \quad (\text{S323})$$

The same calculation, with  $a_0$  and  $D_{\mu_t^J} C_{t,n}^S$ , gives the same right side. Taking the square root of Equation (S323), then applying Equation (S273) with  $q = 4$ , gives

$$\begin{aligned} & \left\| \left[ \{m(\xi_t^P) - \mu_t^J\}(a_1 - a_0) \right] \left[ \{m(\xi_t^P) - \mu_t^J\} D_{\mu_t^J} \mathcal{R}_1^S \right] \right\|_{L^2} \\ & \leq \frac{CG'(\theta)\varrho^{n-t}}{J} \left\{ \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right\}^{1/2}. \end{aligned} \quad (\text{S324})$$

Consequently,

$$\begin{aligned} & \left\| \sum_{n=t}^N \sum_{s=1}^t \alpha^{n-s} \left[ C_{t,n}^{(s)} \{m(\xi_t^P)\} - C_{t,n}^{(s)}(\mu_t^J) - \{m(\xi_t^P) - \mu_t^J\} D_{\mu_t^J} C_{t,n}^{(s)} \right] \right\|_{L^2} \\ & \leq \frac{CG'(\theta)}{J} \left\{ \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right\}^{1/2} \sum_{n=t}^N (\alpha\varrho)^{n-t} \\ & \leq \frac{CG'(\theta)}{J} \left\{ \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right\}^{1/2}. \end{aligned} \quad (\text{S325})$$

For the  $H_{t,n}$  terms and the terms with  $t < s$ , both empirical factors are current functions. Equations (S295), (S307), (S308), and (S309) bound their combined unconditional  $L^2$  norm by  $CG'(\theta)/J$ . Since the  $r = u = 0$  summand again equals one, combining this bound with Equation (S325) gives

$$\|B_{t,i}^J\|_{L^2} \leq \frac{CG'(\theta)}{J} \left\{ \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right\}^{1/2}. \quad (\text{S326})$$

*Step 9: martingale and remainder summation.* Let  $\mathcal{G}_n^P$  be the sigma-field containing all particle-filter randomness through the complete time- $n$  predicted genealogies  $\xi_{n,1:J}^P$ , but not the time- $n$  terminal multinomial-selection uniforms. Then  $R_{m,i}^J$  is  $\mathcal{G}_n^P$ -measurable for  $m < n$ , while conditional independence and centering of the summands in Equation (S245) give

$$\mathbb{E}(R_{n,i}^J \mid \mathcal{G}_n^P) = 0. \quad (\text{S327})$$

The tower property therefore gives, for  $m < n$ ,

$$\begin{aligned} \mathbb{E}(R_{m,i}^J R_{n,i}^J) &= \mathbb{E} [R_{m,i}^J \mathbb{E}(R_{n,i}^J \mid \mathcal{G}_n^P)] \\ &= 0. \end{aligned} \quad (\text{S328})$$

Likewise, expose the combined parent-selection and prediction draws in chronological order. The sigma-field  $\mathcal{F}_{t-1}$  in Step 8 contains every  $\Gamma_{r,i}^J$  with  $r < t$ , and Equation (S321) gives

$$\begin{aligned} \mathbb{E}(\Gamma_{r,i}^J \Gamma_{t,i}^J) &= \mathbb{E} [\Gamma_{r,i}^J \mathbb{E}(\Gamma_{t,i}^J \mid \mathcal{F}_{t-1})] \\ &= 0, \quad r < t. \end{aligned} \quad (\text{S329})$$

These are martingale-difference identities for the particle-sampling randomness, conditional on the fixed observations. They are unrelated to any martingale claim for the observations or for the population discount bias. The two families need not be mutually orthogonal because the selection at time  $n$  also enters the next prediction cloud. Applying  $\|A + B\|_{L^2}^2 \leq 2\|A\|_{L^2}^2 + 2\|B\|_{L^2}^2$  first, and then martingale orthogonality separately within each family, gives

$$\begin{aligned} \left\| \sum_{n=1}^N R_{n,i}^J + \sum_{t=1}^N \Gamma_{t,i}^J \right\|_{L^2}^2 &\leq 2 \sum_{n=1}^N \|R_{n,i}^J\|_{L^2}^2 + 2 \sum_{t=1}^N \|\Gamma_{t,i}^J\|_{L^2}^2 \\ &\leq \frac{CG'(\theta)^2}{J} \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|}. \end{aligned} \quad (\text{S330})$$

For the quadratic terms, Minkowski's inequality gives

$$\begin{aligned} \left\| \sum_{t=1}^N B_{t,i}^J \right\|_{L^2} &\leq \sum_{t=1}^N \|B_{t,i}^J\|_{L^2} \\ &\leq \frac{CG'(\theta)}{J} \sum_{t=1}^N \left\{ \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right\}^{1/2}. \end{aligned} \quad (\text{S331})$$

Combining Equations (S312), (S330), and (S331), and using  $(a + b)^2 \leq 2a^2 + 2b^2$ , yields

$$\mathbb{E}|Z_i^J - z_i^\alpha|^2 \leq CG'(\theta)^2 \left[ \frac{1}{J} \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} + \frac{1}{J^2} \left\{ \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2} \right\}^2 \right]. \quad (\text{S332})$$

Since  $\varrho^{|r-u|} \leq 1$  and  $t \leq N$ ,

$$\sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \leq \left( \sum_{r=0}^{N-1} \alpha^r \right)^2,$$

$$\left\{ \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2} \right\}^2 \leq N^2 \left( \sum_{r=0}^{N-1} \alpha^r \right)^2. \quad (\text{S333})$$

Summing over  $i = 1, \dots, p$  proves Equation (S239). Since  $z_i^\alpha$  is deterministic,

$$\text{Var}(Z_i^J) = \inf_{a \in \mathbb{R}} \mathbb{E}|Z_i^J - a|^2 \leq \mathbb{E}|Z_i^J - z_i^\alpha|^2. \quad (\text{S334})$$

Summing this inequality over the coordinates proves the variance part of Equation (S239).

It remains to evaluate the lag sums. For  $\alpha < 1$  and every  $t$ ,

$$\begin{aligned} \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} &\leq \sum_{r=0}^{\infty} \sum_{u=0}^{\infty} \alpha^{r+u} \varrho^{|r-u|} \\ &= \frac{1}{1-\alpha^2} \left\{ 1 + 2 \sum_{d=1}^{\infty} (\alpha \varrho)^d \right\} \\ &= \frac{1 + \alpha \varrho}{(1-\alpha^2)(1-\alpha \varrho)}. \end{aligned} \quad (\text{S335})$$

Substitution into Equation (S332), followed by summation over the coordinates, proves Equation (S240). At  $\alpha = 1$ ,

$$\begin{aligned} \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \varrho^{|r-u|} &= t + 2 \sum_{d=1}^{t-1} (t-d) \varrho^d \\ &\leq t \frac{1+\varrho}{1-\varrho}, \\ \sum_{t=1}^N t &\leq N^2, \\ \left( \sum_{t=1}^N \sqrt{t} \right)^2 &\leq N^3. \end{aligned} \quad (\text{S336})$$

Equations (S332) and (S336) give the second member of the minimum in Equation (S241). For the first member, Equations (S225) and (S226), together with Equation (S148), give, for every coordinate  $i$ ,

$$\begin{aligned} (1-\alpha) \sum_{t=1}^{N-1} \sum_{k=1}^t \alpha^{t-k} + \sum_{k=1}^N \alpha^{N-k} \\ = \sum_{t=1}^{N-1} (1-\alpha^t) + \sum_{r=0}^{N-1} \alpha^r = N, \end{aligned} \quad (\text{S337})$$

$$\begin{aligned} \left| \partial_{\theta_i} \hat{\ell}^1(\theta) \right| &\leq N G'(\theta), \\ \left| \partial_{\theta_i} \ell^1(\theta) \right| &\leq N G'(\theta), \\ \left| \partial_{\theta_i} \hat{\ell}^1(\theta) - \partial_{\theta_i} \ell^1(\theta) \right|^2 &\leq 4N^2 G'(\theta)^2, \end{aligned}$$

$$\text{Var}\{\partial_{\theta_i} \hat{\ell}^1(\theta)\} \leq N^2 G'(\theta)^2. \quad (\text{S338})$$

Summing Equation (S338) over  $i = 1, \dots, p$  and combining the two valid bounds proves Equation (S241).  $\square$

**Corollary S1** (Particle bias of the DMOP- $\alpha$  score). *Under the conditions of Lemma S13, for  $\alpha \in [0, 1)$ ,*

$$\left\| \mathbb{E} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) - \nabla_{\theta} \ell_n^{\alpha}(\theta) \right\|_2 \leq \frac{C \sqrt{p} G'(\theta)}{J(1-\alpha)}, \quad (\text{S339})$$

$$\begin{aligned} \left\| \mathbb{E} \nabla_{\theta} \hat{\ell}_n^{\alpha}(\theta) - \nabla_{\theta} \ell_n^{\alpha}(\theta) \right\|_2 &\leq \frac{C \sqrt{p} G'(\theta)}{J} \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2} \\ &\leq \frac{CN \sqrt{p} G'(\theta)}{J} \left\{ \frac{1 + \alpha \varrho}{(1 - \alpha^2)(1 - \alpha \varrho)} \right\}^{1/2}. \end{aligned} \quad (\text{S340})$$

*Proof.* Fix a coordinate  $i$  and a conditional-score time  $n$ . Restrict the exact local decomposition in Equation (S312) to this value of  $n$ . The terminal resampling term and every first-order local field have conditional mean zero. The quadratic ratio remainders are the only terms with nonzero expectation. Equations (S267), (S300), and (S231) therefore give

$$\begin{aligned} &\left| \mathbb{E} \partial_{\theta_i} \hat{\ell}_n^{\alpha}(\theta) - \partial_{\theta_i} \ell_n^{\alpha}(\theta) \right| \\ &\leq \frac{CG'(\theta)}{J} \left\{ \sum_{t=1}^n \varrho^{n-t} + \sum_{t=1}^n \sum_{s=1}^t \alpha^{n-s} \varrho^{n-t} + \sum_{t=1}^{n-1} \sum_{s=t+1}^n \alpha^{n-s} \varrho^{s-t} \varrho^{n-s} \right\}. \end{aligned} \quad (\text{S341})$$

The three sums in Equation (S341) satisfy

$$\begin{aligned} \sum_{t=1}^n \varrho^{n-t} &\leq \frac{1}{1-\varrho}, \\ \sum_{t=1}^n \sum_{s=1}^t \alpha^{n-s} \varrho^{n-t} &\leq \left\{ \sum_{a=0}^{\infty} (\alpha \varrho)^a \right\} \left\{ \sum_{b=0}^{\infty} \alpha^b \right\} = \frac{1}{(1-\alpha \varrho)(1-\alpha)}, \\ \sum_{t=1}^{n-1} \sum_{s=t+1}^n \alpha^{n-s} \varrho^{s-t} \varrho^{n-s} &\leq \left\{ \sum_{a=1}^{\infty} \varrho^a \right\} \left\{ \sum_{b=0}^{\infty} (\alpha \varrho)^b \right\} = \frac{\varrho}{(1-\varrho)(1-\alpha \varrho)}. \end{aligned} \quad (\text{S342})$$

Since  $1 - \alpha \varrho \geq 1 - \varrho$ , substitution into Equation (S341) gives

$$\left| \mathbb{E} \partial_{\theta_i} \hat{\ell}_n^{\alpha}(\theta) - \partial_{\theta_i} \ell_n^{\alpha}(\theta) \right| \leq \frac{CG'(\theta)}{J(1-\alpha)}. \quad (\text{S343})$$

Taking the Euclidean norm over the coordinates proves Equation (S339). For the full score, take expectations in the exact decomposition Equation (S312). Equations (S328) and (S329), together with the conditional mean-zero statements that precede them, give

$$\mathbb{E} R_{n,i}^J = 0, \quad \mathbb{E} \Gamma_{t,i}^J = 0. \quad (\text{S344})$$

Therefore Equation (S326) and the triangle inequality give, coordinate by coordinate,

$$\begin{aligned}
|\mathbb{E}Z_i^J - z_i^\alpha| &= \left| \sum_{t=1}^N \mathbb{E}B_{t,i}^J \right| \\
&\leq \sum_{t=1}^N \|B_{t,i}^J\|_{L^2} \\
&\leq \frac{CG'(\theta)}{J} \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2}.
\end{aligned} \tag{S345}$$

Taking the Euclidean norm over  $i = 1, \dots, p$  gives the first inequality in Equation (S340). For  $\alpha < 1$ , Equation (S335) bounds each double sum by  $(1 + \alpha\varrho)/\{(1 - \alpha^2)(1 - \alpha\varrho)\}$ , which proves the second inequality.  $\square$

**Lemma S14** (Population discounting bias). *Under the conditions of Lemma S10, suppose  $\theta = \phi$  and  $\alpha \in [0, 1]$ . Then*

$$\begin{aligned}
\|\nabla_\theta \ell_n^\alpha(\theta) - \nabla_\theta \ell_n(\theta)\|_2 &\leq C\sqrt{p}G'(\theta) \sum_{r=1}^{n-1} (1 - \alpha^r) \varrho^r \\
&\leq C\sqrt{p}G'(\theta) \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}.
\end{aligned} \tag{S346}$$

For the full score, the two finite bounds combine to give

$$\begin{aligned}
&\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\
&\leq C\sqrt{p}G'(\theta) \min \left\{ \sum_{r=1}^{N-1} (N - r)(1 - \alpha^r) \varrho^r, \right. \\
&\quad \left. (1 - \alpha) \sum_{q=0}^{N-2} (N - 1 - q)(\alpha\varrho)^q \right\} \\
&\leq CN\sqrt{p}G'(\theta) \frac{1 - \alpha}{1 - \alpha\varrho}.
\end{aligned} \tag{S347}$$

For  $N = 1$ , both finite sums are empty and the full-score difference is zero.

*Proof. Step 1: exact lagwise difference.* Equation (S162) at  $\alpha$  and at  $\alpha = 1$  gives

$$\begin{aligned}
\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1} (1 - \alpha^r) \Delta_{n,r} \\
&= \sum_{r=1}^{n-1} (1 - \alpha^r) \Delta_{n,r},
\end{aligned} \tag{S348}$$

because  $1 - \alpha^0 = 0$ .

*Step 2: apply forgetting and sum.* Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\begin{aligned}
\|\nabla_{\theta}\ell_n(\theta) - \nabla_{\theta}\ell_n^{\alpha}(\theta)\|_2 &\leq \sum_{r=1}^{n-1} (1 - \alpha^r) \|\Delta_{n,r}\|_2 \\
&\leq C\sqrt{p} G'(\theta) \sum_{r=1}^{n-1} (1 - \alpha^r) \varrho^r \\
&\leq C\sqrt{p} G'(\theta) \sum_{r=1}^{\infty} (1 - \alpha^r) \varrho^r \\
&= C\sqrt{p} G'(\theta) \left\{ \sum_{r=1}^{\infty} \varrho^r - \sum_{r=1}^{\infty} (\alpha\varrho)^r \right\} \\
&= C\sqrt{p} G'(\theta) \left\{ \frac{\varrho}{1 - \varrho} - \frac{\alpha\varrho}{1 - \alpha\varrho} \right\} \\
&= C\sqrt{p} G'(\theta) \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}. \tag{S349}
\end{aligned}$$

*Step 3: exact summation by parts for the full score.* Reindexing Equation (S162) by  $s = n - r$  shows that the contribution of  $h_s$  to the full DMOP- $\alpha$  population score is

$$\begin{aligned}
\rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\
&= \rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \rho_{s+q}(h_s) - \sum_{q=0}^{N-s-1} \alpha^{q+1} \rho_{s+q}(h_s) \\
&= (1 - \alpha) \sum_{q=0}^{N-s-1} \alpha^q \rho_{s+q}(h_s) + \alpha^{N-s} \rho_N(h_s). \tag{S350}
\end{aligned}$$

The contribution of  $h_s$  to the exact score in Equation (S160) is  $\rho_N(h_s)$ . Subtracting Equation (S350) from this contribution gives

$$\begin{aligned}
\rho_N(h_s) - \rho_s(h_s) - \sum_{q=1}^{N-s} \alpha^q \{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\
&= (1 - \alpha) \sum_{q=0}^{N-s-1} \alpha^q \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \tag{S351}
\end{aligned}$$

Summing Equation (S351) over  $s = 1, \dots, N - 1$  yields the exact identity

$$\nabla_{\theta}\ell(\theta) - \nabla_{\theta}\ell^{\alpha}(\theta) = (1 - \alpha) \sum_{s=1}^{N-1} \sum_{q=0}^{N-s-1} \alpha^q \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \tag{S352}$$

*Step 4: contract each partial-smoother comparison.* Fix  $s \leq N - 1$ ,  $0 \leq q \leq N - s - 1$ , and a coordinate  $i \in \{1, \dots, p\}$ . The Markov property implies that observations after time  $s + q$  are conditionally independent of the path through time  $s$  given  $X_{s+q}$ . Hence the following two conditional expectations are the same function of  $x$ :

$$\begin{aligned} \mathbb{E}_{\rho_N}\{h_{s,i} \mid X_{s+q} = x\} &= \mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q} = x\} \\ &= \begin{cases} \bar{h}_{s,i}(x), & q = 0, \\ (B_{s+q}B_{s+q-1} \cdots B_{s+1}\bar{h}_{s,i})(x), & q \geq 1. \end{cases} \end{aligned} \quad (\text{S353})$$

Here  $\bar{h}_{s,i}$  is the conditional projection in Equation (S175); its invariance under later posterior laws is Equation (S178). Also,  $B_t$  is the posterior backward kernel in Equation (S169). Equation (S180) and  $q$  successive applications of Equation (S174) give

$$\begin{aligned} \text{osc}[x \mapsto \mathbb{E}_{\rho_N}\{h_{s,i} \mid X_{s+q} = x\}] \\ \leq 2G'(\theta)(1 - \epsilon)^q \leq 2G'(\theta)\varrho^q. \end{aligned} \quad (\text{S354})$$

Using the tower property under  $\rho_N$  and  $\rho_{s+q}$ , followed by the fact that two probability measures applied to the same bounded function differ by at most its oscillation, therefore gives

$$|\rho_N(h_{s,i}) - \rho_{s+q}(h_{s,i})| \leq 2G'(\theta)\varrho^q. \quad (\text{S355})$$

Summing the squared coordinate bounds and taking square roots yield

$$\|\rho_N(h_s) - \rho_{s+q}(h_s)\|_2 \leq 2\sqrt{p}G'(\theta)\varrho^q. \quad (\text{S356})$$

Insert Equation (S356) into Equation (S352), apply the triangle inequality, and reverse the two finite sums. This gives

$$\begin{aligned} &\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\ &\leq 2\sqrt{p}G'(\theta)(1 - \alpha) \sum_{s=1}^{N-1} \sum_{q=0}^{N-s-1} (\alpha\varrho)^q \\ &= 2\sqrt{p}G'(\theta)(1 - \alpha) \sum_{q=0}^{N-2} (N - 1 - q)(\alpha\varrho)^q \\ &\leq 2N\sqrt{p}G'(\theta)(1 - \alpha) \sum_{q=0}^{\infty} (\alpha\varrho)^q \\ &= 2N\sqrt{p}G'(\theta) \frac{1 - \alpha}{1 - \alpha\varrho}. \end{aligned} \quad (\text{S357})$$

On the other hand, summing the first line of Equation (S346) over  $n = 1, \dots, N$  gives

$$\begin{aligned} &\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\ &\leq C\sqrt{p}G'(\theta) \sum_{n=1}^N \sum_{r=1}^{n-1} (1 - \alpha^r)\varrho^r \\ &= C\sqrt{p}G'(\theta) \sum_{r=1}^{N-1} (N - r)(1 - \alpha^r)\varrho^r. \end{aligned} \quad (\text{S358})$$

Taking the smaller of Equations (S357) and (S358) proves Equation (S347).  $\square$

**Remark 1** (Sharpness of the fixed-observation population order). *The powers of  $N$  and  $1 - \alpha$  in Equation (S347) cannot be improved uniformly under the conditions of Lemma S14. To see this, take state space  $\{0, 1\}$  with counting measure,*

$$X_0 \sim \text{Bernoulli}(3/4), \quad \Pr(X_n = X_{n-1} \mid X_{n-1}) = \frac{3}{4}, \quad (\text{S359})$$

fix  $y_n^* = 1$  for every  $n$ , and, for  $|\theta| \leq 1/4$ , set

$$g_\theta(1 \mid 0) = \frac{1}{4}, \quad g_\theta(1 \mid 1) = \frac{9}{20}(1 + \theta), \quad (\text{S360})$$

with the probabilities at observation zero defined by complementation. At  $\theta = \phi = 0$ ,

$$h_n(x) = x. \quad (\text{S361})$$

If the filtering probability of state one at time  $n - 1$  is  $3/4$ , its one-step prediction and update are

$$\begin{aligned} \Pr(X_n = 1 \mid y_{1:n-1}^*) &= \frac{3}{4} \frac{3}{4} + \frac{1}{4} \frac{1}{4} = \frac{5}{8}, \\ \Pr(X_n = 1 \mid y_{1:n}^*) &= \frac{(5/8)(9/20)}{(5/8)(9/20) + (3/8)(1/4)} = \frac{3}{4}. \end{aligned} \quad (\text{S362})$$

The law of  $X_0$  initializes this induction, so the filtered probability is  $3/4$  at every time. The resulting time-homogeneous posterior backward kernel satisfies

$$B(1 \mid 1) = \frac{(3/4)(3/4)}{5/8} = \frac{9}{10}, \quad B(1 \mid 0) = \frac{(3/4)(1/4)}{3/8} = \frac{1}{2}. \quad (\text{S363})$$

Moreover, the next-observation likelihoods conditional on  $X_{n-1}$  are

$$M_n(G_n)(1) = \frac{3}{4} \frac{9}{20} + \frac{1}{4} \frac{1}{4} = \frac{2}{5}, \quad M_n(G_n)(0) = \frac{1}{4} \frac{9}{20} + \frac{3}{4} \frac{1}{4} = \frac{3}{10}. \quad (\text{S364})$$

Consequently,

$$\begin{aligned} \Pr_{\rho_n}(X_{n-1} = 1) &= \frac{(3/4)(2/5)}{(3/4)(2/5) + (1/4)(3/10)} = \frac{4}{5}, \\ \Pr_{\rho_{n-1}}(X_{n-1} = 1) &= \frac{3}{4}. \end{aligned} \quad (\text{S365})$$

Equation (S363) maps every affine function of the state to an affine function whose slope is multiplied by

$$B(1 \mid 1) - B(1 \mid 0) = \frac{2}{5}. \quad (\text{S366})$$

Applying the  $r - 1$  backward kernels between  $X_{n-r}$  and  $X_{n-1}$  and then using Equation (S365) therefore gives, for  $1 \leq r < n$ ,

$$\begin{aligned} \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}) \\ &= \left(\frac{4}{5} - \frac{3}{4}\right) \left(\frac{2}{5}\right)^{r-1} = \frac{1}{20} \left(\frac{2}{5}\right)^{r-1}. \end{aligned} \quad (\text{S367})$$

All these quantities are nonnegative. Equation (S348), summed over  $n$ , now implies

$$\begin{aligned}\nabla_{\theta}\ell(0) - \nabla_{\theta}\ell^{\alpha}(0) &= \frac{1}{20} \sum_{r=1}^{N-1} (N-r)(1-\alpha^r) \left(\frac{2}{5}\right)^{r-1} \\ &\geq \frac{(N-1)(1-\alpha)}{20}.\end{aligned}\tag{S368}$$

Thus the squared population bias is at least  $(N-1)^2(1-\alpha)^2/400$ . This model satisfies the uniform transition and measurement bounds on  $|\theta| \leq 1/4$ , its simulator is independent of  $\theta$ , and  $h_n$  is a bounded current function. Improving either power uniformly would therefore require an additional signed-cancellation condition, not merely the existing mixing assumptions.

**Theorem S5** (MSE bounds for DMOP- $\alpha$ ). *Under the conditions of Lemmas S13 and S14, for  $\alpha \in [0, 1]$ ,*

$$\begin{aligned}\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 &\lesssim pG'(\theta)^2 \left[ \frac{1}{J} \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right. \\ &\quad \left. + \frac{1}{J^2} \left\{ \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2} \right\}^2 \right. \\ &\quad \left. + \min \left\{ \left\{ \sum_{r=1}^{N-1} (N-r)(1-\alpha^r) \varrho^r \right\}^2, (1-\alpha)^2 \left\{ \sum_{q=0}^{N-2} (N-1-q)(\alpha\varrho)^q \right\}^2 \right\} \right].\end{aligned}\tag{S369}$$

For  $\alpha \in [0, 1)$ , this implies

$$\begin{aligned}\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 &\lesssim pG'(\theta)^2 \left[ \left( \frac{N}{J} + \frac{N^2}{J^2} \right) \frac{1+\alpha\varrho}{(1-\alpha^2)(1-\alpha\varrho)} + N^2 \left\{ \frac{1-\alpha}{1-\alpha\varrho} \right\}^2 \right].\end{aligned}\tag{S370}$$

At  $\alpha = 1$ ,

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^1(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 \min \left\{ N^2, \frac{N^2}{J} + \frac{N^3}{J^2} \right\}.\tag{S371}$$

*Proof.* For reference, direct expansion gives the exact bias-variance identity

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 = \text{Var} \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\} + \left\| \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2,\tag{S372}$$

because

$$\mathbb{E} \left[ \left\{ \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) \right\}^{\top} \left\{ \mathbb{E} \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\} \right] = 0.\tag{S373}$$

For the MSE bound itself, add and subtract  $\nabla_{\theta} \ell^{\alpha}(\theta)$  before squaring. The inequality  $\|a+b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$  gives directly

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2$$

$$\leq 2\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2 + 2 \left\| \nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2. \quad (\text{S374})$$

*Particle error.* Lemma S13 gives the direct particle error

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left[ \frac{1}{J} \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right. \\ & \quad \left. + \frac{1}{J^2} \left\{ \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2} \right\}^2 \right]. \end{aligned} \quad (\text{S375})$$

*Population discounting error.* The full-score bound in Lemma S14 gives

$$\begin{aligned} & \left\| \nabla_{\theta} \ell^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2 \\ & \leq C\sqrt{p}G'(\theta) \min \left\{ \sum_{r=1}^{N-1} (N-r)(1-\alpha^r)\varrho^r, \right. \\ & \quad \left. (1-\alpha) \sum_{q=0}^{N-2} (N-1-q)(\alpha\varrho)^q \right\} \\ & \leq CN\sqrt{p}G'(\theta) \frac{1-\alpha}{1-\alpha\varrho}. \end{aligned} \quad (\text{S376})$$

Insert Equation (S375) and the first three lines of Equation (S376) into Equation (S374). Absorbing the fixed factor 2 into  $\lesssim$  gives

$$\begin{aligned} & \mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left[ \frac{1}{J} \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right. \\ & \quad + \frac{1}{J^2} \left\{ \sum_{t=1}^N \left( \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right)^{1/2} \right\}^2 \\ & \quad \left. + \min \left\{ \left\{ \sum_{r=1}^{N-1} (N-r)(1-\alpha^r)\varrho^r \right\}^2, (1-\alpha)^2 \left\{ \sum_{q=0}^{N-2} (N-1-q)(\alpha\varrho)^q \right\}^2 \right\} \right], \end{aligned} \quad (\text{S377})$$

which proves Equation (S369). For  $\alpha < 1$ , Equation (S335) bounds every finite double sum by

$$\frac{1+\alpha\varrho}{(1-\alpha^2)(1-\alpha\varrho)}. \quad (\text{S378})$$

Therefore the first two particle terms are bounded by

$$\left( \frac{N}{J} + \frac{N^2}{J^2} \right) \frac{1+\alpha\varrho}{(1-\alpha^2)(1-\alpha\varrho)}. \quad (\text{S379})$$

Combining this inequality with the last line of Equation (S376) proves the bound in Equation (S370). The exact bias–variance identity does not create an additional power of  $N$ . Finally, at  $\alpha = 1$ ,  $\nabla_\theta \ell^1(\theta) = \nabla_\theta \ell(\theta)$ . Equation (S241) therefore proves Equation (S371).  $\square$

**Remark 2** (Neutral-model check). *The dependence on  $\alpha$  in Lemma S13 is conservative but cannot be replaced by a bound involving only the current particle cloud. For example, take iid Uniform[0, 1] states, constant weights at  $\theta = \phi = 0$ , and*

$$\Pr_\theta(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{\theta}{4}(2x - 1), \quad y_n^* = 1. \quad (\text{S380})$$

Then  $h_n = (2X_n - 1)/2$  has variance  $\sigma^2 = 1/12$ , and direct conditional multinomial calculation gives

$$\text{Var} \left\{ \nabla_\theta \hat{\ell}^\alpha(\theta) \right\} = \frac{\sigma^2}{J} \left[ N(2 - J^{-1}) + \sum_{r=1}^{N-1} (N - r) \alpha^{2r} (1 - J^{-1})^{r+1} \right]. \quad (\text{S381})$$

Thus the variance has order  $N/J$  for every fixed  $\alpha < 1$ , and its genealogical contribution has order  $N/\{J(1 - \alpha^2)\}$  before horizon saturation. At  $\alpha = 0$ , this agrees with the DMOP-0 bound preceding Theorem S4. The example is consistent with the lag-moment bound proved in Lemma S13.

**Remark 3** (Choice of  $\alpha$  in the fixed-observation upper bound). *The identity  $1 - \alpha^r = (1 - \alpha) \sum_{q=0}^{r-1} \alpha^q$  and  $1 - \alpha^q \leq q(1 - \alpha)$  imply, uniformly in  $N$ ,*

$$\begin{aligned} 0 &\leq (1 - \alpha) \sum_{r=1}^{N-1} (N - r) r \varrho^r - \sum_{r=1}^{N-1} (N - r) (1 - \alpha^r) \varrho^r \\ &\leq N(1 - \alpha)^2 \frac{\varrho^2}{(1 - \varrho)^3}, \\ \sum_{r=1}^{N-1} (N - r) (1 - \alpha^r) \varrho^r &\leq N(1 - \alpha) \frac{\varrho}{(1 - \varrho)^2}. \end{aligned} \quad (\text{S382})$$

No horizon-saturation argument is used for this population term. Combining the sharper particle bound in Equation (S240) with the displayed  $\varrho$ -dependent population upper bound in Theorem S5 gives, with fixed model constants and before the geometric memory reaches the horizon,

$$\begin{aligned} &\mathbb{E} \left\| \nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta) \right\|_2^2 \\ &\lesssim p G'(\theta)^2 \left[ \left( \frac{N}{J} + \frac{N^2}{J^2} \right) \frac{1}{1 - \alpha} + N^2 (1 - \alpha)^2 \right]. \end{aligned} \quad (\text{S383})$$

The factor  $(1 - \alpha)^{-1}$  here is obtained from the exact factor

$$\frac{1 + \alpha \varrho}{(1 - \alpha^2)(1 - \alpha \varrho)}, \quad (\text{S384})$$

and the population term before suppressing fixed model constants is

$$N^2 \left\{ \frac{1 - \alpha}{1 - \alpha \varrho} \right\}^2. \quad (\text{S385})$$

Thus the boundary-comparison argument removes the extra explicit factor  $\varrho^2/(1-\varrho)^2$  from the earlier lagwise simplification. With model constants fixed, it does not change the powers of  $N$ ,  $J$ , or  $1-\alpha$  in the following rate calculation.

Balancing the two terms in Equation (S383) gives

$$1-\alpha \asymp \left( \frac{1}{NJ} + \frac{1}{J^2} \right)^{1/3} \quad (\text{S386})$$

until  $1-\alpha$  reaches order  $N^{-1}$ . Written by regime, choices attaining the resulting upper bounds are

$$\begin{cases} 1-\alpha \asymp J^{-2/3}, & J \lesssim N, \\ 1-\alpha \asymp (NJ)^{-1/3}, & N \lesssim J \lesssim N^2, \\ 1-\alpha \asymp N/J, & J \gtrsim N^2. \end{cases} \quad (\text{S387})$$

and

$$\mathbb{E} \left\| \nabla_{\theta} \hat{\ell}^{\alpha}(\theta) - \nabla_{\theta} \ell(\theta) \right\|_2^2 \lesssim pG'(\theta)^2 \begin{cases} N^2/J^{4/3}, & J \lesssim N, \\ N^{4/3}/J^{2/3}, & N \lesssim J \lesssim N^2, \\ N^2/J, & J \gtrsim N^2. \end{cases} \quad (\text{S388})$$

In the last line, the endpoint  $\alpha = 1$  is already rate-optimal: Equation (S241) gives

$$pG'(\theta)^2 \left( \frac{N^2}{J} + \frac{N^3}{J^2} \right) \lesssim \frac{N^2 pG'(\theta)^2}{J}, \quad J \gtrsim N^2. \quad (\text{S389})$$

Every  $1-\alpha \lesssim J^{-1/2}$  also attains the same order: the nonnegative finite lag sums in Equation (S369) are nondecreasing in  $\alpha$  and hence are bounded by their values at  $\alpha = 1$ , while Equation (S382) bounds the population term by  $CN^2(1-\alpha)^2 \lesssim N^2/J$ .

The explicit finite-sum right side is nevertheless strictly decreased by moving slightly to the left of  $\alpha = 1$ . Indeed, for  $N \geq 2$ ,

$$\begin{aligned} & \left. \frac{\partial}{\partial \alpha} \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} \alpha^{r+u} \varrho^{|r-u|} \right|_{\alpha=1} \\ &= \sum_{t=1}^N \sum_{r=0}^{t-1} \sum_{u=0}^{t-1} (r+u) \varrho^{|r-u|} > 0, \end{aligned} \quad (\text{S390})$$

and the other particle term is also strictly increasing because it is the square of a sum of square roots of these positive increasing sums. In contrast, the squared population term has derivative zero at  $\alpha = 1$ , because its unsquared sum vanishes there.

For completeness, the minimizer of the displayed finite upper bound can be localized when  $J \gg N^2$ . The  $r = 1$  term of the first population bound and the  $q = 0$  term of the second give

$$\min \left\{ \left\{ \sum_{r=1}^{N-1} (N-r)(1-\alpha^r) \varrho^r \right\}^2, (1-\alpha)^2 \left\{ \sum_{q=0}^{N-2} (N-1-q)(\alpha \varrho)^q \right\}^2 \right\} \geq (N-1)^2 \varrho^2 (1-\alpha)^2. \quad (\text{S391})$$

At  $\alpha = 1$ , the same right side has order  $N^2/J + N^3/J^2$ . Hence, if  $\alpha$  minimizes that right side and  $J \gg N^2$ , Equation (S391) implies

$$1 - \alpha = O(J^{-1/2}), \quad N(1 - \alpha) = o(1). \quad (\text{S392})$$

Uniformly in this localized interval, direct differentiation of the particle terms and of either locally active branch of the population minimum gives the one-sided derivative orders

$$\begin{aligned} -\frac{\partial}{\partial(1-\alpha)}\{\text{two particle terms}\} &\asymp \frac{N^3}{J} + \frac{N^4}{J^2}, \\ \frac{\partial}{\partial(1-\alpha)}\{\text{squared population term}\} &\asymp N^2(1-\alpha). \end{aligned} \quad (\text{S393})$$

The population curvature has order  $N^2$ , while the particle curvature is  $O(N^4/J) = o(N^2)$ . Thus the localized minimizer is interior and Equation (S393) gives

$$1 - \alpha \asymp \frac{N^3/J + N^4/J^2}{N^2} \asymp \frac{N}{J}. \quad (\text{S394})$$

The improvement over the endpoint is of lower order; both the interior choice and  $\alpha = 1$  have the  $N^2/J$  upper-bound rate.

In particular, when  $N > J^3$  the first regime applies and the tuned DMOP- $\alpha$  upper bound is  $N^2/J^{4/3}$ , not  $N^2/J$ . Moreover, Equation (S241) and the pathwise bound in Equation (S338) give at  $\alpha = 1$

$$pG'(\theta)^2 \min \left\{ N^2, \frac{N^2}{J} + \frac{N^3}{J^2} \right\} \asymp N^2 pG'(\theta)^2, \quad N > J^3. \quad (\text{S395})$$

Within the regime  $N > J^3$ , the ratio of the two numerical upper-bound expressions is  $J^{4/3}$ . This is a regime-dependent comparison of upper bounds, not a ratio of the actual MSEs.

The power  $J^{-4/3}$  is likewise not a fixed- $N$  or superparametric Monte Carlo rate. It occurs only in the joint regime  $J \lesssim N$ , where the  $N^2/J^2$  quadratic particle-remainder term dominates the  $N/J$  particle-variance term before tuning. For fixed  $N$ , increasing  $J$  eventually enters the last regime, where the displayed rate is the ordinary  $N^2/J$  particle rate.

Oscillation normalization is a separate issue. If a particle theorem is stated for  $\text{osc}(f) \leq 1$  and is applied to a full unnormalized score with oscillation of order  $N$ , its RMSE bound must be multiplied by  $N$ , and its MSE bound by  $N^2$ ; in particular, a norm-one MSE bound of order  $N^2/J$  would become  $N^4/J$ . That rescaling is not used in Lemma S13, which bounds each  $h_s$  before summing its lag covariances. It also must not be applied again to the quadratic asymptotic-variance lower bound of Poyiadjis et al. (2011), which is already stated for an unnormalized additive functional.

This calculation gives rate-balancing choices for the displayed fixed-observation norm upper bound, not the actual MSE. The actual population discount bias is the signed finite sum

$$\nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta) = \sum_{n=1}^N \sum_{r=1}^{n-1} (1 - \alpha^r) \Delta_{n,r}, \quad (\text{S396})$$

whose first-order coefficient can cancel. Consequently, Equation (S388) consists only of upper bounds. It neither identifies the actual MSE rates nor proves equality with the DMOP-1 endpoint. A comparison with the quadratic lower bound of Poyiadjis et al. (2011) additionally requires the nondegeneracy conditions of that result and the identification of its likelihood-ratio estimator with the DMOP-1 estimator in the model under study.

**Remark 4** (Fixed lag and geometric memory). *For  $0 \leq k < N$ , the fixed-lag score and its population truncation error are, respectively,*

$$\sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left( \sum_{s=N-k}^N h_s \right), \quad (\text{S397})$$

$$\sum_{s=1}^{N-k-1} \sum_{u=s+k+1}^N \Delta_{u,u-s}. \quad (\text{S398})$$

Lemma S10 therefore gives

$$\|\text{Bias}_k^{\text{pop}}\|_2 \leq C(N-k-1)\sqrt{p}G'(\theta)\frac{\varrho^{k+1}}{1-\varrho}. \quad (\text{S399})$$

Under the additional current-location premise of Lemma S13, its genealogy calculation also gives the particle bound

$$\begin{aligned} & \max \left\{ \mathbb{E} \left\| \sum_{n=1}^N \sum_{r=0}^{\min(k,n-1)} (\Delta_{n,r}^J - \Delta_{n,r}) \right\|_2^2, \right. \\ & \quad \left. \text{Var} \left( \sum_{n=1}^N \sum_{r=0}^{\min(k,n-1)} \Delta_{n,r}^J \right) \right\} \\ & \lesssim pG'(\theta)^2 \left\{ \frac{N(k+1)}{J} + \frac{N^2(k+1)}{J^2} \right\}. \end{aligned} \quad (\text{S400})$$

Here is the cutoff-specific step needed to obtain this consequence. For  $q \in \{2, 4\}$ , the backward-genealogy calculation in Step 5 of Lemma S13 remains valid when the historical sum is restricted to  $a \leq s \leq b$ . For either conditional path law  $\mu$  appearing there, it gives

$$\begin{aligned} & \mathbb{E} \mu \left[ \left| \sum_{s=a}^b h_{s,i} - \mu \left( \sum_{s=a}^b h_{s,i} \right) \right|^q \right] \\ & \leq CG'(\theta)^q \left\{ \sum_{r=0}^{b-a} \sum_{u=0}^{b-a} \varrho^{|r-u|} \right\}^{q/2} \\ & \leq CG'(\theta)^q (b-a+1)^{q/2}, \end{aligned} \quad (\text{S401})$$

because, for every integer  $L \geq 1$ ,

$$\begin{aligned} \sum_{r=0}^{L-1} \sum_{u=0}^{L-1} \varrho^{|r-u|} &= L + 2 \sum_{d=1}^{L-1} (L-d) \varrho^d \\ &\leq L \frac{1+\varrho}{1-\varrho}. \end{aligned} \quad (\text{S402})$$

Let  $R_{n,i}^{J,k}$ ,  $\Gamma_{t,i}^{J,k}$ , and  $B_{t,i}^{J,k}$  denote, respectively, the terminal-resampling, first-order local, and quadratic local terms obtained from Equations (S310)–(S312) after every coefficient  $\alpha^{n-s}$  is

replaced by  $\mathbf{1}\{0 \leq n - s \leq k\}$ . Coefficient-by-coefficient substitution in that exact decomposition gives

$$\begin{aligned} & \sum_{n=1}^N \sum_{r=0}^{\min(k, n-1)} \{(\Delta_{n,r}^J)_i - (\Delta_{n,r})_i\} \\ &= \sum_{n=1}^N R_{n,i}^{J,k} + \sum_{t=1}^N \Gamma_{t,i}^{J,k} + \sum_{t=1}^N B_{t,i}^{J,k}. \end{aligned} \quad (\text{S403})$$

For terminal resampling at time  $n$ , the retained historical block is  $\sum_{s=\max(1, n-k)}^n h_{s,i}$  and has at most  $k+1$  terms. The conditional empirical second-moment identity and Equation (S401) therefore give

$$\mathbb{E} \left[ \left| m(\xi_n) \left( \sum_{s=\max(1, n-k)}^n h_{s,i} \right) - \Psi_n\{m(\xi_n^P)\} \left( \sum_{s=\max(1, n-k)}^n h_{s,i} \right) \right|^2 \middle| \xi_n^P \right] \leq \frac{CG'(\theta)^2(k+1)}{J}. \quad (\text{S404})$$

The expression inside the absolute value is  $R_{n,i}^{J,k}$ . These terminal terms are martingale differences in  $n$ . At a local particle update time  $t$ , put  $d = n - t$ . If  $d > k$ , the past block is empty. If  $0 \leq d \leq k$ , linearity of the continuation functional gives

$$\sum_{s=\max(1, n-k)}^t D_{\mu_t^J} C_{t,n}^{(s)} = D_{\mu_t^J} C_{t,n}^{(\sum_{s=\max(1, n-k)}^t h_{s,i})}. \quad (\text{S405})$$

This block has at most  $k - d + 1$  terms. Equations (S317)–(S318) and (S401) consequently give, for  $q = 2, 4$ , the corresponding norm bound

$$CG'(\theta) \varrho^d (k - d + 1)^{1/2}. \quad (\text{S406})$$

Summing over the possible values of  $d$  yields

$$\sum_{d=0}^k \varrho^d (k - d + 1)^{1/2} \leq (k+1)^{1/2} \sum_{d=0}^{\infty} \varrho^d = \frac{(k+1)^{1/2}}{1 - \varrho}. \quad (\text{S407})$$

The contributions created after time  $t$  and the current-predictor contributions do not carry the historical block. Their coefficient sums satisfy

$$\begin{aligned} \sum_{s=t+1}^N \sum_{n=s}^{\min(N, s+k)} \varrho^{n-t} &\leq \sum_{a=1}^{\infty} \sum_{d=0}^k \varrho^{a+d} \leq \frac{\varrho}{(1 - \varrho)^2}, \\ \sum_{n=t}^N \varrho^{n-t} &\leq \frac{1}{1 - \varrho}. \end{aligned} \quad (\text{S408})$$

Thus the centered local field at time  $t$  has second moment at most  $CG'(\theta)^2(k+1)/J$ ; explicitly,

$$\mathbb{E}(\Gamma_{t,i}^{J,k})^2 \leq \frac{CG'(\theta)^2(k+1)}{J}. \quad (\text{S409})$$

Repeating the paired-ratio fourth-moment calculation in Equations (S323)–(S325), with Equation (S407) in place of the geometric discount sum, gives

$$\|B_{t,i}^{J,k}\|_{L^2} \leq \frac{CG'(\theta)(k+1)^{1/2}}{J}. \quad (\text{S410})$$

Martingale orthogonality for the terminal and centered local terms, and Minkowski's inequality for the remainders, now give

$$\begin{aligned} & \mathbb{E} \left| \sum_{n=1}^N \sum_{r=0}^{\min(k,n-1)} \{(\Delta_{n,r}^J)_i - (\Delta_{n,r})_i\} \right|^2 \\ & \leq 3\mathbb{E} \left| \sum_{n=1}^N R_{n,i}^{J,k} \right|^2 + 3\mathbb{E} \left| \sum_{t=1}^N \Gamma_{t,i}^{J,k} \right|^2 + 3 \left\| \sum_{t=1}^N B_{t,i}^{J,k} \right\|_{L^2}^2 \\ & = 3 \sum_{n=1}^N \mathbb{E}(R_{n,i}^{J,k})^2 + 3 \sum_{t=1}^N \mathbb{E}(\Gamma_{t,i}^{J,k})^2 + 3 \left\| \sum_{t=1}^N B_{t,i}^{J,k} \right\|_{L^2}^2 \\ & \leq \frac{CNG'(\theta)^2(k+1)}{J} + 3 \left\{ \sum_{t=1}^N \frac{CG'(\theta)(k+1)^{1/2}}{J} \right\}^2 \\ & \leq CG'(\theta)^2 \left\{ \frac{N(k+1)}{J} + \frac{N^2(k+1)}{J^2} \right\}. \end{aligned} \quad (\text{S411})$$

Summing over  $i = 1, \dots, p$  proves Equation (S400); its variance assertion follows because the truncated population target is deterministic. Combining Equations (S399) and (S400) gives

$$\begin{aligned} & \mathbb{E} \left\| \sum_{n=1}^N \sum_{r=0}^{\min(k,n-1)} \Delta_{n,r}^J - \nabla_{\theta} \ell(\theta) \right\|_2^2 \\ & \lesssim pG'(\theta)^2 \left\{ \frac{N(k+1)}{J} + \frac{N^2(k+1)}{J^2} + \frac{N^2 \varrho^{2(k+1)}}{(1-\varrho)^2} \right\}. \end{aligned} \quad (\text{S412})$$

For comparison, in the neutral model of Equation (S381), direct multinomial calculation gives

$$\begin{aligned} & \text{Var} \left\{ \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N) \left( \sum_{s=N-k}^N h_s \right) \right\} \\ & = \sigma^2 \left[ (N-k-1)\{1 - (1-J^{-1})^{k+2}\} + \sum_{a=0}^k \{1 - (1-J^{-1})^{a+2}\} \right]. \end{aligned} \quad (\text{S413})$$

This has order  $N(k+1)/J$  when  $1 \ll k \ll \min(N, J)$ , in agreement with the leading term of Equation (S400).

The scalar geometric-discount bias bound is an exact geometric mixture of the fixed-lag scalar bounds:

$$\sum_{r=1}^{\infty} (1-\alpha^r) \varrho^r = \sum_{q=0}^{\infty} (1-\alpha) \alpha^q \frac{\varrho^{q+1}}{1-\varrho}$$

$$= \frac{\varrho(1 - \alpha)}{(1 - \varrho)(1 - \alpha\varrho)}. \quad (\text{S414})$$

If  $k = \lfloor \alpha/(1 - \alpha) \rfloor < N$ , the ratio of this bound to  $\varrho^{k+1}/(1 - \varrho)$  satisfies

$$\frac{1}{2} < \frac{(1 - \alpha)\varrho^{-k}}{1 - \alpha\varrho} \leq \varrho^{-k}. \quad (\text{S415})$$

Thus the two scalar bounds are within fixed factors when  $k|\log \varrho| = O(1)$ . This comparison does not turn geometric discounting into a hard cutoff: from the exact signed representation,

$$\lim_{\alpha \uparrow 1} \frac{\nabla_{\theta} \ell(\theta) - \nabla_{\theta} \ell^{\alpha}(\theta)}{1 - \alpha} = \sum_{n=1}^N \sum_{r=1}^{n-1} r \Delta_{n,r}, \quad (\text{S416})$$

and the assumptions do not force the right side to vanish. Hence the generic geometric-discount population bias is first order in  $1 - \alpha$ , even though its coefficient contains the forgetting powers  $\varrho^r$ .

## S8 Figures for Bayesian Inference

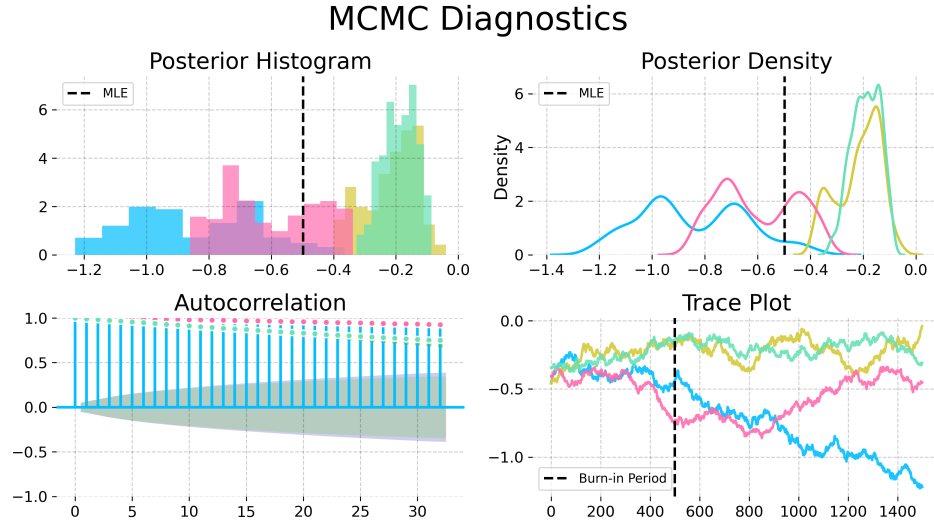


Figure S-1: Convergence diagnostics for the Metropolis-Hastings variant particle MCMC with the random walk proposal and an informative empirical prior from IF2, on the trend parameter in the Dhaka cholera model of King et al. (2008). The poor convergence shown here, using an MCMC algorithm that does not have access to derivatives, contrasts with Fig. 6 in the main text.

## MCMC Diagnostics

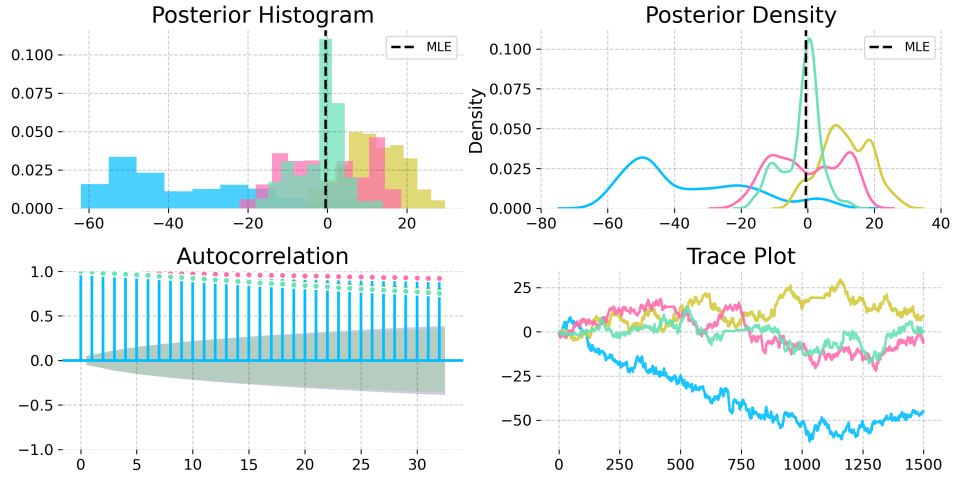


Figure S-2: Convergence diagnostics for a No-U-Turn Sampler (NUTS) with uniform priors on a compact set, on the trend parameter in the Dhaka cholera model of King et al. (2008). The NUTS sampler explores more of the posterior than the Metropolis-Hastings sampler, but fails to converge quickly. This poor convergence, compared with Fig. 6 in the main text, demonstrates the value of the empirical prior for obtaining successful convergence.

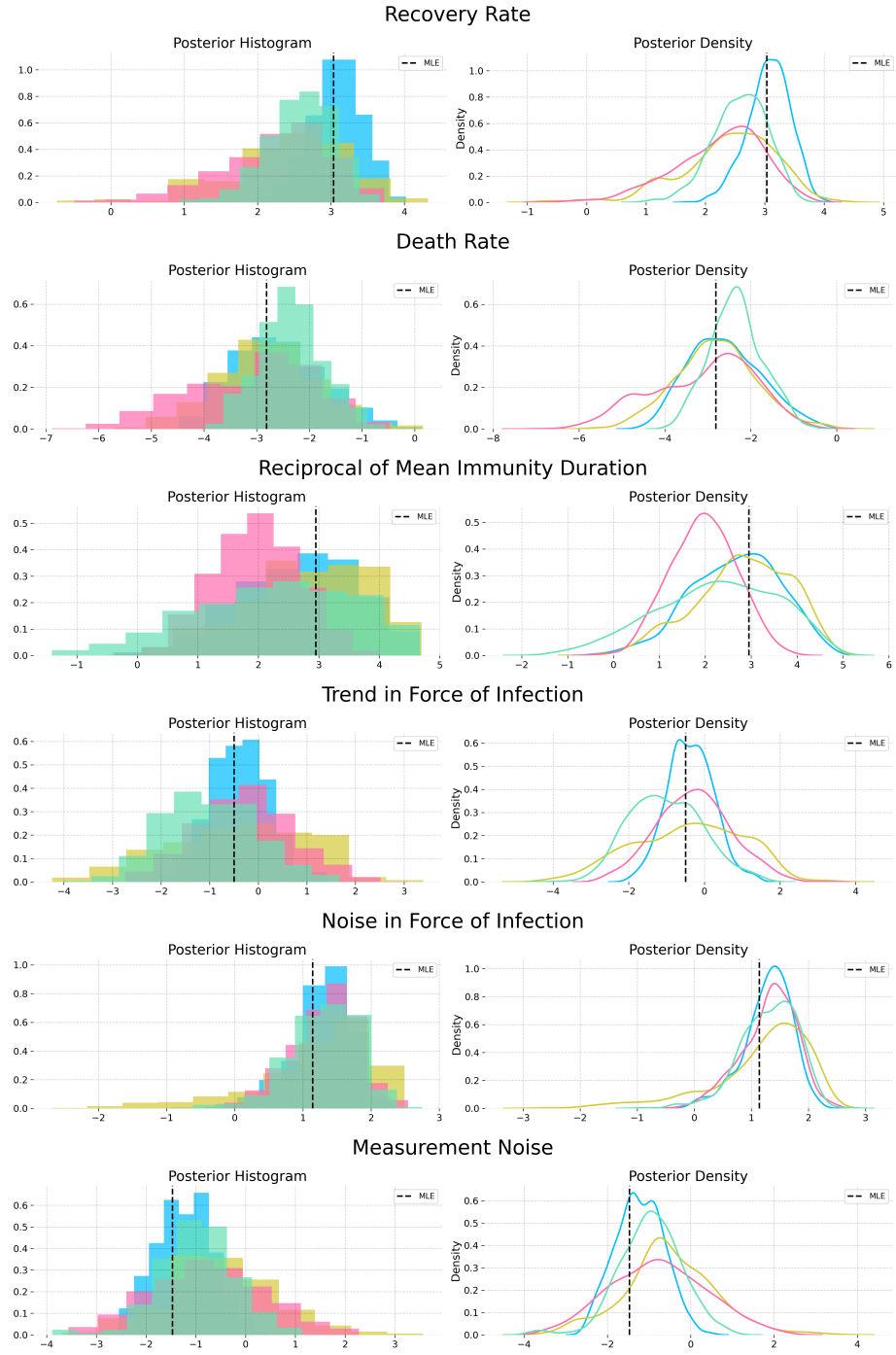


Figure S-3: Posterior estimates from NUTS powered by MOP- $\alpha$  with the informative nonparametric empirical prior obtained from a kernel density estimator on the IF2 parameter cloud. The posterior estimates from each chain are largely in agreement.

## S9 Justification for the Choice of Baseline Comparisons

The reasoning we use in the main text to support the magnitude of our contribution is as follows: (i) iterated filtering methods are the current state of the art, and have been for a decade; (ii) we have demonstrated a substantial advance over iterated filtering. The main text focuses on demonstrating (ii), while supporting (i) via references. Listing a few references amounts only to anecdotal evidence, which is not sufficient to persuade someone properly skeptical about the claim. Therefore, this supplement provides an additional quantitative evaluation.

We count the number of papers published in three top scientific journals (*Science*, *Nature*, and *Proceedings of the National Academy of Sciences of the USA (PNAS)*) that carry out statistical inference for partially observed nonlinear non-Gaussian stochastic dynamic models to address a scientific question by fitting mechanistic models to time series data. For all methods, papers in these high-profile journals are expected to represent only be the tip of the iceberg, but this count provides a concrete measure of the capability of the methods to handle world-class scientific inference problems. We could have considered additional journals, but the three we chose are undoubtedly prestigious and their scope covers the full range of scientific endeavor so there is less room for subfield bias. For iterated filtering, our high-impact application count gives the following: 1. He et al. (2023), 2. Fox et al. (2022), 3. Subramanian et al. (2021), 4. Pei et al. (2021), 5. Li et al. (2020), 6. Giezendanner et al. (2020), 7. Wale et al. (2019), 8. Pons-Salort and Grassly (2018), 9. Ranjeva et al. (2017), 10. Martinez et al. (2016), 11. Becker et al. (2016), 12. Bakker et al. (2016), 13. Laneri et al. (2015), 14. Blake et al. (2014), 15. Blackwood et al. (2013a), 16. Blackwood et al. (2013b), 17. King et al. (2008),

A referee advised consideration of an alternative approach by Gu and Kong (1998) that we found to have a high-impact application count of zero, via Google Scholar.

A referee also recommended comparison with Lindsten et al. (2014), for which the high-impact application count is also zero. Lindsten et al. (2014) has been employed once in *Nature Communications* (Calafat et al., 2018) though that journal did not quite make it onto our elite list. We also found five applications of iterated filtering in *Nature Communications* (Ranjeva et al., 2019; Rodó et al., 2021; Santos-Vega et al., 2022; Briga et al., 2024; Perez-Saez et al., 2025). There is a paper in the Statistics section of *PNAS* by Wang et al. (2022) that demonstrates the method of Lindsten et al. (2014), but here we are counting the demonstrated ability of the methodology to generate high-profile scientific papers, not statistical methodology papers. Otherwise, we would also count Ionides et al. (2006) and Ionides et al. (2015) for iterated filtering.

This reasoning supports iterated filtering as an empirically justified point of reference, in preference to the approaches of Lindsten et al. (2014) and Gu and Kong (1998). Further, the method of Lindsten et al. (2014) does not have the plug-and-play property so it cannot readily be applied to our example. The method of Gu and Kong (1998) would have to be combined with an approach such as Andrieu et al. (2010) that has known scalability problems. Therefore, we focus on comparing with iterated filtering and we do not provide a comparison with Lindsten et al. (2014) or Gu and Kong (1998).

Benchmark tasks are invaluable for making progress on challenging problems (Donoho, 2017). The Dhaka cholera data and model have served as such a benchmark since its introduction by King et al. (2008). Other papers testing methods on this example include Ionides et al. (2015); Fasiolo et al. (2016); Wycoff et al. (2026). Thus, our test is established as a demanding inference problem that challenges the best available methods. We note that this benchmark challenge contains various difficulties that arise commonly in scientific inquiry but are absent from simple SIR models such as

the simulation study considered by Lindsten et al. (2014). Ability to accommodate such features is critical for the methodology to be effective for scientific inquiry.

Iterated filtering has recently been used as a benchmark for new POMP model methodology by Whitehouse et al. (2023). This analysis presented numerical results that claimed to beat iterated filtering, but was subsequently found to contain numerical errors. Corrected results are favorable for iterated filtering methods (Hao et al., 2024; Whitehouse et al., 2025; He et al., 2025).

Recent results of Chen et al. (2025) have extended the previous theoretical understanding of iterated filtering. The ongoing study of iterated filtering as a state-of-the-art approach to inference for POMP models gives additional support for our decision to use iterated filtering as a benchmark for our new methodology.

## S10 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-1. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time  $t_0$ , are treated differently: they are perturbed only at time  $t_0$ , with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

| Hyperparameter              | Standard IF2 | IFAD-0       | IFAD-0.97    | IFAD-1       |
|-----------------------------|--------------|--------------|--------------|--------------|
| Number of Particles ( $J$ ) | 5,000        | 5,000        | 5,000        | 5,000        |
| Warm-start Iterations       | 650          | 60           | 175          | 60           |
| GD Iterations               | —            | 225          | 175          | 225          |
| IF2 Cooling Rate            | 0.8          | 0.5          | 0.5          | 0.5          |
| Initial RWSD                | 0.02         | 0.02         | 0.02         | 0.02         |
| Initial RWSD (IVPs)         | 0.16         | 0.16         | 0.16         | 0.16         |
| Gradient Optimizer          | —            | Adam         | Adam         | Adam         |
| Momentum ( $\beta_1$ )      | —            | 0.0          | 0.9          | 0.9          |
| LR Schedule                 | —            | Cosine Decay | Cosine Decay | Cosine Decay |
| Initial LR ( $\eta_0$ )     | —            | 0.1          | 0.1          | 0.1          |
| Warm-up Steps               | —            | 10           | 10           | 10           |
| Eval. Particles / Reps      | 5,000 / 36   | 5,000 / 36   | 5,000 / 36   | 5,000 / 36   |

Table S-1: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemented by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2014). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not require it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

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